Moon Formation by Triple Phase Transition in the Differentiating Proto-Earth

A dynamical endogeneity: a single mechanism that jointly sets lunar mass, rotation, and angular momentum

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Abstract. The Moon is there. It is a fact, not a hypothesis to be confirmed: our task is to trace back its causes. Just as our own existence attests that the improbable chain of conditions that made it possible did indeed occur, the existence of the Moon — with its mass, its isotopic identity, and the Earth–Moon system angular momentum we observe — imposes a specific set of conditions on the state of the Hadean proto-Earth. This theory does not seek to adjust free parameters to reproduce a desired outcome: it proceeds by *reconstruction*. The values we derive from it (radial density gradient, obliquity, angular momentum input) are not chosen; they are constrained by an outcome that is itself certain. That a window of conditions is narrow is therefore not a fragility, but a retrodictive prediction about the early Earth. The number of steps in the causal chain does not measure an excess of degrees of freedom, but a degree of specification: a long yet fully constrained chain remains more predictive than a model with a few free parameters.

Several well-established characteristics of the Earth–Moon system — the near-isotopic identity of the Earth and Moon[1–4], the crustal dichotomy[5], the residual inclination of $\approx 5^\circ$ of the lunar orbit to the terrestrial orbital plane, and the ≈ 350 Myr delay of the terrestrial dynamo[6] — do not find a fully satisfactory explanation in current lunar formation models. We propose an endogenous alternative, based on a necessary thermodynamic observation: any Earth-mass planet emerges from accretion in a state of near-total melting of the silicate mantle, accretion energy exceeding the fusion energy by a factor ≈ 121[7–9]. The proto-Earth is thus a rapidly rotating ($T_{\rm rot} \approx 3.5$ h) magmatic body, without a stabilizing satellite, whose axis undergoes free nutation of large amplitude in the interval $[40^\circ, 70^\circ]$ measured in its own co-rotating frame — an internal geometric oscillation of the fluid body[10–12].

A single engine — the progressive segregation of metallic iron-nickel toward the forming core — drives three coupled transitions. *(i)* A rheological transition: the magma acquires a Bingham-Herschel yield stress[13,14] and organizes a Coherent Magmatic Torus (CMT) in the intertropical belt $|\phi| < 30^\circ$. *(ii)* A mechanical transition: the CMT undergoes $N = 2$–3 episodes of cohesive hypersonic ejection. Ejection proceeds from a **deterministic dynamical instability** — a parcel leaves the torus when the Solberg–Høiland stability coefficient $\lambda(U)$

becomes positive. The threshold is reached cyclically: angular momentum build-up, ejection of a cohesive magmatic tongue at nutation peaks, loss of momentum, recharge. Each ejection shifts the rotation axis to a new obliquity, and thus the active belt itself: each successive CMT is geometrically distinct, which underpins the observed crustal dichotomy. *(iii)* A magnetic transition: the dynamo starts ≈ 350 Myr after the end of ejections, consistent with Jack Hills zircons[6].

With an ejected mass of 2.2–4.0 $\times 10^{22}$ kg per episode and a capture efficiency of ≈ 0.70, the lunar mass is reconstructed without an external impactor. Isotopic identity and depletion in metallic iron (manifested as a small lunar core) follow as direct mechanical consequences. The angular momentum budget, however, requires a post-formation external sink of $\approx 3 \times 10^{34}$ J s. Reconstruction identifies it: the current angular momentum *and* the lunar inclination of 5° jointly impose an instability episode at the crossing of the Laplace plane transition[15,16], whose activation condition — an inertial obliquity $\varepsilon_{\mathrm{I}} \gtrsim 60°$ — is precisely what the TTP derives independently from the absence of a stabilizer. Two independent paths thus converge on the high Hadean obliquity (Section 12); the evection resonance[16,17] is downgraded to a secondary channel, and quantitative validation of the connection remains required (Limitation L15, Section 20). The distinction between metallic iron and oxidized iron is explicitly addressed in Section 16.4.

The central prediction — key to falsifiability — is the existence of one or more seismic interfaces between 200 and 530 km depth, with impedance contrast $|R| \in [0.01, 0.04]$, testable imminently by **Chang'e 7** (South Pole, August 2026), and then by **FSS**, **LEMS**, and **Artemis III** (2028–2029). Preliminary support comes from Chang'e-6 samples[18,19]: norites dated at 4247 ± 5 Ma and high Fe/Mn ratios in deep olivines, consistent with predictions P3 and P6. This retrodictive framework establishes the coherence and physical plausibility of the scenario; its distinction from the giant impact rests entirely on these observational predictions. If they are disconfirmed, the theory must also be — and we wish, in that case, to one day learn what actually built the Moon.

Interactive simulation: https://orion4622.github.io/moon-formation-triple-phase-transition/Animation_Formation_de_la_Lune_v2.html

Keywords: origin of the Moon · Triple Phase Transition · chaotic axial oscillation · Hadean proto-Earth · Bingham-Herschel · Coherent Magmatic Torus · generalized instability coefficient · parametric elliptic instability · Laplace plane transition · seismic interface · Hadean dynamo · falsifiable predictions · South Pole–Aitken · Chang'e-6.

Contents

1 Introduction: why a new theory of lunar formation?

1.1 Observational constraints open on lunar formation

Although lunar formation has been studied for several decades, a number of observational constraints remain open, which any formation scenario must explain. Five of them are presented here, and directly motivate the approach developed in this work.

1. **Earth–Moon isotopic identity**: the isotopic compositions of oxygen, titanium, tungsten, and chromium are nearly identical between Earth and Moon, to better than 5 ppm[1–4]. Such similarity imposes a strong constraint on the origin of lunar material: either it comes from reservoirs already isotopically identical to Earth's, or a complete homogenization mechanism erased any initial difference between distinct reservoirs. Determining which of these two paths actually occurred remains an open question in lunar cosmochemistry.

2. **Small lunar core**: the Moon has a metallic core of radius ≈ 330 km, significantly smaller than Earth's core relative to the mass of the host planet[20,21]. The reason for this depletion in metallic iron — namely, at what time, and by what mechanism, lunar material lost most of its metallic iron, before or during lunar formation — remains to be specified.

3. **Crustal dichotomy**: the lunar farside is thicker (≈ 54 km) and older than the nearside (≈ 34 km)[5]. The origin of this hemispheric asymmetry, striking on a global scale, is not established with certainty.

4. **Earth's dynamo delay**: Earth's magnetic field is detected in Jack Hills zircons at 4.2 Ga[6], i.e., about 350 Myr after the estimated formation of the Moon (4.55 Ga). The physical mechanism linking lunar formation to the delayed onset of Earth's dynamo remains to be explained.

5. **Angular momentum and lunar orbital inclination**: any scenario in which the proto-Earth rotates rapidly — which includes most modern variants of the giant impact[17,22] as well as the present work — must evacuate an excess angular momentum of the order of the current Earth–Moon system's angular momentum to an external sink. Simultaneously, the lunar orbit retains a residual inclination of $\approx 5^\circ$ to the terrestrial orbital plane, incompatible with assembly in an equatorial disk followed by regular tidal evolution (the "lunar inclination problem,"[11]). These two anomalies, taken together, point retrodictively to the same post-formation dynamical episode — an instability at the Laplace plane transition[15,16] — whose activation condition is a high obliquity of the proto-Earth. This retrodictive thread runs throughout the present work (Sections 3, 9.2, 11.6 and 12).

A recent review highlights the absence, to date, of unambiguous geochemical or isotopic evidence allowing a decisive choice between the different proposed formation scenarios[4]. This situation motivates the exploration of approaches based on the internal dynamics of the proto-Earth.

1.2 The Triple Phase Transition: an endogenous alternative

This work proposes that the Moon did not form by an external impact, but by an endogenous process within the proto-Earth itself during its differentiation. The central thesis is as follows.

Hypothesis H0 (central thesis). The Moon formed through a series of *three coupled transitions* — rheological, mechanical, and magnetic — triggered by the single engine of Fe-Ni segregation toward the forming core. The result is a Moon built layer by layer in two to three episodes of cohesive magmatic ejection, each layer archiving the state of Hadean differentiation at the time of its accretion.

The conceptual core: a dynamical *endogeneity*

The strength of the TTP lies not only in the Moon coming from the Earth, but in that a **single mechanism jointly determines** several quantities that impact scenarios treat as free parameters adjusted a posteriori: lunar mass, initial angular momentum, obliquity, the vector geometry of ejecta, and Earth's rotation rate. Because the third ejection can only occur near a certain energy state of the planet, these quantities are not independent: they are *linked* by the physics of ejections. The initial state of the Earth–Moon system is therefore not a point chosen in a vast parameter space, but would belong to a **reduced region of phase space** toward which the mechanism naturally confines the system. The TTP does not say "we choose a favorable initial state"; it proposes that "the physics of the three ejections may lead to this state." This economy of free parameters — fewer degrees of freedom to explain as many facts — is the model's central argument, and a testable prediction distinct from the free adjustment of impact scenarios.

The mechanism relies on a necessary thermodynamic observation: any Earth-mass planet emerges from accretion in a state of near-total melting of the silicate mantle[7–9]. Accretion energy exceeds the total fusion energy by a factor ≈ 121. The proto-Earth is therefore a rapidly rotating ($T_{\rm rot} \approx 3.5$ h) magmatic body without a stabilizing satellite.

In this rapidly rotating fluid body, the rotation axis undergoes free nutation of large amplitude in the interval $[40°, 70°]$ in the co-rotating frame[10–12] — not an astronomical obliquity, but an internal oscillation of the fluid body. This oblique inclination is the geometric condition that allows the radial Coriolis force ($f_{\rm rad} = 2\Omega U \sin \varepsilon$) to destabilize the flow rather than confine it.

1.3 Organization of the manuscript

The manuscript is organized as follows. Section 2 presents the Hadean physical context (rotation, obliquity, silicate atmosphere). Section 5 presents the intertropical belt as the ejection site. Section 6 responds to the Taylor-Proudman objection. Section 7 inventories the forces. Section 8 explicitly classifies all hypotheses. Section 9 presents the three coupled transitions (rheological, mechanical, magnetic) with the complete resolution of the instability coefficient. Section 10 presents the mass budget. Section 11 presents angular momentum transport. Section 12 identifies the post-formation angular momentum sink — the Laplace plane transition — and closes the retrodictive thread opened by constraint 5 of Section 1. Section 14 gives the 3- to 4-week chronology. Section 15 presents the internal stratification. Section 16 discusses observational constraints and existing evidence. Section 17 presents falsifiable predictions. Section 18 presents validation windows. Section 19 discusses the main objections. Section 20 presents the limitations. Section 21 concludes.

2 The Hadean inferno: physical context

Before diving into the physics of the Triple Phase Transition, it is essential to establish a structural fact, not merely a descriptive one, about the framework in which it operates: no stationary state is physically accessible there. This is not a turbulent period among others in Solar System history — it is a regime where every process that could, in isolation, bring the system back toward equilibrium is itself perturbed, on a timescale shorter than its own relaxation time, by at least one other independent process. It is this structural absence of rest, and not an inventory of individual circumstances, that this section aims to establish: it directly conditions the reading of Section 3 (the axial oscillation is never damped for lack of a stabilizer) and, later, that of Section 9.2 (each ejection episode occurs not on a system at rest, but on a system already and continuously out of equilibrium).

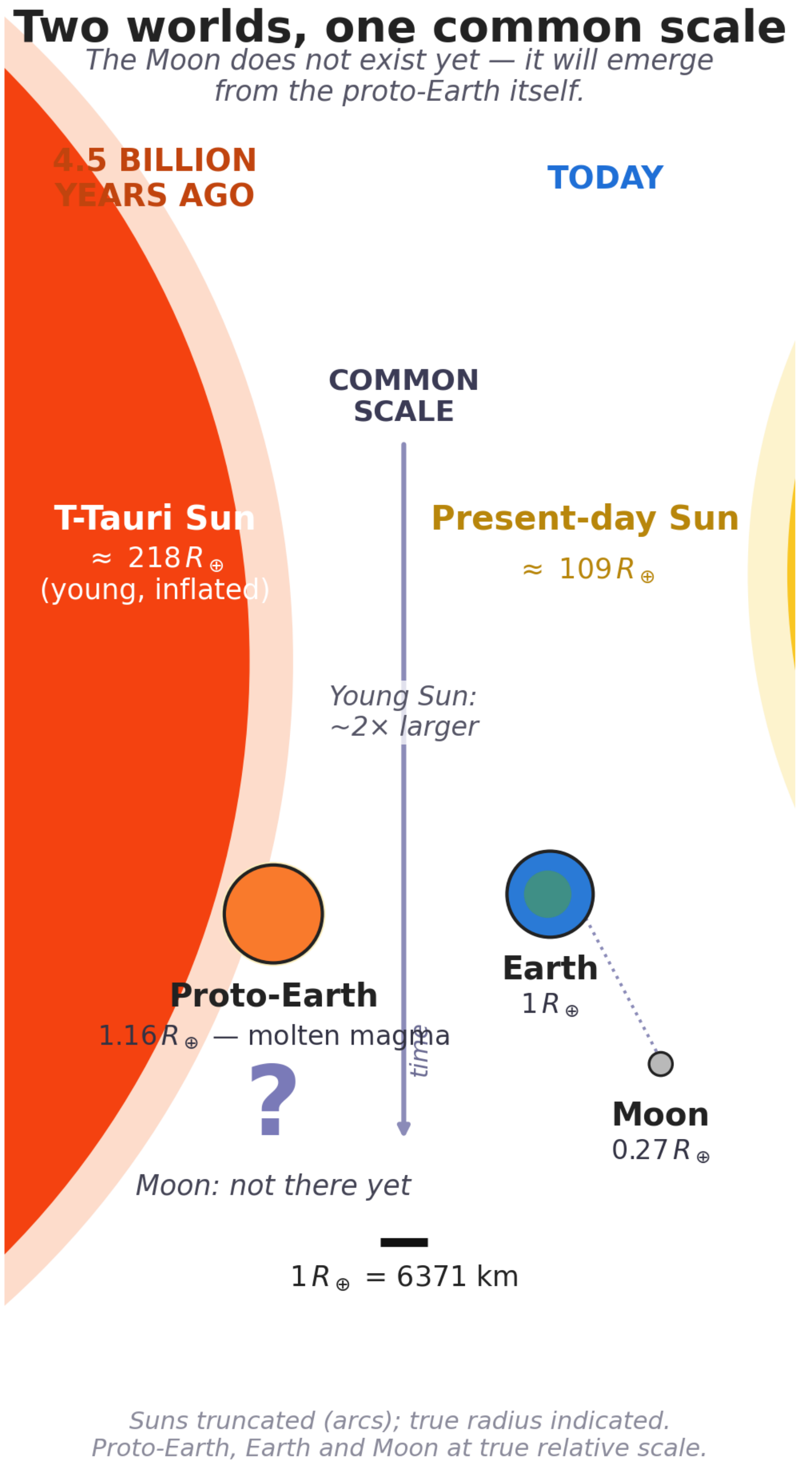

Figure 1: Two worlds, same scale. Left, 4.5 billion years ago: the T-Tauri Sun, young and expanded (≈ 218 $R_\oplus$), and the molten proto-Earth (1.16 $R_\oplus$) — the Moon does not yet exist; it will emerge from the proto-Earth itself. Right, today: the current Sun (≈ 109 $R_\oplus$), Earth (1 $R_\oplus$), and Moon (0.27 $R_\oplus$). Suns truncated (arcs), actual radius indicated; proto-Earth, Earth, and Moon at true relative scale.

2.1 The dynamical context of terminal accretion

The Hadean period (4.568–4.4 Ga) superimposes a set of physical processes whose characteristic timescales overlap without any one dominating long enough to impose a stable regime:

- **Continuous planetesimal accretion**: the protoplanetary disk remains populated with planetesimals and embryos, each major impact occurring every 10^3–10^5 yr[23–25] — an interval much shorter than the characteristic viscous relaxation time of the fluid body (Section 3), so that the effect of one impact never has time to fade before the next.
- **Rapid rotation**: the proto-Earth rotates with a period $\approx$ 3.5 h, a regime in which Coriolis and centrifugal effects dominate the internal dynamics and amplify perturbations rather than damping them.
- **Absence of a tidal stabilizer**: without the Moon, obliquity is free to evolve over a wide chaotic range, with no restoring mechanism toward an equilibrium value.
- **A young and hyperactive Sun**: T-Tauri stars produce flares and coronal mass ejections (CMEs) at a frequency that can reach 0.2–0.4 events per day for superflares with confirmed mass ejection, and more than one event per day for white-light flares alone, based on direct observations of young Sun-like stars ($\sim$ 30–125 Myr)[26], versus about one superflare per few thousand years for the current Sun[27,28] — i.e., a stochastic forcing whose characteristic timescale is daily, whereas that of planetesimal impacts is millennial and that of free nutation is also of the order of a day (Section 3): these three clocks never synchronize; they overlap.
- **Short-lived radioactivity**: isotopes ^{26}Al and ^{60}Fe are still active, injecting heat into the mantle continuously[29,30], with no thermal rest phase.
- **Dense silicate atmosphere**: an envelope of rock vapor at $T \sim 2\,000$–$4\,000$ K and $P \sim 10^5$–10^7 Pa constitutes a thermal blanket and a lateral confinement agent[31–33], but does not itself stabilize anything: it extends the window during which the preceding processes can continue to act.

It is not the list of these processes that matters, but their simultaneity. Taken separately, each could in principle average out over time and allow a quasi-stationary behavior to emerge; taken together, at frequencies that are commensurate on no common timescale, they structurally forbid any such regime from establishing itself. The T-Tauri Sun, although it provides one of the most frequent forcings, is therefore not the cause of Hadean chaos: it is only one contributor among others, whose addition makes any equilibrium inaccessible. It is in this already and continuously out-of-equilibrium context that the ejection episodes themselves will occur (Section 9.2): not as isolated perturbations striking an otherwise calm system, but as additional shocks added to a system that has, at no point in its active history, truly rested.

2.2 The silicate atmosphere: thermal blanket and lateral confinement

At $t = 0$, the proto-Earth is enveloped in a dense silicate atmosphere — rock vapor, gaseous SiO, Mg, and Fe — at temperatures of 2,000 to 4,000 K and pressures of 10^5–10^7 Pa. This cocoon plays two critical roles in the TTP.

First role: thermal insulation. The silicate atmosphere has a high infrared opacity. It reduces the surface radiative cooling by a factor 10^4 to 10^5 [31–33]. Without this atmosphere, the proto-Earth would have cooled in $\sim 10^4$ yr. With it, the melting window extends to 30–50 Myr — exactly the timescale of core formation[34,35]. This is not an ad hoc hypothesis: it is the physical consequence of a rock-vapor atmosphere at 2 000–4 000 K. The same mechanism operates in magma ocean models[31,32].

Second role: lateral confinement. The confinement pressure ($P \approx 10^5$–10^7 Pa) opposes the lateral expansion of the CMT flow during ejection. It guarantees the cohesion of the ejected sheet. The high sound speed in this torrid atmosphere ($c_{\mathrm{son}} \approx 1\,500$ m s^{-1} at $T_{\mathrm{atm}} \approx 3\,000$ K) reduces the effective Mach number of the flow, improving cohesion compared to an ejection into vacuum.

Without this atmosphere, the melting window would close in $\sim 10^4$ yr. With it, the window extends to 30–50 Myr[31–33], which corresponds to the core formation timescale[34,35].

2.3 Continuous planetesimal bombardment

The protoplanetary disk remains populated with planetesimals and embryos during the active TTP window[23–25]. Each impact: (1) reinjects thermal energy into the mantle, sustaining melting; (2) generates a spherical pressure wave propagating through the entire melt pool, perturbing $U(r)$; (3) delivers an impulsive torque on the spin vector, resetting the axial oscillation.

2.4 Short-lived radioactivity: internal heat source

The isotopes ^{26}Al (half-life 0.72 Myr) and ^{60}Fe (half-life 2.6 Myr) are still active during the first million years[29,30], injecting $\approx 3 \times 10^{29}$ J of additional heat into the mantle and contributing to sustained melting throughout the TTP window.

3 Hadean obliquity $\varepsilon \in [40°, 70°]$: a fundamental geometric clarification

The interval $\varepsilon \in [40°, 70°]$ is not a free parameter. It is constrained by five independent and convergent physical arguments. Before enumerating them, a fundamental geometric clarification is necessary.

Geometric clarification. In this work, ε denotes the instantaneous angle between the rotation vector $\mathbf{\Omega}(t)$ and the principal inertia axis of the Maclaurin spheroid, **defined in the body's co-rotating reference frame**. It is *not* the astronomical obliquity relative to the modern ecliptic: this concept has no physical meaning for the Hadean proto-Earth, which possesses neither a Moon nor a stabilized axis. It is a purely internal geometric angle describing the *oscillation* — the free nutation — of the rotation axis relative to the body's own figure. A second distinct angular quantity will intervene after lunar formation: the *inertial obliquity* ε_{I}, defined as the angle between the rotational angular momentum vector $\boldsymbol{L}_p$ and the normal to the proto-Earth's heliocentric orbital plane — two contemporary and locally defined directions, without reference to the current ecliptic. The translation from co-rotating nutation ε to inertial obliquity ε_{I} is established in Section 12.3.1; it is ε_{I} that commands the post-formation angular momentum sink (Section 12).

Definition of the intertropical belt. The Hadean intertropical belt $|\phi| < 30^\circ$ is defined perpendicular to the *instantaneous* rotation axis $\mathbf{\Omega}(t)$, and not with respect to a fixed geographic reference frame. The plane $\phi = 0$ is therefore the *instantaneous dynamical equator* of the body, which oscillates with the free nutation of $\mathbf{\Omega}(t)$ — it is not a fixed geographic equator. The coordinate ϕ is the oblique latitude measured from this instantaneous equator. This is the plane where the radial Coriolis force $f_{\mathrm{rad}} = 2\Omega U \sin\varepsilon$ is maximal and where the effective gravity $g_{\mathrm{eff}}(\phi)$ is minimal (Section 4.3). The belt is therefore a dynamical structure tied to the instantaneous equator, not a fixed latitude.

It follows directly from the absence of any external *stabilizing* reference plane at this epoch — neither modern ecliptic, nor lunar orbit, nor effective restoring torque; the proto-Earth's heliocentric orbital plane exists, but exerts no torque capable of freezing $\mathbf{\Omega}(t)$ — that the very notion of an "intertropical" belt is not geographic but purely relative, defined by this internal axis rather than by a solar or ecliptic reference frame that does not yet exist. Throughout this work, terms evoking ordinary planetary geography ("intertropical," "equator," "latitude") should be read as a formal analogy with the geometry of a rotating spheroid, and not as a reference to an astronomy that had not yet occurred at that epoch.

Argument 1 — Absence of a tidal stabilizer (fundamental). The proto-Earth has no Moon. Without a massive satellite, there is no lunar tidal torque to damp the spin-axis precession and keep it near a stable equilibrium. This argument is logically prior to all others: it is precisely the formation of the Moon that we seek to explain, so invoking it as a stabilizer would be circular. The absence of a tidal stabilizer is therefore a necessary, not contingent, condition of the Hadean proto-Earth.

Argument 2 — Laskar et al. (1993): a chaotic zone without a satellite. Laskar et al. [10] showed that without its Moon, Earth's obliquity would evolve chaotically over a zone extending from nearly $0°$ to $\approx 85°$. Although this result strictly applies to a current Earth rotating in ≈ 24 h, it establishes the general principle: *the Moon is the stabilizer, and without it, large obliquity excursions are the norm*. For the Hadean proto-Earth, this principle applies *a fortiori*.

Argument 3 — Rapid rotation at $T_{\rm rot} = 3.5$ h. The precession constant satisfies $\alpha_{\rm prec} \propto \omega$. At $T_{\rm rot} = 3.5$ h, the precession frequency is $\approx 7\times$ higher than at 24 h, pushing beyond reach the spin-orbit secular resonances identified by Laskar et al.[10]. Li & Batygin[12], working on a Moonless Earth in rapid rotation, found that obliquity remains chaotic but *diffuses slowly* over a wide range. Rapid rotation does not stabilize the axis — it merely changes the nature of the chaos, from resonant to diffusive.

Argument 4 — Continuous bombardment and stochastic chaos. Each major impact during terminal accretion[23] brutally resets the spin vector and the shape of the fluid body. For a low-viscosity fluid ($\eta \sim 10^{1-3}$ Pa s), the viscous relaxation time toward axial alignment, of the form $\tau \sim \eta/(\rho g R_{\rm eq,proto})$[36], is extremely short ($\ll 1$ yr) compared to the interval between major impacts (10^3–10^5 yr): $\tau_{\rm relax} \ll \tau_{\rm pert}$. The system thus realigns between each impact and spends more than 99.9% of the time aligned. Obliquity is therefore not "continuously maintained" in the sense of a permanent sustenance of $\varepsilon > 45°$.

What impacts ensure, on the other hand, is that the system never reaches a stationary state. Each impact resets the initial conditions of the free nutation, and the continuous stochastic forcing (CMEs, bombardment, late accretion) is distributed over timescales that are not commensurate. Obliquity is therefore not stabilized; it is permanently *forced* toward new values. The interval $\varepsilon \in [40°, 70°]$ is not maintained by a slow relaxation mechanism, but by the absence of a stabilizer and the persistence of stochastic forcing that prevents the system from freezing. It is this absence of rest — and not an active maintenance — that makes large obliquity excursions accessible.

Argument 5 — Torques from T-Tauri CMEs. As described in Section 7.4, the impulsive torques from CMEs act on timescales of days to weeks, comparable to $\tau_{\rm pert}$, and constitute a continuous stochastic source of obliquity perturbation, in addition to impacts[27,28].

Argument 6 — Retrodiction from the final state of the Earth–Moon system. This argument proceeds in the opposite direction from the previous five: not from the physics of the proto-Earth, but from what the Moon *is* today, in accordance with the reconstruction approach announced in the introduction. The current angular momentum of the Earth–Moon system and the residual orbital inclination of $\approx 5°$ (constraint 5, Section 1) jointly impose that the Moon passed through an instability episode at the Laplace plane transition, whose activation condition is an inertial obliquity $\varepsilon_{\rm I} \gtrsim 60$–$65°$ at the time of crossing[15,16] — hence an orientation of $\boldsymbol{L}_p$ inherited from strong axial excursions during formation (Sections 12.3.1 and 12). The upper part of the interval $[40°, 70°]$ is thus *required by the final state*, independently of arguments 1 to 5 which derive it from the initial state: two logically disjoint paths converge on the same interval.

The interval $\varepsilon \in [40°, 70°]$ is not an ad hoc hypothesis. It is the predictable consequence of the dynamics of a rapidly rotating fluid body, devoid of a tidal stabilizer, subjected to continuous stochastic perturbations. The simulations of Laskar et al.[10] for a Moonless Earth, and of Li & Batygin[12] for a rapidly rotating Moonless Earth, converge on this interval — the same one adopted by Laskar et al.[10] and Touma & Wisdom[11]. In this oblique geometry, the Coriolis force decomposes into radial and horizontal components (Section 7.3). The radial component $f_{\mathrm{rad}} = 2\Omega \sin\varepsilon \cdot U$ is maximal for $\varepsilon \in [40°, 70°]$.

Section summary

Six convergent arguments support $\varepsilon \in [40°, 70°]$. The absence of a tidal stabilizer is logically necessary. Arguments 2 to 5 provide independent physical justification for a large and sustained misalignment; argument 6, retrodictive, shows that the final state of the Earth–Moon system (angular momentum and 5° inclination) independently requires the upper part of this same interval. The interval $[40°, 70°]$ is adopted as the physically motivated Hadean interval; the mechanism operates for any $\varepsilon > 45°$ (Section 7.3). Obliquity is thus assigned two missions in this work: during formation, it feeds the radial Coriolis force that destabilizes the CMT; after formation, its inertial heritage ε_{I} activates the angular momentum sink (Section 12). The intertropical belt $|\phi| < 30°$ is, throughout this work, tied to the instantaneous dynamical equator $\boldsymbol{\Omega}(t)$, not to a fixed geographic reference frame.

Reference frames in the Hadean era: what exists, what does not

This point governs all subsequent use of ε, $\varepsilon_{\mathrm{dyn}}$, and ε_{I} in this manuscript, and is explicitly stated here. In the Hadean epoch considered ($\approx$ 100 Myr after CAIs, Section 14), *two* reference frames exist, and one is lacking. Exist: the body's own co-rotating reference frame, in which all internal angles of this work are defined; and the proto-Earth's heliocentric orbital plane, which has been orbiting the Sun for tens of millions of years — it is this, not the modern ecliptic (simply the current version of this same plane, since perturbed by planetary interactions), that defines the inertial obliquity ε_{I} used after formation (Section 12). Lacking: any *stabilizer*. The orbital plane exists but exerts no restoring torque on the spin capable of freezing it; no Moon yet exists to damp it. $\boldsymbol{\Omega}(t)$ therefore never stabilizes toward a fixed orientation, in any reference frame, external or internal; it permanently wanders within $\varepsilon \in [40^\circ, 70^\circ]$, reset continuously by bombardment, T-Tauri activity, and the absence of any restoring torque. Every formation angle used in this manuscript — the nutation ε, the CMT colatitude $\varepsilon_{\mathrm{dyn}}$, the intertropical belt itself — is defined purely in the body's own co-rotating reference frame, relative to $\boldsymbol{\Omega}(t)$ at that instant, and must be read as such throughout the text, including in the force derivations of Appendix C. It follows directly that the "tropics" bounding this belt are those of the proto-Earth itself, defined solely by the instantaneous geometry of $\boldsymbol{\Omega}(t)$ and the Maclaurin spheroid it sustains — and not the modern tropics referenced to the current ecliptic. The mechanism developed throughout this work operates on this belt as a whole ($|\phi| < 30^\circ$), and not on a single equatorial line: the instantaneous dynamical equator ($\phi = 0$) is an internal geometric reference for the belt, around which the CMT is centered, not a location that the mechanism must reach nor a state in which it must stabilize.

3.1 Relaxation time and rheological strengthening

The viscous relaxation time toward axial alignment for an isolated fluid body, $\tau_{\mathrm{relax}} \sim \eta/(\rho g R_{\mathrm{eq,proto}})$ [36], is very short ($\ll$ 1 yr) at low viscosity, as established in Section 3. The Bingham-Herschel rheology locally slows this relaxation. The rheological response time to a stress perturbation is:

$$\tau_{\mathrm{BH}} \sim \frac{\tau_y}{\mu \dot{\gamma}_{\mathrm{typ}}} \approx 10^2\text{–}10^4\ \mathrm{s}, \tag{1}$$

with $\tau_y \sim 10^2$–10^4 Pa, $\mu \sim 1$–100 Pa s, and $\dot{\gamma}_{\mathrm{typ}} \sim 1\ \mathrm{s}^{-1}$ [37], where μ is the dynamic viscosity and $\dot{\gamma}_{\mathrm{typ}}$ the typical shear rate. This time remains much shorter than the impact interval τ_{pert}, which means that mass redistribution is locally frozen by the yield stress between perturbations, delaying local shape relaxation without affecting the conclusion of Argument 4: it is the absence of rest, due to the frequency of perturbations and the absence of a stabilizer, that maintains the system out of equilibrium.

4 Initial state of the proto-Earth

4.1 Initial rotation period: $T_{\rm rot} \approx 3.5$ h

The initial rotation period of the proto-Earth is not a free parameter: it is directly constrained by the literature on terrestrial planet formation to $T_{\rm rot} \approx$ 3–5 h.

1. T-Tauri magnetic torque. The young, rapidly rotating Sun ($\approx$ 3 days, Bouvier et al.[38]) magnetically couples to the inner protoplanetary disk[39,40], transferring angular momentum to material in the inner orbit and spinning up the proto-Earth.

2. N-body simulations of terrestrial planet formation. N-body simulations show that the final spin periods of Earth-mass planets lie in the 3–5 h interval[23,24,41]. Agnor et al.[23] find that protoplanetary embryos rotate rapidly during the final stages of accretion, during their first collision, suggesting that the proto-Earth may have rotated rapidly before any subsequent major impact. Kokubo & Genda[24], using N-body simulations of protoplanet formation under realistic accretion conditions, find final rotation periods of 3–5 h for Earth-mass bodies, consistent with Chambers[41]. The value $T_{\rm rot} \approx 3.5$ h adopted here lies within this independently established interval, and is of the same order of magnitude as the rapid pre-impact rotations considered in giant impact models by Ćuk & Stewart[17] (2.3–2.7 h) and Canup[22] (a pre-impact rotation of 3 h in one simulated case) — although none of these specific values is identical to 3.5 h, and the TTP does not rely on the giant impact hypothesis itself.

Why rapid rotation requires an acceleration mechanism. Dones & Tremaine[42] show analytically that orderly accretion from a uniform or slowly varying disk of small bodies cannot produce a rotation as rapid as that of Earth or Mars, regardless of the velocity dispersion of the disk: the prograde and retrograde contributions of individual planetesimals cancel almost by symmetry. Rapid spin speeds of terrestrial planets are therefore most naturally explained by one or a few late collisions between large bodies — a requirement that the TTP satisfies through the continuous planetesimal bombardment of Section 2, independently of whether or not a single dominant impactor is invoked (as in the giant impact hypothesis). Consistently, Kokubo & Ida[43], using N-body simulations of the fusion phase of accretion embryos, find that planetary spin angular velocities follow a Maxwellian distribution with substantial dispersion: there is no single privileged rotation period predicted by accretion physics alone, but only a broad statistical interval within which $T_{\rm rot} \approx 3.5$ h is a plausible value.

3. Skater effect: Fe-Ni segregation. The migration of Fe-Ni toward the forming core reduces the moment of inertia as segregation proceeds. With $I_f/I_0 \approx 0.82$[9], this contributes an additional acceleration of about 18% in rotation during the TTP window, in addition to the rotation speed already acquired by the proto-Earth during accretion. This effect is qualitative and contributory: it is not used here to derive $T_{\mathrm{rot}} \approx 3.5$ h from an assumed pre-segregation value, since no such value is independently constrained.

4.2 Why rapid rotation: a dedicated subsection

The rapid rotation of the proto-Earth ($T_{\mathrm{rot}} \approx 3.5$ h) is not a decorative hypothesis. It is required by accretion physics and by the TTP mechanism itself. This subsection summarizes the five independent reasons constraining this value.

1. **Orderly accretion is insufficient.** Dones & Tremaine[42] show analytically that symmetric accretion from a disk of small bodies cannot produce rapid rotation: prograde and retrograde contributions almost cancel. A rapid spin requires a late acceleration mechanism.
2. **N-body simulations impose 3–5 h.** Kokubo & Ida[43] find that planetary spin angular velocities follow a Maxwellian distribution with substantial dispersion. 3.5 h is a plausible value within this interval.
3. **Fe-Ni segregation spins up the rotation.** Migration of metallic iron toward the core reduces the moment of inertia by about 18%[9], accelerating rotation via the skater effect.
4. **The ejection threshold requires rapid rotation.** The condition $\Omega > \Omega_{\mathrm{crit}}$ (Section 9.2) is reachable only for $T_{\mathrm{rot}} \lesssim 4$ h. A slower proto-Earth could never eject material.
5. **The proto-Earth is not a synestia.** The corotation limit (CoRoL) of Lock & Stewart[44] is at ≈ 1.8 h for proto-Earth parameters. At 3.5 h, Ω reaches only $\approx 50\%$ of this limit: the body remains a coherently rotating spheroid, not a distended structure.

On the initial rotation period

The value $T_{\mathrm{rot}} \approx 3.5$ h is directly constrained by the literature on terrestrial planet formation (Agnor et al.[23], Kokubo & Genda[24], Chambers[41]; 3–5 h), is of the same order of magnitude as the rapid rotations considered in giant impact models[17,22] without depending on the giant impact hypothesis, and is consistent with the theoretical requirement that a rapid telluric spin requires a late accretion acceleration mechanism[42], within the broad statistical interval found for planetary spin in N-body simulations[43]. It is not derived from an assumed initial period via a contraction argument. The Coriolis compensation parameter b', a simple ratio between the critical ejection speed U_{crit} and the geostrophic equilibrium speed of the CMT, provides an independent cross-check: Fe-Ni segregation increases b', ejections decrease it, and the equilibrium point $b' = 1$ is a stable attractor that naturally selects $T_{\mathrm{rot}} \approx 3.5$ h and $\varepsilon \approx 55^\circ$. The full derivation is given in Appendix B.

4.3 Proto-Earth geometry: the Maclaurin spheroid

A rapidly rotating fluid mass adopts the shape of a Maclaurin spheroid[45]. The equatorial radius is $R_{\mathrm{eq,proto}} \approx 7.4$ Mm, combining reduced density (+4%), thermal expansion (+7.5%), and equatorial bulge (+7.5%). The dynamical flattening, given by $J_2 = q/[2\Phi(\bar{C}) - 3]$ where $q = \Omega^2 R^3_{\mathrm{eq,proto}}/(GM_\oplus)$ is the rotation parameter and $\bar{C}$ the normalized axial moment of inertia[46], reduces to $J_2 \approx q/2$ for a strictly homogeneous body ($\bar{C} = 0.4$):

$$J_2 \approx \frac{1}{2} \cdot \frac{\Omega^2 R^3_{\mathrm{eq,proto}}}{GM_\oplus} \approx 0.13, \tag{2}$$

a value consistent with the interval $J_2 \sim 0.10$–0.13 expected for a body near homogeneity[46], giving an equatorial bulge $\delta R \approx 600$–650 km. The effective gravity varies with latitude ϕ (measured from the instantaneous dynamic equator, as defined in Section 3) according to:

$$g_{\mathrm{eff}}(\phi) = g_{\mathrm{eff}}^{(0)} \left(1 + \tfrac{3}{2} J_2 \sin^2\phi\right) - \Omega^2\, r(\phi) \cos^2\phi, \tag{3}$$

where $r(\phi)$ is the local radius. Effective gravity is minimal within the belt $|\phi| < 30°$: this is the geometric attractor of the CMT.

The free nutation (Euler) period, $\omega_E = \Omega(C - A)/A$ where C and A are the polar and equatorial moments of inertia of the spheroid[46], with $(C - A)/A \equiv f \approx 0.075$ for a near-homogeneous body at the dynamical flattening J_2 adopted here (Eq. 2), is:

$$\omega_E = \Omega f \approx 3.74 \times 10^{-5}\,\mathrm{rad\,s^{-1}}, \qquad T_{\mathrm{nut}} \approx 46.7\,\mathrm{h}. \tag{4}$$

This frequency plays a central role in the parametric elliptic instability (Section 7.6).

Stability of the Maclaurin spheroid at this rotation rate. A rapidly rotating self-gravitating fluid body is not necessarily constrained to adopt the axisymmetric Maclaurin shape: beyond a critical rotation rate, the Maclaurin sequence bifurcates to the triaxial Jacobi sequence. The bifurcation criterion compares two characteristic timescales: the self-gravitation timescale, $t_g \sim 1/\sqrt{G\bar{\rho}}$, and the rotation period, $t_{\mathrm{rot}} = 2\pi/\Omega$; their ratio defines the dimensionless number $\Omega^2/(\pi G\bar{\rho})$. This criterion is derived analytically by Chandrasekhar[45] (Chapter 4, full demonstration; see also Chapters 3 to 5 for Maclaurin spheroids and Chapters 6 and 7 for Jacobi ellipsoids and the bifurcation itself) as the point where the family of Jacobi ellipsoids branches off from the Maclaurin sequence — and not as an empirical approximation — and is presented in historical perspective by Jeans[47], taken up in the stellar context by Tassoul[48], the theory being formally identical for any self-gravitating fluid; Lamb[49] gives the classical hydrodynamic treatment, and Durand[50] the ellipsoidal integrals on which Chandrasekhar's demonstration rests. This bifurcation occurs at $\Omega^2/(\pi G\bar{\rho}) \approx 0.374$, Maclaurin spheroids becoming truly unstable only beyond

$\Omega^2/(\pi G\bar{\rho}) \approx 0.440$.

The ρ in this criterion is the uniform density of the homogeneous fluid in the theoretical model. Its application to a real planetary body substitutes for this uniform density the mean density $\bar{\rho} = M/(\frac{4}{3}\pi R^3)$, an approximation explicitly discussed by Chandrasekhar[45] (Chapters 1 and 5) and Tassoul[48] (Chapter 2). For the Hadean proto-Earth, $\bar{\rho}$ is not directly measurable: it depends on the degree of differentiation already achieved, which itself is not precisely known. Available estimates place the mean density of a homogeneous Earth of the same mass, before complete core segregation, around 4 100–4 500 $\mathrm{kg\,m^{-3}}$, versus 5 510 $\mathrm{kg\,m^{-3}}$ for the fully differentiated current Earth[51–54]. These values are themselves derived from mineralogical models (chondritic or pyrolitic composition, equations of state) rather than direct measurement, and thus constitute a modeling interval rather than a physical constant.

For the proto-Earth parameters adopted here ($R_{\mathrm{eq,proto}} \approx 7.4$ Mm, mean density $\bar{\rho} \approx 4\,300\ \mathrm{kg\,m^{-3}}$, the median value of this interval, for a still weakly differentiated body), $T_{\mathrm{rot}} \approx 3.5$ h gives $\Omega^2/(\pi G\bar{\rho}) \approx 0.276$ — below the Jacobi bifurcation point (0.374), with a margin of about 26%. At this density, the bifurcation threshold itself corresponds to $\Omega_c = \sqrt{0.374\,\pi G\bar{\rho}} \approx 5.8 \times 10^{-4}\ \mathrm{rad\,s^{-1}}$, i.e., $T_{\mathrm{rot}} \approx 3.0$ h, and the strict dynamical instability threshold to $T_{\mathrm{rot}} \approx 2.8$ h. The axisymmetric Maclaurin geometry adopted throughout this work is therefore consistent with the TTP rotation rate, and does not require recourse to a triaxial or otherwise more complex equilibrium figure; the available margin remains sensitive to the exact value of $\bar{\rho}$ adopted, which is not known to better than the modeling interval cited above.

Distinction from the synestia regime. Lock & Stewart[44] define the corotation limit (CoRoL) as the rotation speed at which a body's equatorial velocity equals the local Keplerian orbital speed; bodies exceeding this limit form a distended, partially vaporized structure they call a synestia, rather than a simply rigidly rotating body. For the proto-Earth equatorial radius derived above ($R_{\mathrm{eq,proto}} \approx 7.4$ Mm), the corotation limit corresponds to a period of about 1.8 h. At $T_{\mathrm{rot}} \approx 3.5$ h, Ω reaches only $\approx 50\%$ of this limit: the proto-Earth described by the TTP therefore lies well in the regime of a coherently rotating rigid body, and not a synestia.

4.4 Near-total melting: a thermodynamic necessity

The formation of an Earth-mass planet releases a considerable amount of gravitational energy. In the homogeneous sphere approximation[55], the total accretion energy is:

$$E_{\mathrm{grav}} = \frac{3\,G\,M_\oplus^2}{5\,R_{\mathrm{eq,proto}}} \approx 1.93 \times 10^{32}\ \mathrm{J}, \tag{5}$$

where G is the gravitational constant, $M_\oplus$ Earth's mass, and $R_{\mathrm{eq,proto}} \approx 7.4$ Mm the equatorial radius of the undifferentiated proto-Earth.

Equation (5) is a thermodynamic bound, not an instantaneous formation model.

It calculates the *total* gravitational energy released during accretion, independent of its duration. This is the standard approach in the literature[7–9]. The manuscript explicitly cites Kleine et al.[34], Yin et al.[35] as evidence that accretion lasted ~ 30 Myr.

The energy required to melt the entire silicate mantle[56]:

$$E_{\text{fus}} \approx M_{\text{mantle}} \times L_{\text{fus}} \approx 4 \times 10^{24}\,\text{kg} \times 4 \times 10^{5}\,\text{J}\,\text{kg}^{-1} \approx 1.6 \times 10^{30}\,\text{J}, \tag{6}$$

where M_{mantle} is the silicate mantle mass and L_{fus} the specific latent heat of fusion.

The resulting energy ratio is:

$$E_{\text{grav}}/E_{\text{fus}} \approx 121. \tag{7}$$

This precise value follows from the parameters adopted here ($R_{\text{eq,proto}} \approx 7.4$ Mm, $M_{\text{mantle}} \approx 4 \times 10^{24}$ kg, $L_{\text{fus}} \approx 4 \times 10^{5}$ J kg^{-1}), explicitly stated above; it is not a universal constant of Earth. Other equally defensible parameter choices in the literature — a radius close to Earth's current radius, a lower mantle mass, or a higher latent heat — give different numerical values, typically between ≈ 80 and ≈ 190 depending on the assumptions adopted. What remains robust over this entire range, and what constitutes the consensus result in the literature[7–9], is the qualitative conclusion: accretion energy exceeds the fusion energy of the silicate mantle by more than two orders of magnitude, making near-total melting of the proto-Earth a robust thermodynamic consequence rather than a result contingent on a particular choice of parameters.

Why melting was sustained despite radiative cooling

Accretion lasted ~ 30 Myr[34,35]. During this period, radiative cooling occurred. The relevant physical question is not whether the *total* accretion energy exceeds the *total* fusion energy (Eq. 7) — a necessary but not sufficient condition — but whether the *rate* of heat input keeps pace with the *rate* of radiative loss throughout the accretion history, so that melting is sustained rather than merely possible in integrated terms. This rate balance is established, quantitatively and independently of the present work, by coupled magma ocean/atmosphere thermal evolution models[31–33]: for a silicate vapor atmosphere of the pressure and opacity considered here ($P \approx 10^5$–10^7 Pa), these models explicitly solve the time-dependent energy balance $\mathrm{d}E/\mathrm{d}t = Q_{\text{in}}(t) - Q_{\text{rad}}(t)$ and find magma ocean lifetimes of order 10–10^2 Myr — directly comparable to, and consistent with, the accretion timescale of ~ 30 Myr required here. Without this atmospheric blanket, these same models show that the surface would instead cool and solidify on a timescale of $\sim 10^4$ yr: four to six orders of magnitude shorter than the accretion timescale, and thus radically insufficient to sustain melting. It is this reduction factor of 10^4 to 10^5 in the *rate* of radiative cooling — and not a simple comparison of integrated energies — that constitutes the operating mechanism, quantitatively modeled, sustaining melting throughout accretion. Three additional and independent heat sources (Mechanism 2 below) further reinforce this rate balance, particularly during the late stages of accretion when the rate of gravitational energy input itself declines.

Mechanism 1 — The silicate atmosphere as a thermal blanket. The dense silicate atmosphere ($P \approx 10^5$–10^7 Pa, $T \approx 2\,000$–$4\,000$ K) has high infrared opacity. It reduces surface radiative cooling by a factor 10^4 to 10^5 [31–33]. The effective cooling time increases from $\sim 10^4$ yr to 30–50 Myr.

Mechanism 2 — Additional heat sources. Four heat sources prolong melting beyond the initial accretion energy:

- **Fe-Ni segregation heat:** $\approx 3 \times 10^{30}$ J [57,58].
- **Short-lived radioactivity:** ^{26}Al (half-life 0.72 Myr) and ^{60}Fe (half-life 2.6 Myr) inject $\approx 3 \times 10^{29}$ J [29,30].
- **Continuous planetesimal bombardment:** thermal energy reinjection[23,24].
- **T-Tauri irradiation:** maintains surface temperatures above 2,000 K[27,28].

Mechanism 3 — The TTP does not require melting of the entire mantle. The Triple Phase Transition requires melting only within the Hadean intertropical belt $|\phi| < 30^\circ$ (upper and middle mantle). The factor 121 thermodynamically guarantees this partial melting, even if the deep mantle was not entirely molten[59].

Mechanism 4 — The active TTP window is 30–50 Myr. The TTP occurs *during* the 30–50 Myr window during which melting is sustained by the above mechanisms. Core formation over about 30 Myr[34,35] is subsequent to, or simultaneous with, this window — it is not a contradiction. Near-total melting constitutes the obligatory starting point. The objection based on the $\sim$ 30 Myr accretion timescale does not contradict the mechanism: the TTP occurs *during* the melting window sustained by the silicate atmosphere, additional heat sources, and continuous bombardment.

Hypothesis H1 — Total volumetric melting. At the onset of Fe-Ni segregation, the entire silicate mantle lies above the liquidus, without a stable internal solid interface during the active window 4.568–4.4 Ga. This is a self-gravitating fluid volume, and not a magma ocean resting on a substrate: the dynamics, rheology, and response to perturbations are physically distinct.

Caveat: Nomura et al.[59] suggest that the deep lower mantle may not have been entirely molten, due to the elevation of the solidus with pressure. This does not affect the TTP, which requires melting only within the Hadean intertropical belt $|\phi| < 30^\circ$ (upper and middle mantle) — not at the planet's center nor in the deep lower mantle — a requirement that the factor 121 thermodynamically guarantees by construction (Appendix A).

5 The intertropical belt $|\phi| < 30°$: the ejection site

A fundamental and often misunderstood point: the CMT is not an equatorial structure. It is a coherent zonal belt extending over $\pm 30°$ of oblique latitude — about one-third of the proto-Earth's surface. This extension results from three independent and convergent mechanisms.

As established in Section 3, the belt $|\phi| < 30°$ is defined perpendicular to the instantaneous rotation axis $\mathbf{\Omega}(t)$, and not with respect to a fixed geographic reference frame. This is the plane where the radial Coriolis force is maximal and effective gravity is minimal.

Mechanism 1 — Maclaurin geometry. Effective gravity is minimal within the belt $|\phi| < 30°$ (Eq. 3). Molten material accumulates there by centrifugal effects, independently of any other argument. This is a consequence of the ellipsoidal shape imposed by rapid rotation.

Mechanism 2 — Inverse geostrophic cascade. The Rossby number of the CMT, with $U_{\text{turb}} \approx 0.8\ \text{km}\,\text{s}^{-1}$ (Eq. 19), is:

$$\text{Ro} = \frac{U_{\text{turb}}}{2\,\Omega\,R_{\text{eq,proto}}} \approx 0.108 \ll 1, \tag{8}$$

In this regime, turbulent energy cascades toward large scales (inverse cascade)[60]. The Rhines scale:

$$L_\beta = \sqrt{\frac{U_{\text{turb}}}{\beta}}, \qquad \beta = \frac{2\Omega\cos\varepsilon}{R_{\text{eq,proto}}}, \tag{9}$$

gives $L_\beta \approx 0.51\,R_\oplus$ — corresponding to the harmonic mode $\ell = 1$, $m = 1$: the largest coherent structure possible on a sphere. The CMT is the statistical attractor of this cascade[60,61].

Mechanism 3 — Radial Coriolis force. $f_{\text{rad}} = 2\Omega U \sin\varepsilon$ concentrates the flow within the oblique intertropical belt and maintains it coherently there.

Section summary

These three mechanisms — Maclaurin geometry, inverse geostrophic cascade, and radial Coriolis force — are independent and converge on the same belt $|\phi| < 30°$. Their simultaneous convergence constitutes an argument for the structural robustness of the theory.

6 The CMT as a purely toroidal flow: response to the Taylor-Proudman objection

Any geophysical fluid dynamicist confronted with the CMT will naturally raise the following objection: in a rapidly rotating fluid (Ro $\ll 1$), the Taylor-Proudman theorem imposes $(\boldsymbol{\Omega} \cdot \nabla)\boldsymbol{u} = 0$, generating columnar structures parallel to $\boldsymbol{\Omega}$, rather than a single oblique belt $|\phi| < 30°$. The objection is legitimate and deserves a precise response.

6.1 Poloidal/toroidal decomposition

Any velocity field within a spherical shell decomposes into a poloidal part (radial and meridional components) and a toroidal part (purely azimuthal component):

$$\boldsymbol{u} = \boldsymbol{u}_{\mathrm{pol}} + \boldsymbol{u}_{\mathrm{tor}}, \qquad \boldsymbol{u}_{\mathrm{tor}} = \nabla \times (\psi \hat{\boldsymbol{r}}). \tag{10}$$

The Taylor-Proudman theorem, in its classical form, constrains the poloidal field (which carries vorticity along $\boldsymbol{\Omega}$) and produces Taylor columns. It does not constrain the purely toroidal field.

For a purely azimuthal flow independent of longitude ($u_r = u_\theta = 0$ on average, $u_\phi = u_\phi(r, \phi)$), the Taylor-Proudman constraint is automatically satisfied:

$$(\boldsymbol{\Omega} \cdot \nabla) u_{\mathrm{tor}} = 0 \tag{11}$$

for any toroidal field, without additional conditions. The equilibrium is cyclostrophic, not geostrophic: the centrifugal pressure gradient balances the Coriolis force within the azimuthal plane.

6.2 Why curved columns do not extend beyond the active belt

In a spherical shell in oblique rotation ($\varepsilon \neq 0$), Taylor columns are curved along the projected field lines of $\boldsymbol{\Omega}$. They tend to channel flow away from its source — unless a cutoff mechanism stops them.

This is precisely the role of the yield stress τ_y. In a Bingham-Herschel fluid, flow propagates only in regions where stress exceeds τ_y. As soon as columns penetrate a region where $\tau < \tau_y$, they are truncated: the magma solidifies there[62,63]. The belt $|\phi| < 30°$ is precisely the region where effective gravity is minimal (Maclaurin) and where Coriolis stress is maximal ($\varepsilon \approx 55°$): it is there, and only there, that $\tau > \tau_y$. Curved columns emerging from the active belt are thus naturally truncated at its boundaries.

Section summary

The CMT is a purely toroidal flow. The Taylor-Proudman objection does not apply. What remains to be demonstrated numerically — identified here as Limitation L1 (Section 20) — is whether a single torus forms within the belt $|\phi| < 30°$ rather than multiple alternating bands. This is a question for simulation, not a theoretical inconsistency.

7 Complete inventory of forces

All forces, projections, and equations are expressed in the Hadean instantaneous co-rotating reference frame of the body, with ε defined as in Section 3.

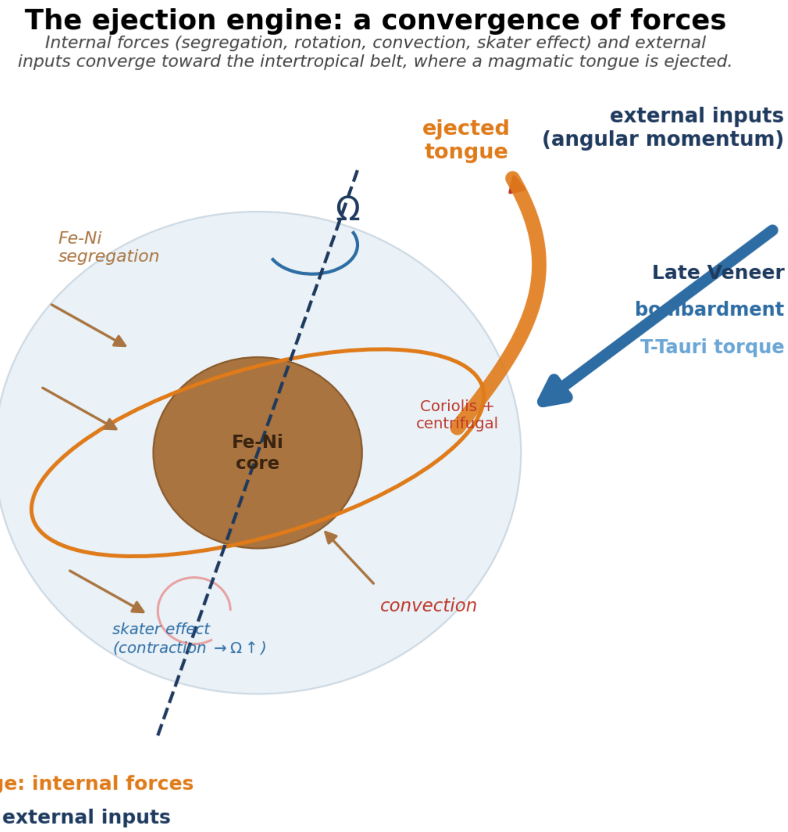

Figure 2: The ejection engine: a convergence of forces. Internal forces (Fe-Ni segregation toward the core, rotation Ω, convection, skater effect: contraction $\to \Omega \uparrow$) and external angular momentum inputs (Late Veneer, bombardment, T-Tauri torque) converge toward the intertropical belt, where a magmatic tongue is ejected under the combined action of Coriolis and centrifugal forces. Orange: internal forces; blue: external inputs.

7.1 Complete force inventory: summary table

The CMT is a toroidal flow within a rapidly rotating fluid body, subject to a set of forces. Table 1 summarizes all forces, their physical role, and their mathematical expression.

Table 1: Complete inventory of forces acting on the CMT.

Force	**Role**	**Expression**
Effective gravity	Radial confinement	$g_{\text{eff}}(\phi) = g_{\text{eff}}^{(0)}(1 + \frac{3}{2}J_2 \sin^2\phi) - \Omega^2 r(\phi)\cos^2\phi$
Centrifugal force	Lateral ejection, flattening	$F_{\text{cent}} = \Omega^2 r(\phi)\cos^2\phi$
Radial Coriolis	Destabilization, outward ejection	$F_{\text{cor,rad}} = 2\Omega U \sin\varepsilon$
Horizontal Coriolis	Lateral confinement	$F_{\text{cor,hor}} = 2\Omega U \cos\varepsilon$
Taylor-Proudman	Vertical columns, toroidal flow	$(\boldsymbol{\Omega}\cdot\nabla)\boldsymbol{u}_{\text{pol}} = 0$; $\boldsymbol{u}_{\text{tor}}$ unconstrained
Planetesimal bombardment	Stochastic perturbation, thermal reinjection	$\delta\boldsymbol{\Omega} = \sum_i \delta\boldsymbol{\Omega}_i$, $\delta U = \sum_i \delta U_i$
T-Tauri (CME)	Stochastic perturbation, thermal maintenance	$\boldsymbol{F}_{\text{CME}}(t) = \sum_j \boldsymbol{F}_{\text{CME},j}\,\delta(t - t_j)$
Solar tide	Low-frequency obliquity forcing	$F_{\text{tide,sol}} \sim 10^{-4} g_0 \cos(\omega_{\text{sol}} t)$
Proto-lunar tide	Mathieu resonance (episodes 2+)	$F_{\text{tide,moon}}(t) = A_m^{(k)}(t)\cos(n_{\text{Moon}} t)$
Euler acceleration	Axial nutation	$\boldsymbol{a}_{\text{Euler}} = \dot{\boldsymbol{\Omega}} \times \boldsymbol{r}$
Atmospheric pressure	Lateral confinement	$P_{\text{atm}} \sim 10^5 - 10^7$ Pa
BH yield stress	Plasticity threshold	$\tau = \tau_y + k\dot{\gamma}^n$, $\tau > \tau_y$
Viscous stress	Dissipation	$\tau_{\text{visc}} = \eta\nabla^2\boldsymbol{u}$

7.2 Oscillation as a direct force source

Axial oscillation implies $\dot{\boldsymbol{\Omega}} \neq \boldsymbol{0}$. Poincaré[64] showed that this generates an Euler acceleration on each fluid parcel:

$$\boldsymbol{a}_{\text{Euler}} = \dot{\boldsymbol{\Omega}} \times \boldsymbol{r}, \tag{12}$$

where $\boldsymbol{r}$ is the position vector. The radial projection:

$$\boldsymbol{a}_{\text{Euler}} \cdot \hat{r} = |\dot{\Omega}|\, r(\phi)\, \sin(\varepsilon + \phi), \tag{13}$$

is maximal within the belt $|\phi| < 30°$ where $r(\phi)$ is largest.

7.3 Radial Coriolis force and geometric threshold

The Coriolis force on a fluid parcel moving at azimuthal speed U decomposes, in the oblique co-rotating frame, into:

$$f_{\mathrm{rad}} = 2\,\Omega\,\sin\varepsilon \cdot U \quad \text{(radial, outward, destabilizing)}, \tag{14}$$

$$f_{\mathrm{horiz}} = 2\,\Omega\,\cos\varepsilon \cdot U \quad \text{(horizontal, confining)}, \tag{15}$$

with the ratio:

$$\frac{f_{\mathrm{rad}}}{f_{\mathrm{horiz}}} = \tan\varepsilon. \tag{16}$$

Geometric threshold $\varepsilon = 45°$ — necessary condition for radial ejection. For $\varepsilon > 45°$: $\tan\varepsilon > 1$, the destabilizing radial component dominates the confining horizontal component. For $\varepsilon < 45°$: the horizontal component dominates and the Taylor-Proudman constraint inhibits radial ejection. This threshold follows solely from Coriolis kinematics: no adjustable parameter intervenes. At $\varepsilon = 55°$: $f_{\mathrm{rad}}/f_{\mathrm{tot}} = \sin 55° \approx 0.82$, so 82% of the total Coriolis force is directed radially outward.

7.4 The T-Tauri Sun: catalyst and perturber

The Sun in its T-Tauri phase is not a driving force in the same sense as gravity or the Coriolis force — it does not act continuously on each fluid parcel of the CMT. Nevertheless, it plays three essential roles within the TTP window.

(1) Rheological modulator. T-Tauri UV/X-ray radiation maintains T_{surf} in the interval 1 800–4 200 K (Stefan-Boltzmann law, albedo $A_0 \approx 0.1$), which keeps the yield stress τ_y of the silicate magma in the interval 10^2–10^4 Pa[37] — the optimal window for CMT cohesion. Below $T^* \approx 1\,800$ K, τ_y increases to 5.5×10^3–8.0×10^4 Pa (Eq. 25) and cohesive ejection becomes progressively more difficult. The T-Tauri Sun thus acts as a *guardian of the rheological window* via surface radiative forcing transmitted to the mantle by conduction through the silicate atmosphere.

(2) Source of stochastic perturbations. CMEs (10^{27}–10^{29} J per event, several per day) and radiative pulsations provide a continuous source of perturbation.

(3) Obliquity perturber. CMEs acting on the proto-atmosphere and magnetosphere deliver impulsive torques on the spin vector, contributing to the maintenance of axial oscillation $\varepsilon \in [40°, 70°]$ (Section 3). The T-Tauri phase also provides an independent shutdown mechanism: once the Sun reaches the main sequence ($\approx$ 4.4 Ga), irradiation declines, τ_y increases, and the TTP window closes.

The T-Tauri Sun is therefore not the *engine* of the mechanism — gravity, Coriolis force, centrifugal force, Fe-Ni segregation, and Taylor-Proudman columns are — but it is the *catalyst* that keeps the window open and sustains the axial oscillation that drives obliquity to its maxima.

7.5 Tidal forces

Solar tide. Relative amplitude $\approx 10^{-4} g_0$; contributes as a low-frequency resonant forcing on obliquity.

Growing proto-lunar tide — Mathieu resonance. From Episode 2 onward, the proto-satellite exerts a tidal force that modulates CMT stability:

$$\ddot{\xi}_r = \left[\lambda + A_m^{(k)}(t)\cos(n_{\text{Moon}}\, t)\right]\xi_r, \tag{17}$$

which is Mathieu's equation[65]. Parametric resonance bands exist for $n_{\text{Moon}} \approx 2\sqrt{|\lambda|}$ even when $\lambda < 0$: the growing proto-lunar tide is the dominant trigger for Episodes 2 through N.

Tides from giant planets. Primarily Jupiter: a low-frequency perturbation on obliquity, sustaining oscillation on timescales of 10^5–10^6 yr.

7.6 Parametric elliptic instability as a CMT amplifier

In a precessing fluid ellipsoid, precessional flows can resonate with the fluid's inertial modes (Poincaré waves, frequencies $\omega_{\text{in}} = 2\Omega m/n$). This mechanism — the parametric elliptic instability — was established theoretically by Malkus[66], formalized by Kerswell[67], and demonstrated experimentally by Lacaze et al.[68].

For the Maclaurin spheroid geometry at $T_{\text{rot}} = 3.5$ h and $f \approx 0.075$, the free nutation frequency is $\omega_E = \Omega f \approx 3.74 \times 10^{-5}$ rad s^{-1} (Eq. 4). This frequency is commensurate with the fluid's inertial modes by geometric construction — and not by parameter adjustment. The linear growth rate is:

$$\sigma_{\text{res}} = \frac{\Omega f}{2} \approx 1.87 \times 10^{-5}\,\text{rad}\,\text{s}^{-1}. \tag{18}$$

The CMT speed grows exponentially:

$$U_{\text{CMT}}(t) = U_{\text{turb}}\, e^{\sigma_{\text{res}} t}, \qquad U_{\text{turb}} \approx 0.8\,\text{km}\,\text{s}^{-1}, \tag{19}$$

where U_{turb} is the characteristic speed at the Rhines scale[60]. The time to reach $U_{\text{crit}} \approx 9.8$ km s^{-1} is:

$$t_{\text{acc}} = \frac{\ln(U_{\text{crit}}/U_{\text{turb}})}{\sigma_{\text{res}}} \approx 37\,\text{h}, \tag{20}$$

with $\ln(U_{\text{crit}}/U_{\text{turb}}) = \ln(9.8/0.8) \approx 2.51$.

The CMT reaches the bifurcation speed in ≈ 37 hours. This result is entirely determined by $f \approx 0.075$ and $\Omega = 4.99 \times 10^{-4}$ rad s^{-1}, imposed by $T_{\text{rot}} = 3.5$ h and Maclaurin spheroid physics. Elliptic instability thus acts as a powerful amplifier, bringing the CMT to the bifurcation speed in ≈ 37 hours — a timescale fully consistent with TTP chronology.

As Fe-Ni segregation progresses, the CMT density decreases:

$$\rho_{\mathrm{CMT}}(t) = \rho_0 - \xi_{\mathrm{Fe}}(t)(\rho_0 - \rho_f), \quad \rho_0 \approx 4\,500\,\mathrm{kg\,m^{-3}},\ \rho_f \approx 3\,200\,\mathrm{kg\,m^{-3}}, \tag{21}$$

which amplifies all destabilizing terms via the factor $\rho_0/\rho_{\mathrm{CMT}}(t)$ in the instability equation (Section 9.2).

7.7 Geostrophic β force and the Rhines scale

The parameter β, which measures the latitudinal gradient of the Coriolis parameter:

$$\beta = \frac{2\Omega\cos\varepsilon}{R_{\mathrm{eq,proto}}} \approx 7.7\times10^{-11}\,\mathrm{m^{-1}\,s^{-1}} \quad (\varepsilon = 55^\circ,\ R_{\mathrm{eq,proto}} = 7.4\times10^{6}\,\mathrm{m}), \tag{22}$$

arrests the inverse cascade at the Rhines scale[60]:

$$L_\beta = \sqrt{\frac{U_{\mathrm{turb}}}{\beta}} \approx 3.2\times10^{6}\,\mathrm{m} \approx 0.51\,R_\oplus, \tag{23}$$

where U_{turb} is the turbulent velocity scale. This scale $L_\beta \approx 0.51\,R_\oplus$ corresponds to the harmonic mode $\ell = 1$, $m = 1$: the largest coherent structure possible on a sphere, and the spatial expression of the CMT.

8 Hierarchy of hypotheses

The following table distinguishes established physical consequences (Level 1), well-constrained but not yet numerically validated hypotheses (Level 2), and speculative hypotheses awaiting simulation (Level 3).

Table 2: Explicit hierarchy of hypotheses underpinning the TTP.

ID	Hypothesis / Statement	Level	Depends on
H0	Central thesis: the Moon formed by three coupled transitions triggered by Fe-Ni segregation	2 (constrained)	Thermodynamic bound $E_{\mathrm{grav}}/E_{\mathrm{fus}} \approx 121$
H1	Total volumetric melting: the entire silicate mantle above the liquidus during 4.568–4.4 Ga	1 (established)	Factor $E_{\mathrm{grav}}/E_{\mathrm{fus}} \approx 121$[7–9]

ID	Hypothesis / Statement	Level	Depends on
H2	$\varepsilon \in [40°, 70°]$: sustained free nutation in the body's co-rotating frame	2 (constrained)	Absence of Moon + 5 convergent arguments[10–12]
H3	CMT confinement to $\lvert\phi\rvert < 30°$ admits an effective description as the fundamental mode of a nonlinear Schrödinger equation with potential $V_{\text{eff}}(\phi)$	3 (speculative)	Reduction from potential vorticity equation[69–71]; conditions $\tau_y \lesssim 10^2$ Pa, $\varepsilon > 45°$; quantitative validity awaiting simulation (Limitation L12)
H4	The $\ell = 1$ mode is selected by the combination of BH rheology (suppression of modes $\ell < \ell_{\text{min}}$) and the inverse cascade	3 (speculative)	3D SPH simulation with full BH rheology (L1)
H5	$\tau_y(T)$ follows an Arrhenius law with $T^* \approx$ 1 800 K and $\tau_y \lesssim 10^2$ Pa for $T \gtrsim 4\,000$ K	1 (established)	Laboratory rheology[37]
H6	The ejection threshold is defined by the crossing of $\lambda(U) = 0$; the generalized instability coefficient governs the transition between confinement and ejection	2 (constrained)	Coupled system (Eqs. 26, 28); P22
H7	The effective yield stress τ_y ensures the cohesive integrity of the ejected toroidal flow via the Bi_{KH} criterion	2 (constrained)	Frigaard & Nouar[62]; three independent confinement mechanisms

ID	Hypothesis / Statement	Level	Depends on
H8	$f_{\text{cap}} \approx 0.70$ and $\partial f_{\text{cap}} / \partial M_{\text{Moon}} > 0$	3 (speculative)	3D SPH simulation (L2); lower bound $f_{\text{cap}}^{\min} \gtrsim 0.45$ (Appendix G)
H9	Dynamo delay = 350 ± 50 Myr, sum of three independent phases	2 (constrained)	Hf-W [34,35] + zircons [72,73] + Tarduno et al. [6]
H10	Seismic preservation window: 200–530 km	2 (constrained)	Ilmenite cumulate overturn [74] + mare volcanism [75,76]
H11	Biphasic *'aā*-like structure of the CMT: mobile plastic core + fragmented clinker crust	3 (speculative)	3D SPH simulation (L1, L3)
H12	$\text{Bi}_{\text{KH}} \gg 1$ suppresses Kelvin-Helmholtz fragmentation at large and intermediate scales	2 (constrained)	Frigaard & Nouar [62]; three independent confinement mechanisms
H13	$b' = 1$ is a dynamical attractor of the coupled segregation–ejection system	2 (constrained)	Geostrophic equilibrium + angular momentum conservation (Appendix B)
H14	The inverse cascade is arrested at the Rhines scale $L_\beta \approx 0.51\, R_\oplus$; the peak at $\ell \approx 14$ is redistributed to $\ell = 1$ by nonlinearity and BH rheology	3 (speculative)	3D SPH simulation (L1)
H15	The ejected flow transits through three phases: ballistic expansion $\rightarrow$ orbital insertion $\rightarrow$ cohesive accretion beyond R_{Roche}	3 (speculative)	3D SPH simulation (L6)

ID	Hypothesis / Statement	Level	Depends on
H16	Parametric elliptic instability amplifies the CMT with a growth rate $\sigma_{\text{res}} = \Omega f/2$	2 (constrained)	Malkus[66], Kerswell[67], Lacaze et al.[68]; Maclaurin geometry
H17	The T-Tauri phase maintains $\tau_y \lesssim 10^2$ Pa via irradiation, making the $\ell = 1$ mode possible	2 (constrained)	Stefan-Boltzmann law + BH rheology; testable by simulation

9 The three coupled transitions

The single engine is the progressive segregation of Fe-Ni toward the forming core. It simultaneously governs three regime changes: the rheological transition opens the window, the mechanical transition produces the ejection episodes, and the magnetic transition closes the loop with the terrestrial dynamo.

9.1 Transition 1 — Rheological: the Bingham-Herschel law

A Newtonian fluid flows under any applied stress. A Bingham-Herschel (BH) fluid, by contrast, resists like a solid as long as the applied stress does not exceed a plasticity threshold τ_y[13,14]:

$$\tau = \tau_y + k\dot{\gamma}^n \quad (\tau > \tau_y), \tag{24}$$

where τ is the applied shear stress, τ_y the yield stress, k the consistency index, $\dot{\gamma}$ the shear rate, and n the flow index. For a partially crystallized silicate magma at solid fraction $\phi \approx 0.1$–0.3: $n = 1.2$ (shear-thickening, Caricchi et al.[77]), $\tau_y \in [10^2, 10^4]$ Pa at $T = 3\,000$–$3\,500$ K[37].

The yield stress depends exponentially on temperature (Arrhenius law):

$$\tau_y(T) = \tau_y^{(0)} \exp\left(+\frac{E_a}{RT}\right), \qquad E_a \approx 150\text{–}250\,\text{kJ}\,\text{mol}^{-1}, \tag{25}$$

where R is the gas constant and E_a the activation energy — the positive sign in the exponent reflects the fact that τ_y *increases* as T *decreases*, as for liquid viscosity[37]. At $T = 3\,000$ K: $\tau_y \approx 10^2$ Pa — the flow moves easily. At $T^* \approx 1\,800$ K, depending on the value adopted for E_a within the interval 150–250 kJ mol^{-1}, τ_y increases to 5.5×10^3–8.0×10^4 Pa — an increase of one to three orders of magnitude from

$T = 3\,000$ K, sufficient to make cohesive ejection progressively more difficult as the T-Tauri Sun exits its active phase and the surface cools.

This is the natural thermal shutdown mechanism at $T^* \approx 1\,800$ K: the TTP window closes irreversibly by the Arrhenius law alone, without additional assumptions. This is not an adjusted parameter: it is a thermodynamic consequence of the exponential temperature dependence of τ_y.

The 'aā structure of the CMT

The toroidal flow exhibits a biphasic architecture: a mobile plastic core at high temperature surrounded by a fragmented clinker crust. This is not a cohesion problem: it is its guarantee. The outer crust protects the core against Kelvin-Helmholtz instabilities at the flow/atmosphere interface. A shell of tempered glass (analogous to the vitreous margins of pillow lavas, Hekinian et al.[78]), forming in ≈ 7 minutes upon contact with the atmosphere, completes this rheological shield. This structure is consistent with the work of Frigaard & Nouar[62] and Balmforth & Craster[63].

9.2 Transition 2 — Mechanical: dynamical instability and ejection

9.2.1 *Principle: a dynamical instability*

Ejection of material from the CMT is a *deterministic dynamical instability*: when the resultant of radial accelerations acting on a fluid parcel becomes repulsive, the parcel accelerates outward and leaves the torus. A parcel undergoing an infinitesimal radial displacement ξ_r obeys the Solberg-Høiland parcel displacement equation[79,80]:

$$\ddot{\xi}_r = \lambda\,\xi_r, \qquad \begin{cases} \lambda < 0 & \text{: oscillation, stable confinement,} \\ \lambda > 0 & \text{: exponential growth, ejection.} \end{cases} \tag{26}$$

The ejection threshold is therefore the condition $\lambda = 0$. This threshold is a well-defined surface in the parameter space $(\Omega, \varepsilon, \alpha, \beta)$, which gives the mechanism a *predictive* character: under given conditions, ejection occurs or not in a univocal way.

9.2.2 The stability coefficient λ

ε_{dyn} is instantaneous, never a fixed reference

Each occurrence of ε_{dyn} in the coefficient below is the *instantaneous* colatitude of the CMT relative to the dynamic equator defined by $\mathbf{\Omega}(t)$ at that instant (Section 3). No fixed axis, no ecliptic, no external Solar System reference plane exists at this epoch. As $\mathbf{\Omega}(t)$ undergoes free nutation within $\varepsilon \in [40°, 70°]$, ε_{dyn} is re-evaluated continuously; it is never fixed to a particular value, and the destabilizing Eötvös term below grows and declines with this permanent oscillation rather than being evaluated at a stabilized state. The mechanism itself operates on the CMT as it occupies the intertropical belt $|\phi| < 30°$ (Section 5), not at a single equatorial point; ε_{dyn} parametrizes the torus position within this belt at each instant.

In solid-body rotation ($U = \Omega r$), with $U \propto r^{\alpha}$ the azimuthal velocity profile, the stability coefficient gathers all radial accelerations acting on the parcel:

The instability coefficient

$$\lambda = \underbrace{-\frac{g_{\text{eff}}}{r} - N^2}_{\text{constant terms}} + \Omega^2 \underbrace{\left(\overbrace{2\sin\varepsilon_{\text{dyn}}}^{\text{Eötvös}} - \overbrace{2(1+\alpha)}^{\text{Rayleigh}} + \overbrace{\beta}^{\text{centrifugal buoyancy}} \right)}_{D} \tag{27}$$

Each term has a precise physical origin — and a simple history:

- $-g_{\text{eff}}/r$: effective gravitational confinement, always *stabilizing*; this is the term to overcome.
- $-N^2$: *thermal* buoyancy (Brunt–Väisälä frequency squared,[81,82]). The magma is cooled from above: the thermal gradient makes the medium superadiabatic, so $N^2 < 0$ and $-N^2 > 0$; this thermal part is *destabilizing* and sustains convection. The *compositional* part of the density gradient is carried separately by β (Ledoux-type separation, Appendix C).
- $2\sin\varepsilon_{\text{dyn}}$: the *non-traditional* component of the Coriolis force (Eötvös effect): any prograde flow is radially lightened by $2\Omega U \cos\phi$, where ϕ is the latitude. Here ε_{dyn} is the *colatitude of the CMT relative to the instantaneous dynamic equator* ($\sin\varepsilon_{\text{dyn}} = \cos\phi$) — no orbital reference enters the internal dynamics. Neglected in the geophysical "traditional approximation," this component must be retained at low latitudes[83]: this is precisely the regime of the intertropical belt. *Destabilizing* term, maximal when the torus overlaps the dynamic equator ($\varepsilon_{\text{dyn}} \to 90°$).
- $-2(1+\alpha)$: Rayleigh term[84]; the epicyclic frequency $\kappa^2 = r^{-3}\,\mathrm{d}[(rU)^2]/\mathrm{d}r = 2(1+\alpha)U^2/r^2$ is always *stabilizing*, hence the negative sign. This centrifugal charge gradient contains *by itself* the entire centrifugal effect: no separate

"centrifugal pressure" term should be added, under penalty of double counting (Appendix C).

- $+\beta$: *centrifugal buoyancy* on the density gradient, $\beta \equiv -\mathrm{d}\ln\rho/\mathrm{d}\ln r$. In the CMT, $\beta > 0$: Fe-Ni segregation makes the body denser at depth. With respect to gravity, this stratification is stable; but the centrifugal force is a gravity pointing *outward*, and for it, the same stratification is a Rayleigh-Taylor-like configuration: dense fluid is pushed outward. *Destabilizing* term — and essential: this is the fuel of the engine. Fe-Ni segregation thus serves a dual purpose: it accelerates rotation by contraction (skater effect) and it creates the stratification whose centrifugal buoyancy fuels the ejection.

9.2.3 *The rotation threshold*

Setting $\lambda = 0$, we isolate the critical angular speed:

$$\Omega_{\mathrm{crit}}^2 = \frac{g_{\mathrm{eff}}/r + N^2}{2\sin\varepsilon_{\mathrm{dyn}} - 2(1+\alpha) + \beta}. \tag{28}$$

A finite threshold exists only if the denominator D is positive, i.e., if $\beta > 2(1 + \alpha) - 2\sin\varepsilon_{\mathrm{dyn}}$. The driving role of β is read directly here: with nominal values, $2\sin\varepsilon_{\mathrm{dyn}} - 2(1+\alpha) = 1.879 - 3 = -1.12$ — without centrifugal buoyancy, no rotation would ever trigger ejection. With $\beta = 2.5$ ($\varepsilon_{\mathrm{dyn}} = 70^\circ$ at the nutation peak, $\alpha = 0.5$, $N^2 = -1.93 \times 10^{-7}\ \mathrm{s}^{-2}$), the denominator is $D = 1.379$ and the threshold $\Omega_{\mathrm{crit}} \approx 1.27\,\Omega_0$, reachable by angular momentum input without approaching the critical rotation regime.

The sensitivity of this threshold to the rotation profile α is direct: for $\alpha \lesssim 1.0$, the threshold remains accessible via angular momentum input from Appendix H; beyond $\alpha \approx 1.19$, the denominator D vanishes and the threshold diverges — ejection ceases to be reachable through this channel, which physically bounds the mechanism to moderately super-rigid profiles. The nominal value ($\alpha = 0.5$, $\Omega_{\mathrm{crit}}/\Omega_0 = 1.27$) operates near — precisely 1.60% above — the strict dynamical instability limit of the Maclaurin spheroid independently derived in Section 4.3 ($\Omega/\Omega_0 = 1.25 = 3.5/2.8$, corresponding to $T_{\mathrm{rot}} \approx 2.8$ h). This proximity may reflect a physical link between the two phenomena — the instability triggering ejection being potentially a local manifestation of the same shape transition — rather than a fortuitous coincidence, but the derivation of λ (Appendix C) assumes an axisymmetric Maclaurin geometry whose strict validity in the immediate vicinity of this threshold remains to be confirmed by a simulation allowing triaxial deformations.

9.2.4 *Status of the threshold: a bifurcation surface with two-dimensional closure*

It is worth specifying what the threshold we have just obtained actually represents. The critical speed U_{crit} is not a fundamental speed of the problem: it is only the translation, on the velocity axis, of a deeper regime change — the passage of the growth rate λ through zero. The ejection threshold is therefore, properly, the vanishing locus

$$\boxed{\lambda(\mathbf{X};\alpha,\beta)=0,} \tag{29}$$

i.e., a *bifurcation surface* in state space, and not an evolution equation: it answers the question “does the system change regime?”, not “how does it evolve?”.

All terms of λ (Eq. 27) are derived from state variables and the adopted constitutive laws, except for two coefficients: α, which parametrizes the differential rotation profile, and β, which parametrizes the efficiency of centrifugal buoyancy. As long as the corresponding submodels are not constructed, neither is presented as derived: the residual closure of λ is thus, in its current state, *two-dimensional*, (α,β). The profile α could be closed by an independent model of internal angular momentum transport (Section 19.2, Limitation L5); after this closure, β would remain the sole phenomenological coefficient of the surface. It follows that $U_{\rm crit}$ is not an independent parameter, but a *diagnostic* quantity obtained by solving Eq. 29; it should not be swept separately from (α,β), but is derived from them as a projection.

9.2.5 *The episode cycle*

The mechanism is a **deterministic relaxation cycle**:

1. Ω grows slowly through net angular momentum input (skater effect of Fe-Ni segregation, supplemented by continuous accretion inputs).
2. The axis nutates (period $\approx$ 46.7 h): the effective obliquity ε oscillates in $[40°,70°]$. Since the CMT is confined to the intertropical belt $|\phi|<30°$, its instantaneous colatitude $\varepsilon_{\rm dyn}$ relative to the dynamic equator oscillates accordingly in $[60°,90°]$. At the nutation peak, the threshold $\Omega_{\rm crit}$ is minimal.
3. A major Late Veneer impact (body $\gtrsim$ 750 km) occurs during the post-ejection plateau window (Appendix I). The impact's impulsive torque provides a jump $\Delta\Omega$, and ε is already near its maximum. The combination of the two lowers $\Omega_{\rm crit}$ and crosses the threshold.
4. A tongue of magma stretches, pinches off by Rayleigh–Plateau instability[85,86], delayed by Bingham cohesion, and detaches tangentially: one ejection episode.
5. The ejection carries away a fraction η of the angular momentum; Ω falls back below the threshold, then the system recharges until the next major impact.

9.2.6 *Role of cohesion: the* $\mathrm{Bi}_{\rm KH}$ *criterion*

The Bingham yield stress τ_y **does not enter** the ejection threshold (Eq. 28): at the torus scale, it is negligible compared to gravity. Its role is distinct: it maintains the cohesion of the ejected flow, which propagates as a *continuous tongue*, via the

Bingham-Kelvin-Helmholtz criterion $\mathrm{Bi}_{\mathrm{KH}} \gg 1$ (Appendix D). Flow cohesion and ejection threshold are thus two physically separate questions.

9.2.7 *Critical speed and maximum ejectable mass*

The critical ejection speed, derived from the energy escape condition, is:

$$U_{\mathrm{crit}}^{(0)} = \sqrt{2\, g_{\mathrm{eff}}^{(0)}\, R_{\mathrm{eq,proto}}} \approx 9.8\,\mathrm{km\,s^{-1}}, \tag{30}$$

which is consistent with the escape energy scale. The Mach number at ejection, using the sound speed in biphasic magma $c_s \approx 0.9$ km s^{-1} [87], is $\mathrm{Ma} = U_{\mathrm{crit}}/c_s \approx 10.9$: the ejection is **hypersonic**.

The maximum ejectable mass per episode is the fraction of the CMT with $U > U_{\mathrm{crit}}$:

$$M_{\mathrm{ej}}^{\mathrm{max}} \approx f_{\mathrm{inst}}\, M_{\mathrm{CMT}}, \tag{31}$$

where f_{inst} is the unstable fraction of the velocity distribution. For a typical rotation profile, this fraction is ≈ 0.1, giving:

$$M_{\mathrm{ej}}^{\mathrm{max}} \approx 2.0\text{–}4.0 \times 10^{22}\,\mathrm{kg}, \tag{32}$$

consistent with the mass budget (Section 10). This is why $N = 2$–3 episodes suffice.

9.2.8 *Cohésion of the ejected flow: the* $\mathrm{Bi}_{\mathrm{KH}}$ *criterion*

For a free flow, the Weber number $We \approx 1.4 \times 10^{15}$ would imply fragmentation by atomization. However, the CMT is not a free flow: it is a collective, self-gravitating toroidal flow, confined by toroidal symmetry, self-gravity, and atmospheric pressure.

The correct criterion is the Bingham-Kelvin-Helmholtz number, which compares the yield stress to the Kelvin-Helmholtz shear stress: for $\mathrm{Bi}_{\mathrm{KH}} \gg 1$, the yield stress suppresses the KH instability and cohesion is maintained [62]. Three independent conditions reinforce cohesion: (1) atmospheric confinement ($P \approx 10^5$–10^7 Pa); (2) internal crystal network ($\phi \approx 0.1$–0.3, multiplying fragmentation resistance by a factor 3 to 10); (3) density ratio $\rho_{\mathrm{flow}}/\rho_{\mathrm{atm}} \approx 200$ (Rayleigh-Taylor suppression). The full demonstration of the $\mathrm{Bi}_{\mathrm{KH}}$ criterion is given in Appendix D.

Effective cohesion reinforcements

The effective cohesion is the product of the three reinforcements:

$$\mathrm{Bi}_{\mathrm{KH}}^{\mathrm{eff}} = \mathrm{Bi}_{\mathrm{KH}} \times f_{\mathrm{atm}} \times f_{\mathrm{crist}} \times f_{\mathrm{RT}} \tag{33}$$

with:

- $f_{\text{atm}} \sim 10$–100 (atmospheric confinement) - $f_{\text{crist}} \sim 3$–10 (internal crystal network) - $f_{\text{RT}} \sim 10$–50 (Rayleigh-Taylor suppression)

With nominal values, $\text{Bi}_{\text{KH}} \sim 3 \times 10^{-2}$ and the product of the three reinforcements is ~ 300, giving $\text{Bi}_{\text{KH}}^{\text{eff}} \sim 9$, much greater than 1. Cohesion is therefore ensured.

Acknowledged limitation: this calculation is a nominal estimate. A 3D SPH simulation with full BH rheology is required to validate the exact values, in particular the nonlinear interaction between the three effects (Limitation L14, Section 20).

9.2.9 *Thermal resistance of the ejected tongue during hypersonic flight*

Ejection occurring at Ma ≈ 10.9 (Eq. 30), the magmatic tongue traverses the confining silicate atmosphere in a detached shock regime, analogous — inverted — to that of a body in atmospheric reentry. The stagnation-point shock heat flux follows the Sutton-Graves law, $q \sim k\sqrt{\rho_{\text{atm}}/R}\,U^3$, where R is the characteristic radius of the body. For the tongue geometry ($R \sim 10^3$ km, derived from the volume $V = M_{\text{ej}}^{\text{max}}/\rho_{\text{flow}}$), this flux remains modest ($q \sim 10^5$–10^6 W m^{-2}) compared to classical atmospheric reentry heat shields, because the specific flux decreases with the square root of the frontal radius.

Over the flight duration ($t_{\text{ej}} \approx 0.6$ h), the ablated mass — estimated by integrating this flux over the entire body surface and comparing it to a typical silicate ablation enthalpy ($\sim 10^7$ J kg^{-1}) — remains below 10^{-7} of $M_{\text{ej}}^{\text{max}}$: completely negligible for the mass budget (Section 10). This result is dominated by the large size of the ejected body (the surface-to-volume ratio, and thus ablation efficiency, decreasing with R), rather than by an active protection mechanism: in the same way that large bolides survive atmospheric entry proportionally better than small meteors, the ejected tongue, of planetary dimension, ablates negligibly over its flight time.

The tempered glass crust (Section 9.1), already formed before ejection to ensure flow cohesion against the Kelvin-Helmholtz instability, nevertheless provides additional insurance against local heat flux heterogeneities — at the leading edge, or in wake turbulence — not captured by this average shock flux estimate.

9.2.10 *Post-ejection transient precession and crustal dichotomy*

The ejection of $M_{\text{ej}} \approx 2$–4×10^{22} kg from the intertropical belt is not dynamically negligible. The asymmetric mass loss modifies the inertia tensor, producing: (1) a decrease $\Delta\Omega/\Omega \approx -10$–$15\%$, lowering g_{eff} and U_{crit} for the next episode; (2) shape oscillations as the spheroid contracts toward a rounder sphere; (3) a reorganization of obliquity toward a new value within $[40°, 70°]$.

This last point has a direct geometric consequence, not merely a numerical one: since the intertropical belt $|\phi| < 30°$ is defined perpendicular to the instantaneous axis $\mathbf{\Omega}(t)$ (Section 3), a new value of ε shifts the position of the dynamic equator $\phi = 0$

itself, and hence the entire active belt, relative to the body. The intertropical belt hosting the second episode is therefore not, in general, the same region of the body as that hosting the first: same angular width ($\pm 30°$), same definition relative to $\mathbf{\Omega}(t)$, but a different location on the spheroid. It is this geometric shift of the active belt, episode after episode, that constitutes the direct cause of the crustal asymmetry described below, and not merely the change in the numerical value of ε.

This shift of the active belt, combined with the shape oscillations of point (2) above, has an additional consequence for the CMT itself: the torus that reforms for the next episode does not reproduce an identical copy of the previous one simply shifted. It forms on a spheroid whose proper shape has slightly changed, along a belt whose position has itself changed. The next CMT is therefore ejected toroidally in a manner distinct from the previous one, both in shape and in orientation. This remains, at this stage, a qualitative statement: no quantitative derivation of the shape or angular offset between successive episodes is attempted here.

The second episode ejects a cohesive toroidal flow onto a still-hot and deformable proto-POB, within a limited solid angle (corresponding to the projected intertropical belt), which becomes the current farside. Three mechanisms amplify the resulting crustal asymmetry: early gravitational locking ($\tau_{\text{lock}} \sim 10^4$–$10^5$ yr[5]), the residual obliquity of the proto-POB, and irreversible BH plastic welding. The thickness ratio $\approx 2{:}1$ (nearside ≈ 34 km versus farside ≈ 54 km, Wieczorek et al.[5]) is a qualitative prediction of the TTP.

Vector recoil budget: the inherited inertial obliquity. The recoil formalism developed above produces a second observable, independent of crustal dichotomy. Each ejection carries an angular momentum $\boldsymbol{L}_{\text{ej},k}$ whose direction is fixed by the CMT geometry at the time of firing — geometry that reorients from one episode to the next, since the active belt follows $\mathbf{\Omega}(t)$. The rotational angular momentum vector of the proto-Earth after N episodes is:

$$\boldsymbol{L}_p^{\text{final}} = \boldsymbol{L}_p^{\text{init}} - \sum_{k=1}^{N} \boldsymbol{L}_{\text{ej},k}, \tag{34}$$

and the inherited inertial obliquity at the start of tidal recession is $\varepsilon_{\text{I}}^{(0)} = \angle\big(\boldsymbol{L}_p^{\text{final}}, \hat{\boldsymbol{n}}_{\text{orb}}\big)$, where $\hat{\boldsymbol{n}}_{\text{orb}}$ is the normal to the heliocentric orbital plane. The *same* mechanism — the ejection recoil — thus simultaneously governs the active belt shift (crustal dichotomy, above) and the final orientation of $\boldsymbol{L}_p$ (inertial obliquity). This second consequence determines whether the post-formation angular momentum sink activates (condition $\varepsilon_{\text{I}} \gtrsim 60$–$65°$ at the Laplace plane transition crossing, Sections 12 and 12.3.1); its quantitative evaluation, which applies Eq. 34 to the nutation distribution $[40°, 70°]$, falls under Limitation L15.

Monte Carlo evaluation (adverse hypothesis). Eq. 34 is evaluated by random sampling (4×10^5 realizations). On this specific point — where the result conditions the activation of the nominal channel — the TTP deliberately adopts the most *adverse* hypothesis: the azimuthal phases of the three ejections are drawn uniformly over $[0, 2\pi]$ (*free* azimuths, uncorrelated), and the instantaneous obliquity of the axis at the time of each firing varies over a nutation amplitude of $\pm 30°$ — i.e., twice the value invoked elsewhere. Ejected masses are drawn over $M_{\rm ej} \in [2, 4] \times 10^{22}$ kg, under the constraint $|\boldsymbol{L}_{{\rm ej},k}| \leq 0.955\, M_{\rm ej} r_{\rm ej} U_{\rm crit}$ (Section 11.6), and the initial inertial obliquity $\varepsilon_{\rm I}^{\rm init}$ is swept over $[60°, 70°]$.

The result (Fig. 3) survives this severe treatment. The distributions of $\varepsilon_{\rm I}^{(0)}$ are *wide* (P1–P99 covering $\approx 18°$, a direct consequence of free azimuths), but remain above the activation threshold $\approx 60°$ in **97 to 100% of draws** depending on $\varepsilon_{\rm I}^{\rm init}$. A remarkable fact emerges: recoils *systematically increase* obliquity (median raised from 60° to 69°, from 65° to 73°, from 70° to 76°), because each recoil removes angular momentum preferentially along the axis, tilting $\boldsymbol{L}_p$ further. Azimuthal orientation therefore need not be correlated for the upper part of the Hadean interval $[40°, 70°]$ to be *preserved*: the result is generic, not tuned.

Two honest caveats remain. First, the model remains two-dimensional (planar nutation); the full three-dimensional version falls under Limitation L15. Second, the margin above the *upper* threshold of 65° narrows for $\varepsilon_{\rm I}^{\rm init}$ at the bottom of the range: at $\varepsilon_{\rm I}^{\rm init} = 60°$, 82% of draws exceed 65° (versus > 99% for $\varepsilon_{\rm I}^{\rm init} = 70°$). Since the exact value of the activation threshold itself is debated in the literature — $\approx 60°$ in the integrations of Ćuk et al.[16], up to 68.9° for the strict instability of the Laplace surface of Tremaine et al.[15] — the result is robust for the low threshold and becomes conditional for the high threshold. Constraining this threshold is an open question that the TTP inherits from the mechanism, not a flaw of its own.

9.3 Transition 3 — Magnetic: the progressive dynamo

The delay of about 350 Myr between the inferred time of lunar formation (≈ 4.55 Ga) and the first geological detection of Earth's magnetic field (≈ 4.2 Ga, Tarduno et al.[6]) is not an adjusted parameter in the TTP. It is proposed as the sum of three independently constrained durations:

$$\Delta t_{\rm dynamo} = \underbrace{\Delta t_{\rm ej}}_{\approx 60\,\rm Myr} + \underbrace{\Delta t_{\rm refr}}_{\approx 140\,\rm Myr} + \underbrace{\Delta t_{\rm em}}_{\approx 150\,\rm Myr} \approx 350\,\rm Myr. \tag{35}$$

Phase 1 ($\approx$ 60 Myr) — Active Fe-Ni segregation. Constrained by the hafnium-tungsten (Hf-W) chronometer[34,35]: the core grows rapidly but the thermal gradient remains subadiabatic; no dynamo is possible during this phase.

This timescale of ≈ 60 Myr concerns the *global* core segregation — the cumulative

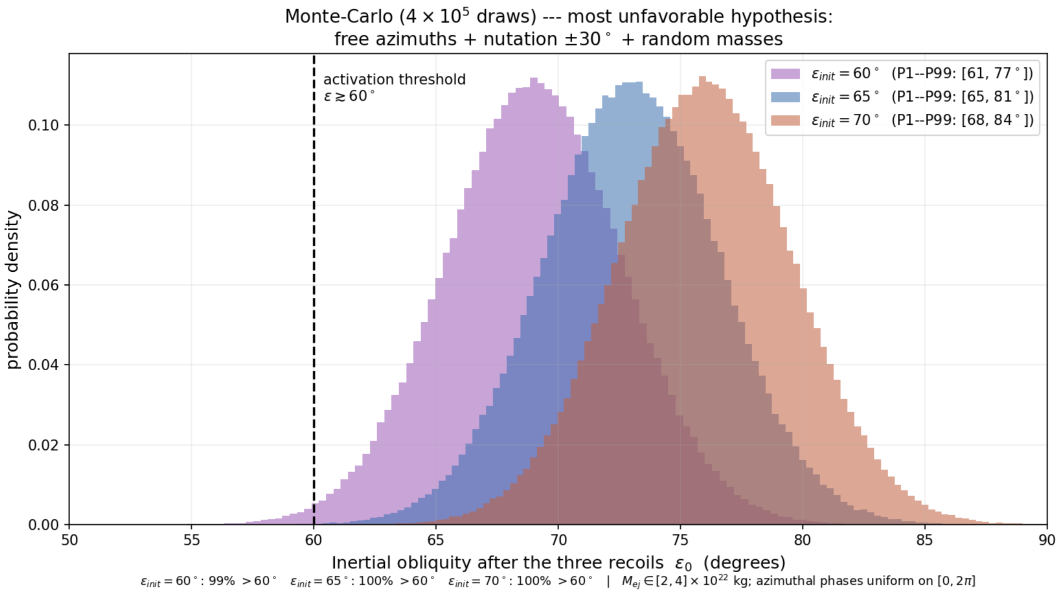

Figure 3: Distribution of post-ejection inertial obliquity $\varepsilon_{\mathrm{I}}^{(0)}$ from the vector recoil budget of the three ejections (Eq. 34), for 4×10^5 Monte Carlo draws, under the *most adverse* hypothesis: free azimuthal phases (uniform over $[0, 2\pi]$, uncorrelated), nutation $\pm 30^\circ$, random ejected masses. Despite wide distributions ($\approx 18^\circ$ between P1 and P99, a consequence of free azimuths), each case remains above the activation threshold $\varepsilon_{\mathrm{I}} \gtrsim 60^\circ$ (dashed) in 97–100% of draws, and the medians are *higher* than the initial obliquities: recoils preserve, or even strengthen, the inherited obliquity. The margin above 65° narrows for $\varepsilon_{\mathrm{I}}^{\mathrm{init}} = 60^\circ$ (82%), making the exact threshold value a point to be constrained (Limitation L15).

accumulation of metallic iron at the center of the entire body — and not the resupply of each ejection episode, which operates at a much smaller spatial and temporal scale. The local density stratification within the active belt, that which feeds the β term of Eq. 27, is renewed by Stokes sedimentation ($\tau_{\text{Stokes}} \approx 254$ s, Appendix F) and by convective stirring of the CMT ($U_{\text{turb}} \approx 0.8$ km s^{-1}, Eq. 19) over a characteristic distance largely exceeding the active belt thickness during the recharge window ($\approx$ 10–12 days). The local resupply required for each ejection cycle is therefore not limited by the slow global core segregation: these are two distinct clocks, interlocked with one another.

Phase 2 ($\approx$ 140 Myr) — Protocrust formation. The growing Moon stabilizes obliquity to $\approx 23.5°$; protocrust forms around 4.4 Ga, as attested by Jack Hills zircons[72,73].

Phase 3 ($\approx$ 150 Myr) — Gradual emergence of the dynamo. Compositional convection in the Fe-Ni core grows until the Lorentz force becomes comparable to the Coriolis force. The Elsasser number Λ balances these two forces. The threshold $B_{\text{crit}} \approx 2.6$ mT is derived in Appendix E. Modern geodynamo simulations[88,89] show that the transition is gradual, not abrupt: $\Lambda \sim 1$ is an order-of-magnitude reference, not a strict threshold.

Section summary

The $\approx$ 350 Myr delay involves no free parameter: Δt_{ej} is constrained by the Hf-W chronometer[34,35]; Δt_{refr} is constrained by Jack Hills zircons at 4.4 Ga[72,73]; and Δt_{em} is constrained by Tarduno et al.[6] at 4.2 Ga. These are three independent observations, not an interpolation. The associated uncertainty is ± 50 Myr (Limitation L4, Section 20).

10 Mass budget: $N = 2$–3 episodes

10.1 Corrected lunar target mass

A fundamental point: the currently observed lunar mass $M_{\text{Moon}} = 7.34 \times 10^{22}$ kg is *not* the mass that the TTP mechanism must produce. The Moon accreted additional mass after the ejection episodes: a differentiated body about 260 km in diameter impacted the Moon at the South Pole–Aitken basin and *remained there*, contributing $M_{\text{SPA}} \approx 3.2 \times 10^{19}$ kg[90] — this material is now part of the observed lunar mass — and documented late accretion contributed an additional mass ΔM_{late}[91]. The target mass for the TTP mechanism is therefore:

$$M_{\text{TPT}} = M_{\text{Moon}} - M_{\text{SPA}} - \Delta M_{\text{late}} < 7.34 \times 10^{22} \text{ kg}. \tag{36}$$

10.2 Mass budget equation

The mass budget equation

$$M_{\text{Moon}} = N \cdot f_{\text{cap}} \cdot M_{\text{ej}}^{\max} + M_{\text{SPA}} + \Delta M_{\text{late}}, \tag{37}$$

with $N = 2$–3 (nominal), $f_{\text{cap}} \in [0.65, 0.85]$ (nominal 0.70), $M_{\text{ej}}^{\max} \approx 2.0$–$4.0 \times 10^{22}$ kg per episode.

f_{cap}	$M_{\text{TPT}}^{(N=2)}$	$M_{\text{TPT}}^{(N=3)}$	Fraction of M_{Moon}
0.50	2.5×10^{22} kg	3.7×10^{22} kg	34–50%
0.70	3.5×10^{22} kg	5.2×10^{22} kg	48–71%
0.85	4.3×10^{22} kg	6.4×10^{22} kg	59–87%

The remainder is accounted for by M_{SPA} and documented late accretion. The budget is robust over the entire f_{cap} interval.

f_{cap} is an increasing state variable: $\partial f_{\text{cap}}/\partial M_{\text{Moon}} > 0$, because the larger the proto-Moon, the more efficiently it captures subsequent ejecta. This positive feedback, combined with the decrease of $g_{\text{eff}}^{(k)}$ at each episode, defines a dynamical attractor that reinforces the mechanism.

Feedback during TTP episodes. This growth of f_{cap} with already accreted mass can be formalized, to first order, by a linear law:

$$f_{\text{cap}}(M_{\text{proto}}) = f_0 + \alpha_f \, M_{\text{proto}}, \qquad \alpha_f > 0, \tag{38}$$

where M_{proto} is the proto-Moon mass already in place before the episode considered. In steady state (after the last episode, $M_{\text{proto}} \simeq M_{\text{Moon}} - \Delta M_{\text{late}}$), the mass budget (37) closes as an implicit equation in ΔM_{late}, whose sensitivity to the nominal value f_0 is $\partial \Delta M_{\text{late}}/\partial f_0 = -N M_{\text{ej}}^{\max}/(1 - N M_{\text{ej}}^{\max} \alpha_f)$. With $N M_{\text{ej}}^{\max} \sim 7.5 \times 10^{22}$ kg, a variation of f_0 of 0.1 changes ΔM_{late} by about 7.5×10^{21} kg, i.e., $\sim 10\%$ of the lunar mass. This scaling law does not predict α_f *ab initio* — its value falls under 3D SPH simulation (Limitation L2) — but it formalizes the feedback already qualitatively established above and bounds α_f physically by the condition $N M_{\text{ej}}^{\max} \alpha_f < 1$, necessary for the budget to remain well-posed.

This feedback effect *during* the episodes is distinct from, and cumulative with, the geometric scaling law of late accretion *after* the end of ejections ($\Delta M_{\text{late}} \propto M_{\text{TPT}}^{2/3}$, Section 10.3): the former acts in real time on the capture of successive ejecta from the mechanism itself, the latter on the subsequent capture of external material by the already-formed body. The final budget results from the sum of these two contributions.

A sensitivity analysis varying f_{cap} and $M_{\text{ej}}^{\max}$ simultaneously over their respective ranges (Section 10) confirms $N = 2$–3 for the nominal case as well as for any favorable combination of the two parameters. The most adverse corner of this range — f_{cap} at its lower bound combined with $M_{\text{ej}}^{\text{max}}$ at its lower bound — formally pushes N beyond this range, with a corresponding higher ΔM_{late} deficit to be filled by late accretion.

This increased deficit is not freely compensable, however, because the capture capacity of late accretion itself depends on the mass of the capturing body: the geometric cross-section grows as $M_{\text{TPT}}^{2/3}$, so $\Delta M_{\text{late}} \propto M_{\text{TPT}}^{2/3}$ at fixed impact speed and external mass flux. Applying this scaling law between the nominal case ($f_{\text{cap}} = 0.70$, $N = 3$, $M_{\text{TPT}} \approx 5.2 \times 10^{22}$ kg, $\Delta M_{\text{late}} \approx 2.13 \times 10^{22}$ kg) and the most adverse corner of the official range ($f_{\text{cap}} = 0.65$, $M_{\text{ej}}^{\text{max}} = 2.0 \times 10^{22}$ kg, $N = 3$, $M_{\text{TPT}} \approx 3.9 \times 10^{22}$ kg, requiring $\Delta M_{\text{late}} \approx 3.45 \times 10^{22}$ kg), the late accretion mass effectively available in this adverse corner reduces to $\Delta M_{\text{late}} \approx 1.76 \times 10^{22}$ kg — leaving a residual deficit of about 1.7×10^{22} kg ($\approx 23\%$ of M_{Moon}) unfilled. This adverse parameter combination is thus doubly penalized by this mechanism, rather than compensated by a simple adjustment of late accretion: low capture efficiency simultaneously reduces the direct contribution of the TTP mechanism and the subsequent capacity of the body to accrete the missing material. $N = 2$–3 remains the nominal and most parsimonious result of the model; the most adverse corner of the parameter range remains, under this scaling law, physically insufficient to reach the observed lunar mass without an additional, as-yet undocumented mass source.

10.3 f_{cap} as a testable prediction, not a free parameter

The position adopted here is that f_{cap} should not be treated as an adjustable parameter whose broad range would make the mass budget uninformative, but as a quantity that the TTP predicts quantitatively and that can, in principle, falsify the mechanism.

Physical basis: the classical accretion disk range. Reference N-body simulations of lunar accretion from an impact-generated debris disk[92] establish that the efficiency of incorporation of disk material into a final Moon is 10 to 55%, this efficiency increasing with the initial specific angular momentum of the disk. This result, robust over a wide range of initial conditions and numerical resolutions, constitutes the external reference to which f_{cap} must be compared.

The deviation as a potential signature. The nominal value $f_{\text{cap}} \approx 0.70$ adopted in the TTP is significantly higher than the upper bound of the dispersed-disk accretion range (55%,[92]). Within the TTP framework, this deviation is not accidental: it follows from the coherence of the ejected toroidal flow and its orbital concentration within a limited solid angle (Appendix G), as opposed to accretion from a debris disk dispersed over 4π steradians. If this interpretation is correct, the deviation between the f_{cap} value adopted by the TTP and the disk accretion range becomes a discriminating test rather than merely a parametric latitude.

Falsifiable prediction — P17 — Capture efficiency beyond the classical debris-disk accretion range

The orbital coherence of the toroidal flow ejected by the TTP pushes the effective capture efficiency beyond the classical 10–55% range established for accretion from a dispersed debris disk[92]. The TTP predicts $f_{\mathrm{cap}} \gtrsim 0.65$–$0.70$.

Falsification: if a value of f_{cap} derived from a dedicated 3D SPH simulation of the CMT geometry (BH rheology, atmospheric confinement, orbital concentration within the intertropical belt) falls within the classical 10–55% range rather than above, this aspect of the theory is falsified. A f_{cap} measured in the classical range would not necessarily invalidate the TTP as a whole (the mass budget would remain reachable via M_{SPA} and late accretion), but it would deprive f_{cap} of its status as a distinctive signature of toroidal ejection geometry.

Acknowledged limitation: this prediction rests on the assumption that the orbital concentration of the TTP flow (Appendix G) produces a qualitatively different accretion regime from the classical dispersed disk. This assumption is itself Level 3 (not quantitatively demonstrated), so this becomes a robust test only once the 3D SPH simulation is available. In the meantime, this remains a qualified prediction: the deviation from the classical range is consistent with the TTP, but does not independently confirm it.

Status of the hypothesis: plausible-high, not conservative. For methodological consistency, it is important to distinguish hypotheses in this work according to their position within their plausibility interval. On some points — notably the vector recoil budget (Section 9.2, Fig. 3) — the TTP deliberately adopts the most *adverse* hypothesis (free azimuths, nutation $\pm 30°$), and the result still holds. The value $f_{\mathrm{cap}} = 0.70$ has a different status: it is a *high but physically grounded* value, not a conservative hypothesis. It is justified by a real mechanism — the toroidal tongue remains cohesive for about half its mass, with the remainder recovered by re-accretion of co-orbital fragments, which share the proto-Moon's orbit and coalesce instead of escaping[93]. This mechanism makes $f_{\mathrm{cap}} = 0.70$ achievable, but does not make it adverse: the physical lower bound of f_{cap} remains ≈ 0.45–0.55 (Appendix G), independent of the cohesion hypothesis. The TTP therefore does not claim systematically to adopt the strictest hypotheses; it adopts the adverse hypothesis where the result is sensitive and the parameter free, and a plausible-high grounded value where a physical mechanism supports it — documenting in both cases the opposite bound.

11 Angular momentum transport by ejecta

In the present framework, angular momentum loss is not treated as a single-speed ballistic process, but as a distributed transport mechanism resulting from a spectrum of ejected material generated during successive phases driven by differentiation.

11.1 Ejecta distribution function

We describe the mass distribution of ejecta as a function of speed and ejection angle:

$$dM = \rho(v, \theta)\, dv\, d\theta, \tag{39}$$

where v is the ejection speed, θ is the angle relative to the local tangential direction, and $\rho(v, \theta)$ is the ejecta distribution function depending on the phase.

The distribution $\rho(v, \theta)$ is not arbitrary. It follows from three physical processes: the CMT speed spectrum is not monokinetic, turbulent fluctuations at the Rhines scale (Section 7.6) producing a speed spread $\delta v_{\mathrm{CMT}} \sim U_{\mathrm{turb}} \approx 0.8\ \mathrm{km\,s^{-1}}$; and the finite torus extent produces an angular spread $\rho_\theta(\theta) \propto \cos\theta$ for $|\theta| < \theta_{\max}$, with $\theta_{\max} \approx 30^\circ$.

For analytical tractability, we adopt the separable form:

$$\rho(v, \theta) = M_{\mathrm{ej}}\, \mathcal{N}_v\, \mathcal{N}_\theta\, \exp\left[-\frac{(v-\bar{v})^2}{2\sigma_v^2}\right] \cos\theta, \quad v > v_{\mathrm{esc}},\ |\theta| < \theta_{\max}, \tag{40}$$

where $v_{\mathrm{esc}} = \sqrt{2 g_{\mathrm{eff}} R_p}$. No free parameter is introduced beyond those already present in the theory developed in previous sections.

11.2 Specific angular momentum

The specific angular momentum of an ejecta element is $h = r\, v \sin\theta$, where r is the effective emission radius. This radius is not simply the planetary radius R_p: the silicate atmosphere (Section 2) extends the effective surface to $r_{\mathrm{atm}} \approx R_p + H$, where the scale height $H \approx 200$ km for $T \approx 3000$ K and $g \approx 6\ \mathrm{m/s^2}$; and the toroidal flow is elevated above the surface by centrifugal effects, the CMT center being at $r_{\mathrm{CMT}} \approx R_p + U_{\mathrm{CMT}}^2/(2 g_{\mathrm{eff}}) \approx R_p + 500$ km. We therefore adopt $r \approx R_p + \Delta r$, with $\Delta r \approx 300$–500 km. This correction introduces only a few percent uncertainty in the angular momentum budget, well within the overall error margin.

The silicate atmosphere mass itself ($M_{\mathrm{atm}} \lesssim 10^{21}$ kg, obtained from $M_{\mathrm{atm}} \sim 4\pi R_p^2 P/g$ for $P \sim 10^5$–10^7 Pa) is negligible in both budgets: it represents less than 2% of the lunar target mass in the mass budget (Section 10), and its contribution to the moment of inertia ($I_{\mathrm{atm}} \sim \frac{2}{3} M_{\mathrm{atm}} R_p^2$) and to the initial angular momentum L_{init} remains, in both cases, below 0.1% of the corresponding quantities.

11.2.1 Assembly beyond the Roche radius

An apparent paradox deserves to be resolved here, as it conditions the very birth of the Moon. The maximum specific angular momentum an ejecta can carry at launch, $h_{\max} \approx 7.74 \times 10^{10}\ \mathrm{m^2\,s^{-1}}$, corresponds to a circular orbit of radius $r_{\mathrm{circ}} = h_{\max}^2/(GM_p) \approx 1.5 \times 10^7$ m, i.e., about 15,000 km — *below* the Roche radius ($\approx 22,200$ km), where tidal forces prevent direct accretion of a self-gravitating body[94]. Taken in isolation, no captured ejecta can therefore become a Moon: the material seems condemned to orbit as a ring, torn apart by tides.

The resolution of this paradox is known, and requires no new hypothesis: the captured torus is not a set of particles fixed on independent Keplerian orbits, but a dense disk where inelastic collisions and gravitational interactions generate an effective viscosity. This viscosity does what any disk viscosity does: it transports angular momentum outward and mass in both directions, so that a fraction of the disk is pushed beyond the Roche radius — where accretion becomes permissible again and where the proto-Moon assembles, fed by the disk's edge. This spreading mechanism is precisely the one that builds the Moon in post-impact disk models: in the particulate regime, assembly occurs in a few months to a year[95]; in the fluid disk regime, where radiative cooling regulates viscosity, it extends over $\sim 10^2$–10^3 yr[96]. In both cases, assembly is complete well before the encounter with solar resonant sinks (evection resonance at 10^3–10^5 yr, Section 11.6; Laplace plane transition at 10^5–10^7 yr, Section 12): the Moon exists as a single body when the angular momentum sink engages.

This result has an important conceptual consequence for the TTP: the cohesion of the ejected flow (Bingham-Kelvin-Helmholtz number, Appendix D, Limitation L14) governs the *launch and transport* phase — how much material reaches orbit, and with what angular momentum — but it is not required for the *assembly* phase, which viscous spreading accomplishes alone. The critical question posed to the SPH simulation is therefore not "does the flow remain perfectly cohesive to orbit?" but "what fraction of the ejected mass is captured in stable orbit?" — i.e., the determination of f_{cap} (Limitation L2).

11.2.2 Ejection energy budget

With the angular momentum budget now established, the symmetric question arises naturally: does the proto-Earth have the *energy* to launch three magmatic tongues at escape speed? The total kinetic energy bill for the ejecta is:

$$E_{\mathrm{kin,ej}} = \frac{1}{2} N M_{\mathrm{ej}}\, \bar{v}^2 \approx \frac{1}{2} \times 3 \times 2.5 \times 10^{22} \times (9.8 \times 10^3)^2 \approx 3.6 \times 10^{30}\ \mathrm{J}, \qquad (41)$$

most of which is spent climbing the potential well ($\bar{v}$ being close to escape speed, the residual kinetic energy at infinity is small). The available reservoir is the proto-

Earth's rotational energy:

$$E_{\text{rot}} = \frac{1}{2}\, I_p\, \Omega_p^2 \approx \frac{1}{2} \times 1.32 \times 10^{38} \times (4.99 \times 10^{-4})^2 \approx 1.6 \times 10^{31}\,\text{J}. \tag{42}$$

The ratio $E_{\text{kin,ej}}/E_{\text{rot}} \approx 0.22$: the three episodes consume about one-fifth of the spin reservoir.

The total angular momentum carried *at launch* by all ejecta, whether subsequently captured or escaped, is:

$$L_{\text{ej}} = M_{\text{ej}}\, r\, \bar{v}\, \langle \sin\theta \rangle \approx 7.5 \times 10^{22} \times 7.9 \times 10^{6} \times 9.8 \times 10^{3} \times 0.25 \approx 1.5 \times 10^{33}\ \text{J} \cdot \text{s}, \tag{43}$$

where $\langle \sin\theta \rangle \approx 0.25$ is the average geometric factor accounting for the angular spread of the torus ($\theta_{\max} \approx 30°$). This quantity L_{ej} is distinct from L_{loss} as formally defined in Eq. 46 (Section 11.3): L_{ej} covers all ejecta at launch time, L_{loss} covers only the non-captured fraction. Weighting by the escaped fraction $(1 - f_{\text{cap}}) = 0.30$, we obtain a mean estimate of L_{loss} consistent with this formal definition:

$$L_{\text{loss}} \approx (1 - f_{\text{cap}})\, L_{\text{ej}} \approx 0.30 \times 1.5 \times 10^{33} \approx 4.5 \times 10^{32}\ \text{J} \cdot \text{s}. \tag{44}$$

This value is numerically below the conservative bound $L_{\text{esc}}^{\max} \approx 1.74 \times 10^{33}$ J s (Section 11.6), which instead assumes a purely tangential ejection ($\sin\theta = 1$) for the escaped material only — a deliberately generous upper bound, not a central value. The escape channel thus covers an even more minor fraction of the deficit ΔL than the conservative bound alone suggests.

The associated rotational energy loss for L_{ej} is:

$$\Delta E_{\text{rot}} \approx 2\, \frac{L_{\text{ej}}}{L_{\text{init}}}\, E_{\text{rot}} \approx 2 \times \frac{1.5 \times 10^{33}}{6.59 \times 10^{34}} \times 1.6 \times 10^{31} \approx 7.3 \times 10^{29}\,\text{J}, \tag{45}$$

to which is added, and dominates, the Fe-Ni segregation heat ($\approx 3 \times 10^{30}$ J, Section 9.2), which constitutes the main source covering the ejection energy bill and also sustains the molten state and convection. The energy budget is therefore covered to order unity by reservoirs already inventoried in the model: ejection is energetically feasible without a hidden source, and without compromising the rotational stability of the host body.

11.3 Angular momentum loss

The total angular momentum carried away by non-accreted material is:

$$L_{\rm loss} = \int\int \left(1 - f_{\rm cap}(v,\theta)\right) \; r\, v \sin\theta\, \rho(v,\theta)\, dv\, d\theta. \tag{46}$$

The capture efficiency $f_{\rm cap}(v,\theta)$ is not a constant: it depends on the orbital fate of each ejecta element,

$$f_{\rm cap}(v,\theta) = \frac{1}{2}\left[1 + \mathrm{erf}\left(\frac{v\sin\theta - v_{\rm crit}}{\sigma_f}\right)\right], \tag{47}$$

where $v_{\rm crit} \approx v_{\rm circ} = \sqrt{g_{\rm eff} R_p}$ is the circular orbital speed and σ_f a smoothing width representing the finite transition between captured and lost material. Rather than treating $f_{\rm cap}$ as a single ad hoc value, we express it here as a function $f_{\rm cap}(v,\theta)$ of orbital mechanics. The nominal value $f_{\rm cap} \approx 0.70$, independently estimated from the cohesive fraction of the CMT and the re-accretion of co-orbital fragments, is consistent with the mass-weighted average of this function over the proposed ejecta spectrum.

11.4 Discrete formulation and Monte Carlo implementation

For numerical implementation, the expression can be discretized as:

$$L_{\rm loss} = \sum_i m_i\, r_i\, v_i \sin\theta_i \;\left(1 - f_{{\rm cap},i}\right). \tag{48}$$

A stochastic treatment can implement this discrete formulation by Monte Carlo sampling of $\rho(v,\theta)$: sample N particles from $\rho(v,\theta)$, assign each particle a mass $m_i = M_{\rm ej}/N$, compute $h_i = r_i v_i \sin\theta_i$, compute $f_{{\rm cap},i}$ from Eq. 47, and sum $L_{\rm loss} = \sum_i m_i h_i (1 - f_{{\rm cap},i})$. This Monte Carlo approach is recommended for SPH 3D validation (Limitation L2): it provides a robust check of analytical estimates.

In a multi-phase formation scenario, ejecta are additionally grouped into k distinct formation phases:

$$L_{\rm loss} = \sum_k \sum_i m_{k,i}\, r_{k,i}\, v_{k,i} \sin\theta_{k,i} \left(1 - f_{{\rm cap},k,i}\right), \tag{49}$$

each phase k having its own mean ejection speed $\bar{v}_k$ (decreasing as the proto-Earth loses mass and angular momentum), its own angular spread $\theta_{{\rm max},k}$ (which may increase as the CMT reorganizes between episodes), and its own mean capture efficiency $f_{{\rm cap},k}$ (which increases as the proto-Moon grows, the positive feedback of Section 10.3).

11.5 Conservative angular momentum partitioning during ejections

Before writing the global budget, a bookkeeping ambiguity must be resolved. The placing of lunar material into orbit does not constitute a *creation* of angular momentum: it is a *redistribution* of the proto-Earth's initial angular momentum among several reservoirs — the residual Earth spin, the orbital angular momentum of the captured material, its own spin, and the momentum carried away by material permanently lost.

Each ejection is represented by a jump operator $J_k : \mathbf{X}^{(k-)} \mapsto \mathbf{X}^{(k+)}$, where $k-$ and $k+$ denote the states immediately before and after. Over the brief duration of the event and in the absence of significant external torque, the total angular momentum of the complete system is conserved through the jump:

$$\boxed{\boldsymbol{L}_{\text{tot}}^{(k-)} = \boldsymbol{L}_{\text{tot}}^{(k+)}.} \tag{50}$$

Decomposing the post-jump state, the redistribution law associated with the jump is:

$$\boldsymbol{S}_{\oplus}^{(k+)} = \boldsymbol{S}_{\oplus}^{(k-)} - \boldsymbol{L}_{\text{orb},k}^{+} - \boldsymbol{S}_{k}^{+} - \boldsymbol{L}_{\text{esc},k}^{+} - \Delta\boldsymbol{L}_{\text{int},k}, \tag{51}$$

where $\boldsymbol{S}_{\oplus}$ is the Earth's spin, $\boldsymbol{L}_{\text{orb},k}^{+}$ the orbital angular momentum of the bound fraction, $\boldsymbol{S}_{k}^{+}$ its own spin, $\boldsymbol{L}_{\text{esc},k}^{+}$ the momentum carried by escaped material, and $\Delta\boldsymbol{L}_{\text{int},k}$ the change in unresolved internal reservoirs (negligible compared to $\boldsymbol{L}_{\text{esc},k}$ once internal re-equilibration conserves mantle angular momentum).

The essential point is that the lunar orbit is *drawn from* the initial reservoir, not added to it. A writing such as $L_{\text{proto}}^{0} + L_{\text{orb,L}}$ would therefore count the same quantity twice. After the three ejections, the angular momentum of the bound Earth–Moon subsystem is:

$$\boxed{\boldsymbol{L}_{\text{lie}}^{+} = \boldsymbol{L}_{\text{proto}}^{0} - \sum_{k=1}^{3} \boldsymbol{L}_{\text{esc},k}.} \tag{52}$$

Two levels of conservation must be distinguished: the complete system conserves $\boldsymbol{L}_{\text{tot}}$ through each jump, while the bound subsystem $\boldsymbol{L}_{\text{lie}} = \boldsymbol{S}_{\oplus} + \boldsymbol{L}_{\text{orb,L}} + \boldsymbol{S}_{\text{L}}$ loses the fraction carried away when some ejecta permanently escape.

Scalar coplanar approximation. Relation (52) is vectorial. When vectors are approximately collinear and lunar spin is negligible, it reduces in norm to $L_{\text{lie}}^{+} \simeq S_{\oplus}^{+} + L_{\text{orb,L}}^{+}$, from which post-formation Earth rotation follows as a direct consequence of the partition:

$$S_{\oplus}^{+} \simeq L_{\text{lie}}^{+} - L_{\text{orb,L}}^{+}, \qquad T_{\text{rot}}^{+} = \frac{2\pi I_p}{S_{\oplus}^{+}}. \tag{53}$$

This scalar reduction is legitimate only under the coplanar hypothesis above; the vector recoil budget (Eq. 34) constitutes its full treatment.

Numerical application. With nominal values $L^0_{\rm proto} = 6.59 \times 10^{34}$ J s $\simeq 1.90\, L_{\rm TL}$ and a lunar orbital angular momentum at $a_0 = 3.8\, R_\oplus$ of $L_{\rm orb,L}(a_0) \simeq 7.2 \times 10^{33}$ J s, the conservative redistribution gives $S^+_\oplus \simeq 5.87 \times 10^{34}$ J s, i.e., $T^+_{\rm rot} \simeq 3.9$ h for $I_p = 1.32 \times 10^{38}$ kg m^2: formation slows the proto-Earth from ≈ 3.5 h to ≈ 3.9 h. After subtracting the permanently carried momentum ($\sum_k L_{{\rm esc},k} \lesssim L^{\rm max}_{\rm esc} = 1.74 \times 10^{33}$ J s, Eq. 56), the net value of the bound subsystem lies in the interval:

$$\boxed{L^+_{\rm lie} \simeq 1.85\text{–}1.90\, L_{\rm TL},} \tag{54}$$

the upper bound corresponding to negligible losses and the lower bound to $L_{\rm esc}$ taken at its maximum. It is this time-stamped, loss-corrected quantity — and not the sum $I_p\Omega_p + L_{\rm orb}$ — that constitutes the physically relevant initial condition for the orbital evolution prior to the Laplace transition.

11.6 Closure of the angular momentum budget

The total angular momentum budget is:

$$L_{\rm init} = L_{\rm TL} + L_{\rm esc} + L_{\rm tidal} + L_{\rm other}, \tag{55}$$

where:

- $L_{\rm init} = 6.59 \times 10^{34}$ J s is the initial angular momentum of the proto-Earth at $T = 3.5$ h;
- $L_{\rm TL} = 3.47 \times 10^{34}$ J s is the current angular momentum of the Earth–Moon system (Earth spin + lunar orbit);
- $L_{\rm esc}$ is the angular momentum carried away by ejecta that permanently escape the system;
- $L_{\rm tidal}$ is the angular momentum evacuated to heliocentric orbit by solar resonant sinks: the Laplace plane transition (nominal channel, Section 12) and the evection resonance (secondary channel);
- $L_{\rm other}$ groups losses through other channels (magnetic torque, solar wind, interaction with the protoplanetary disk).

Conservative bound on $L_{\rm esc}$. For ejecta launched from $r \approx 7.9 \times 10^6$ m at $v \approx 9.8$ km/s (pre-shock speed), the maximum specific angular momentum is $h_{\rm max} = r\bar{v} = 7.74 \times 10^{10}$ m²/s. With a total ejected mass $M_{\rm ej} = 7.5 \times 10^{22}$ kg and a capture efficiency $f_{\rm cap} = 0.70$, the escaped mass is $M_{\rm esc} = 2.25 \times 10^{22}$ kg. The upper bound on $L_{\rm esc}$ is therefore:

$$L^{\rm max}_{\rm esc} = M_{\rm esc} h_{\rm max} \approx 1.74 \times 10^{33}\ {\rm J \cdot s}. \tag{56}$$

This bound is computed with the pre-shock speed. Since ejection is hypersonic (Ma ≈ 11, Section 9.2), a detached shock forms at the CMT interface, reducing the tangential speed of ejecta. The actual value of L_{esc} is therefore likely below this bound. It should be understood as a conservative estimate — an upper limit, not a central value.

Deficit and hierarchy of resonant sinks. The deficit relative to the current state is:

$$\Delta L = L_{\text{init}} - L_{\text{TL}} - L_{\text{esc}}^{\text{max}} \approx 6.59 \times 10^{34} - 3.47 \times 10^{34} - 1.74 \times 10^{33} \approx 2.95 \times 10^{34}\ \text{J} \cdot \text{s}, \tag{57}$$

i.e., about 44.8% of the initial angular momentum.

With $L_{\text{esc}} \leq 1.74 \times 10^{33}$ J s, the escape channel covers only a minor fraction of the deficit — and Section 12.1 will show that it is additionally energetically impossible to enlarge it. The bulk of ΔL must therefore be evacuated to heliocentric orbit, necessarily via resonant channels (direct solar tides are too slow by three orders of magnitude, Section 12.1). Two resonant channels exist, hierarchized in this work: the *nominal* channel is the Laplace plane transition[15,16], crossed by the Moon at 15–25 $R_\oplus$ and whose activation condition is purely geometric — the inertial obliquity $\varepsilon_{\text{I}} \gtrsim 60$–$65°$ inherited from the ejection recoils (Eq. 34); it is the subject of Section 12. The *secondary* channel is the evection resonance[17], encountered earlier ($a \approx 4.6\ R_\oplus$), either as a strict capture or as a limit cycle[97]; its efficiency depends on the global quality factor Q_{global} of the proto-Earth, which is controlled by dissipation in the entire body, not by the local rheology of the CMT. The remainder of this subsection quantifies this dependence on Q_{global}, which conditions the contribution of the secondary channel — but no longer the closure of the budget.

11.6.1 *Distinction between local and global quality factors*

For a rotation period of about 3.5 h, the characteristic tidal frequency is $\omega \simeq 5 \times 10^{-4}$ s^{-1}. In a Maxwell viscoelastic model, maximum dissipation is obtained when $\omega\tau \simeq 1$, i.e., $\tau \simeq 2 \times 10^3$ s. For an effective shear modulus of order a few GPa, this corresponds to a viscosity of about 10^{13} Pa·s. The viscosities expected for a magma ocean, even partially crystallized ($\phi \sim 0.1$–0.3), are several orders of magnitude below this value. These regions are therefore far from the viscoelastic dissipation maximum and should not constitute the main contributors to global tidal dissipation.

This point can be made quantitative. Using the *molecular* viscosity of silicate magma, derived from the Arrhenius law ($\eta \sim 10^6$–10^8 Pa s over the relevant cooling range, Eq. 25), in the Maxwell relation $Q^{-1} = \omega\eta/\mu$, one obtains a tidal evolution time $\tau_{\text{tide}} \sim \mu/(\omega^2\eta) \sim 10$–$10^3$ yr — far too short to sustain the resonance over the 10^3–10^5 yr needed for capture[17,98], and incompatible with the cooling window itself, of several Myr. Molecular magma viscosity, whatever its exact value within the range imposed by cooling, cannot therefore be the operating dissipation channel: it is either too fluid (well below the Maxwell peak, as for the local CMT rheology, Section 11.6 above), or, even at its most viscous Hadean values, still far from the $\sim 10^{13}$ Pa s needed to approach this peak. This reinforces, rather than contradicts, the turbulent convection argument developed above: it is the effective viscosity of large-scale convective stirring ($\eta_{\rm turb} \sim 10^{13}$ Pa s, derived from $U_{\rm turb}$ and L_β), and not the molecular viscosity of magma at rest at any temperature, that can plausibly approach the Maxwell dissipation peak and provide a $Q_{\rm global}$ compatible with the capture timescale of the evection resonance.

The TTP is thus compatible with a $Q_{\rm global}$ sufficiently high to allow the evection resonance to act as a secondary channel. The quantitative determination of $Q_{\rm global}$ requires a dedicated thermo-rheological model, coupling the thermal evolution of the magma ocean to the viscoelastic response of the mantle. It no longer conditions the closure of the budget, however, now ensured by the nominal channel of the Laplace plane transition (Section 12), whose activation condition is geometric and not rheological: it only fixes the fraction of the deficit pre-evacuated before the transition crossing. This modeling remains a testable prediction of the model (Limitation L15).

11.6.2 A common solar clock: T-Tauri mass loss and the resonance encounter window

The synchronization between the favorable dissipation window and the encounter with the evection resonance is not necessarily fortuitous: it could be set by a single and same clock, the evolution of the T-Tauri Sun itself.

Stellar mass loss by wind during the T-Tauri phase is documented at a rate of order $\dot M_{\rm wind} \sim 0.1\,\dot M_{\rm accr}$, where $\dot M_{\rm accr} \sim 10^{-8}\,M_\odot\,{\rm yr}^{-1}$ is the typical accretion rate observed for classical T-Tauri stars[99,100], i.e., $\dot M_{\rm wind} \sim 10^{-9}\,M_\odot\,{\rm yr}^{-1}$[101,102]. Integrated over the active lifetime of the protoplanetary disk (1–10 Myr,[103]), this rate gives a fractional mass loss:

$$\frac{\Delta M}{M_\odot} \sim 10^{-3}\text{–}10^{-2}, \tag{58}$$

consistent with the virial upper bound on total pre-main-sequence mass loss (a few tenths of $M_\odot$,[104]).

By conservation of orbital angular momentum during slow central star mass loss ($a\,M_\star \approx$ const), this mass loss adiabatically shifts the Earth's semi-major axis, and thus the duration of the sidereal year that defines the evection resonance condition (lunar perigee precession equal to one year):

$$\frac{\Delta T_{\rm year}}{T_{\rm year}} \sim \frac{\Delta M}{M_\odot} \sim 10^{-3}\text{–}10^{-2}, \tag{59}$$

obtained by combining Kepler's third law ($T^2 \propto a^3/M_\star$) and the relation $\Delta a/a \approx$

$\Delta M/M_\star$.

This drift remains slow compared to the capture/crossing time of the resonance itself (10^3–10^5 yr,[17,98]): it therefore does not perturb an ongoing capture, but it determines *at what time*, within the 30–50 Myr active window, the resonance condition presents itself. Simultaneously, the declining luminosity of the same Sun as it exits its T-Tauri phase drives the cooling and progressive stiffening of the proto-Earth, a thermal-orbital coupling already initiated by Rufu & Canup[105] in their variable-Q tidal evolution model. If the resonance encounter — 10^3–10^5 yr after lunar formation — occurs early in the 30–50 Myr window, while a good dissipative regime is still active, the closure of the angular momentum budget becomes more plausible than a static examination would suggest. This synchronization remains qualitative in its current state: it rests on the comparison of compatible timescales, not on a complete coupled dynamical calculation, and constitutes a proposed extension of Limitation L15.

State of the budget. In the current state of the manuscript, the angular momentum budget is not closed for the nominal value of Q_{CMT} associated with the partially molten belt ($Q_{\mathrm{CMT}} \sim 10^4$). However, this does not imply a contradiction, because Q_{global} is a distinct quantity that can be much larger. The tension identified in earlier versions of this work is therefore resolved by the distinction between local and global quality factors. The budget will be definitively closed as soon as a dedicated thermo-rheological simulation provides a robust value of Q_{global} for the Hadean proto-Earth. This is the highest-priority numerical validation task (Limitation L15, Section 20).

The minimum period compatible with the carry-away bound, given for indication, is:

$$T_{\mathrm{init}} \gtrsim \frac{2\pi I_p}{L_{\mathrm{TL}} + L_{\mathrm{esc}}^{\max}} \approx \frac{2\pi(1.32 \times 10^{38})}{3.47 \times 10^{34} + 1.74 \times 10^{33}} \approx 6.33 \text{ h}. \quad (60)$$

This path is excluded by the mechanism itself: the ejection threshold Ω_{crit} (Section 9.2) is an absolute quantity, and a proto-Earth at $T = 6.33$ h could not reach it.

11.7 Scaling laws and dimensionless form

The angular momentum transport problem admits a natural non-dimensionalization that reveals the essential physical dependencies and simplifies the parameter space. We define the characteristic velocity scale $U_0 = U_{\mathrm{crit}} \approx 9.8\ \mathrm{km\,s^{-1}}$, the length scale $R_0 = R_p \approx 7.4 \times 10^6$ m, the mass scale $M_0 = M_{\mathrm{ej}} \approx 2.5 \times 10^{22}$ kg, and the angular momentum scale $H_0 = M_0 R_0 U_0 \approx 1.8 \times 10^{36}\ \mathrm{kg\,m^2\,s^{-1}}$. The dimensionless variables $\tilde{v} = v/U_0$, $\tilde{r} = r/R_0$, $\tilde{m} = m/M_0$, and $\tilde{h} = h/(R_0 U_0) = \tilde{r}\tilde{v}\sin\theta$ reformulate the ejecta distribution as:

$$\tilde{\rho}(\tilde{v},\theta) = \tilde{M}_{\rm ej}\,\tilde{\mathcal{N}}_v\,\tilde{\mathcal{N}}_\theta\,\exp\left[-\frac{(\tilde{v}-\bar{\tilde{v}})^2}{2\tilde{\sigma}_v^2}\right]\cos\theta, \quad \tilde{v} > \tilde{v}_{\rm esc},\ |\theta| < \theta_{\rm max}, \tag{61}$$

with $\tilde{M}_{\rm ej} = 1$ by definition, $\bar{\tilde{v}} = \bar{v}/U_0 \approx 1$, $\tilde{\sigma}_v = \sigma_v/U_0 \approx 0.05$–$0.10$, and $\tilde{v}_{\rm esc} = v_{\rm esc}/U_0 \approx 1/\sqrt{2}$. The dimensionless capture efficiency takes the same erf form as Eq. 47, with $\tilde{v}_{\rm crit} \approx 1/\sqrt{2}$, and the dimensionless lost angular momentum, $L_{\tilde{\rm loss}}$, recovers the physical quantity via $L_{\rm loss} = H_0\,L_{\tilde{\rm loss}}$.

This dimensionless form reveals that the problem depends only on four dimensionless parameters: the mean ejection speed relative to $U_{\rm crit}$ ($\bar{\tilde{v}} \approx 1$), the speed dispersion ($\tilde{\sigma}_v \approx 0.05$–$0.10$), the angular extent of the torus ($\theta_{\rm max} \approx 30°$), and the capture threshold ($\tilde{v}_{\rm crit} \approx 1/\sqrt{2}$). This low-dimensional parameter space keeps the model testable: each parameter can in principle be independently constrained by future simulations or observations.

The dimensionless form admits three asymptotic regimes. In the monokinetic limit ($\tilde{\sigma}_v \to 0$), the distribution reduces to a single speed and recovers the single-speed ballistic approximation used in previous sections. In the broad-spectrum limit ($\tilde{\sigma}_v \gg 1$), the speed distribution becomes uniform over the available range, maximizing the angular momentum carried by the fastest ejecta. In the sharp-cutoff limit ($\tilde{\sigma}_f \to 0$), capture efficiency becomes a step function, representing the idealized case where all material above the capture threshold is retained.

Falsifiable prediction — P20 — Asymptotic regime indicator

The actual ejecta spectrum of the TTP lies between the monokinetic and broad-spectrum limits, with $\tilde{\sigma}_v \approx 0.05$–$0.10$. The sharp-cutoff limit provides an upper bound on $L_{\rm loss}$, while the monokinetic limit provides a lower bound. The actual value, accessible by Monte Carlo simulation, should lie between these bounds.

Test: Monte Carlo sampling of $\rho(v,\theta)$ (Section 11.4) compared to the analytical monokinetic and broad-spectrum limits derived above.

The dimensionless form also allows the derivation of scaling laws for $L_{\rm loss}$ as a function of fundamental proto-Earth parameters: $L_{\rm loss} \propto M_p R_p^2 \Omega_p$ (the standard scaling law for the angular momentum of a rotating body), $L_{\rm loss} \propto \sqrt{\sin\varepsilon}$ (from the obliquity dependence of the ejection speed threshold), and $\delta L_{\rm loss}/L_{\rm loss} \approx -\delta f_{\rm cap}/(1 - f_{\rm cap})$. For $f_{\rm cap} = 0.70$, $1 - f_{\rm cap} = 0.30$, so a 10% variation in $f_{\rm cap}$ gives a 33% variation in $L_{\rm loss}$ — a sensitivity that underscores the importance of precisely determining $f_{\rm cap}$ in future simulations. Table 3 summarizes the sensitivity of $L_{\rm loss}$ to each input parameter.

Table 3: Sensitivity of L_{loss} to input parameters.

Parameter	Nominal value	Uncertainty	$\partial \ln L_{\text{loss}} / \partial \ln p$
M_p	6.04×10^{24} kg	$\pm 5\%$	≈ 1
R_p	7.4×10^{6} m	$\pm 3\%$	≈ 2
Ω_p	4.99×10^{-4} s^{-1}	$\pm 10\%$	≈ 1
ε	55°	$\pm 10^\circ$	≈ 0.5
f_{cap}	0.70	± 0.05	≈ -3

The budget is most sensitive to f_{cap} (due to the $1/(1 - f_{\text{cap}})$ factor) and to R_p (due to the R_p^2 scaling law). This sensitivity motivates the priority given to Limitation L2 (numerical determination of f_{cap}).

Physical bound on L_{esc}. Independently of the exact form of the distribution, physics imposes an absolute upper bound — the same as that established in Eq. 56 (Section 11.6), re-derived here in dimensionless form:

$$L_{\text{esc}} \leq M_{\text{esc}}\, r_{\text{emission}}\, \bar{v} \approx 2.25 \times 10^{22} \times 7.9 \times 10^{6} \times 9.8 \times 10^{3} \approx 1.74 \times 10^{33}\ \text{J s}. \tag{62}$$

With nominal parameters ($f_{\text{cap}} = 0.70$, $M_{\text{ej}} = 7.5 \times 10^{22}$ kg, $I_p = 1.32 \times 10^{38}$ kg m^2, $\Omega_p = 4.99 \times 10^{-4}$ s^{-1}), the fraction of the initial angular momentum that can be carried away by escaped ejecta is at most:

$$\frac{L_{\text{esc}}}{L_{\text{init}}} \leq \frac{M_{\text{esc}}\, r_{\text{emission}}\, \bar{v}}{I_p \Omega_p} \approx \frac{1.74 \times 10^{33}}{6.59 \times 10^{34}} \approx 0.026. \tag{63}$$

With $T_{\text{init}} = 3.5$ h, the post-ejection Earth–Moon system would therefore retain at least 97.4% of the initial angular momentum, whereas the current state retains only 52.6%. The resulting deficit, $\Delta L \approx 2.95 \times 10^{34}$ J s (44.8% of L_{init}), and the minimum compatible period $T_{\text{init}} \gtrsim 6.33$ h are derived in Section 11.6 (Eqs. 57 and 60) and identified as a major limitation (Limitation L15, Section 20).

Falsifiable prediction — P21 — Numerical closure of the angular momentum budget

A Monte Carlo simulation of the ejecta spectrum (Section 11.4), with parameters sampled over their uncertainty ranges, should reproduce the compatibility of the angular momentum budget (Eqs. 56 and 57, Section 11.6). The mean specific angular momentum of escaped ejecta should satisfy the bound:

$$\langle h \rangle_{\text{esc}} = \frac{L_{\text{esc}}}{M_{\text{esc}}} \leq h_{\max} = r\bar{v} \approx 7.74 \times 10^{10}\ \text{m}^2\,\text{s}^{-1}. \tag{64}$$

Given the angular spread of the distribution ($\theta_{\max} \approx 30^\circ$, $\langle|\sin\theta|\rangle \approx 0.25$), the Monte Carlo value of $L_{\rm esc}$ should lie significantly *below* the bound $L_{\rm esc}^{\max}$ — which makes the deficit ΔL (Eq. 57, Section 11.6) a conservative estimate (lower bound on the actual deficit).

Falsification: a Monte Carlo value $L_{\rm esc} > M_{\rm esc}\, r\, \bar{v}$ would indicate an error in the distribution $\rho(v,\theta)$ or in the bookkeeping; a deviation greater than $\sim 5\%$ between the Monte Carlo $L_{\rm esc}$ and the analytical estimate weighted by $\langle h \rangle$ would invalidate the discrete formulation of Section 11.4.

The Monte Carlo implementation of this model is simple and computationally inexpensive; it is recommended as a first step in the SPH 3D validation program, ahead of full hydrodynamic simulations.

12 The angular momentum sink: the Laplace plane transition

Section 11.6 established the deficit $\Delta L \approx 2.95 \times 10^{34}$ J s ($\approx 45\%$ of $L_{\rm init}$) and hierarchized the two solar resonant sinks. This section develops the nominal channel — the *Laplace plane transition*[15,16] — and closes the retrodictive thread opened in Section 1: the current angular momentum and the lunar inclination of 5° jointly impose this episode, whose only activation condition is a high inertial obliquity — precisely what the TTP derives independently (Section 3).

Conventions. Orbital distances are expressed in megameters (Mm), with their equivalent in current Earth radii ($R_\oplus = 6\,371$ km, a simple constant length unit) in parentheses. The body's physics (flattening J_2, Roche radius, tidal torque in R^5) uses the real equatorial radius $R_{\rm eq,proto} \approx 7\,900$ km, which decreases as rotation slows and cooling proceeds. Inertial obliquity $\varepsilon_{\rm I}$ denotes the angle between the rotational angular momentum vector $\boldsymbol{L}_p$ and the normal to the proto-Earth's heliocentric orbital plane — two contemporary directions, without reference to the current ecliptic.

12.1 Closure of non-resonant channels

Two channels involving no resonant dynamics must be definitively closed.

The escape channel cannot be enlarged. The maximum specific angular momentum of an ejecta launched from the surface is $h_{\max} \approx 7.74 \times 10^{10}$ m^2 s^{-1} (Section 11.6). Evacuating ΔL by pure ejection would require escaping a mass:

$$M_{\rm esc}^{\rm required} = \frac{\Delta L}{h_{\max}} \approx 3.8 \times 10^{23} \text{ kg} \approx 5 \text{ lunar masses}, \tag{65}$$

for a kinetic cost $\frac{1}{2} M_{\rm esc}^{\rm required} \bar{v}^2 \approx 1.8 \times 10^{31}$ J, i.e., $\approx 115\%$ of the total rotational reservoir $E_{\rm rot} \approx 1.6 \times 10^{31}$ J (Section 11.2.2). The escape channel is therefore *energetically forbidden* at the required scale: ejection from the surface is structurally poor in angular momentum ($\sin\theta \leq 1$, $r \approx R_{\mathrm{eq,proto}}$) but costly in energy, most of the bill being absorbed by the potential well. This impossibility is a property of any fission scenario, independent of TTP tunings, and is further reinforced by the delayed re-accretion of heliocentric ejecta on orbits close to Earth's.

Direct solar tides are too slow. The solar tidal torque on Earth's spin:

$$N_\odot \simeq \frac{3}{2}\frac{k_2}{Q}\frac{GM_\odot^2 R_{\mathrm{eq,proto}}^5}{a^6} \approx 1 \times 10^{18}\left(\frac{k_2}{Q}\right)\ \mathrm{N\,m}, \tag{66}$$

gives, even for $k_2/Q \sim 0.1$, an evacuation time $\Delta L/N_\odot \sim 3 \times 10^{17}$ s $\approx$ 10 Gyr. Transfer to heliocentric orbit must therefore necessarily pass through resonant amplification — which justifies the hierarchy of the two resonant channels in Section 11.6.

12.2 Physics of the Laplace plane transition

A satellite's orbit precesses around an equilibrium direction — the Laplace plane — set by the competition between the equatorial bulge torque (dominant near the planet, precession around the equatorial plane $\perp \boldsymbol{L}_p$) and the solar torque (dominant far away, precession around the heliocentric orbital plane). The transition radius is the Laplace radius[15]:

$$r_{\mathrm{L}} = \left[2\,J_2\,R^2\,a^3\,\frac{M_p}{M_\odot}\right]^{1/5}, \tag{67}$$

i.e., for the current Earth, $r_{\mathrm{L}} \approx 62$ Mm ($\approx 9.8\,R_\oplus$).

For low obliquity, the crossing of r_{L} by a receding satellite is smooth. But Tremaine et al.[15] showed that for obliquity greater than $\approx 68.9°$ the circular Laplace surface becomes unstable, and Ćuk et al.[16] established by numerical integration that, for a rapidly rotating proto-Earth with obliquity ≈ 60–$80°$, crossing this zone triggers an episode of high eccentricity and inclination during which the solar torque massively extracts angular momentum from the Earth–Moon system to heliocentric orbit. In their integrations, the system enters with an angular momentum near $2\,L_{\mathrm{TL}}$ and exits near L_{TL}, with Earth's obliquity lowered to moderate values — precisely the relaxation required by the budget of Section 11.6. The mechanism also yields a remarkable by-product: the Moon exits on an inclined orbit, whose subsequent tidal damping leaves the residual inclination of $\approx 5°$ observed today — the "lunar inclination problem," which no equatorial disk assembly naturally explains.

Three properties distinguish this mechanism from the strict evection resonance: *(i)* it does not require fine capture — the instability is crossed robustly over a wide range of tidal parameters, whereas evection capture requires delicate tuning of Q_{global}[17,97,98]; *(ii)* it operates late (10^5–10^7 yr after formation, at 15–25 $R_\oplus$), thus decoupled from the early thermal window; *(iii)* its activation condition is geometric (obliquity) and not rheological (dissipation). For the TTP, whose post-formation state is a highly dissipative magma ocean — unfavorable for evection capture — this third property is decisive.

12.3 The required obliquity is a native prediction of the TTP

In the TTP, the obliquity interval $[40°, 70°]$ is derived by six convergent arguments (Section 3), of which the sixth is precisely the retrodiction from the final state that we close here. The Laplace plane transition requires an inertial obliquity $\varepsilon_{\mathrm{I}} \gtrsim 60$–$65°$ at crossing[15,16]; this activation condition covers the upper part of the derived interval. The TTP therefore does not need to import a new hypothesis, but to demonstrate that the inertial obliquity *at crossing* remains in this upper part.

12.3.1 From co-rotating nutation to inertial obliquity

Three axes must be distinguished, whose behaviors diverge during and after the TTP:

(i) **the figure axis** of the spheroid, seat of the large-amplitude free nutation $[40°, 70°]$ in the co-rotating frame (Section 3), engine of the CMT elliptic instability;

(ii) **the instantaneous rotation axis $\boldsymbol{\Omega}(t)$**, which wobbles around the third during nutation;

(iii) **the rotational angular momentum vector $\boldsymbol{L}_p$**, which, in the absence of external torque, is *fixed in inertial space* — a conservation law. Free nutation, whatever its amplitude, is a motion of axes (i) and (ii) *around* $\boldsymbol{L}_p$; only external torques (solar tide, ejection recoils) reorient it.

The obliquity that commands the transition is $\varepsilon_{\mathrm{I}} = \angle(\boldsymbol{L}_p, \hat{\boldsymbol{n}}_{\mathrm{orb}})$. Two consequences follow.

Robustness to residual nutation. The lunar orbit precession responds to the quadrupolar potential *averaged* over nutation, whose period ($\sim$ days) is short compared to orbital precession periods. Now the time average of the quadrupole of a body in free nutation is axisymmetric around $\boldsymbol{L}_p$: the effective equator felt by the Moon is the plane $\perp \boldsymbol{L}_p$, independent of the wobble amplitude. The mechanism therefore requires *no* prior damping of nutation — ε_{I} is defined and conserved even at the height of oscillations.

Amplitude relaxation after the TTP. Free nutation never completely disappears: its equilibrium amplitude results from the excitation–dissipation tug-of-war (the current Earth retains a Chandler wobble of amplitude $\sim 0.1''$, continuously re-excited). After the third episode, dissipation in a magmatic body damps the free mode in a time short compared to recession[36], while excitation sources (ejections, large impacts) cease. The amplitude therefore relaxes to an equilibrium several orders of magnitude below the Hadean interval — without the mechanism depending on it, by the averaging property above.

The heritage. The value of ε_I at the start of recession is fixed by the final orientation of $\boldsymbol{L}_p$ after the N recoils, already given by Eq. 34 (Section 9.2). The full causal chain is: co-rotating nutation $[40°, 70°]$ → orientation of $\boldsymbol{L}_p$ after N recoils → amplitude relaxation → $\varepsilon_\mathrm{I}^{(0)}$ frozen → activation of the transition if $\varepsilon_\mathrm{I} \gtrsim 60$–$65°$ at crossing. The critical budget question thus reformulates as a solid mechanics calculation entirely internal to the theory (L15).

12.4 A crossing by convergence

The crossing of the transition is not a positional accident: it results from the convergence of two opposing migrations.

The initial Laplace radius is large. At $T_\mathrm{rot} = 3.5$ h, the rotational parameter $q = \Omega^2 R^3_\mathrm{eq,proto}/GM_p \approx 0.25$ places the spheroid near its bifurcation limit (Section 4.3), with a Maclaurin flattening $J_2 \sim 0.13$, two orders of magnitude above the current value. With $R = R_\mathrm{eq,proto} = 7\,400$ km, Eq. 67 gives:

$$r_\mathrm{L}^\mathrm{init} \approx 171\ \mathrm{Mm} \quad (\approx 27\, R_\oplus). \tag{68}$$

Then r_L migrates inward while the Moon migrates outward. As recession proceeds, slowing ($J_2 \propto \Omega^2$) and cooling ($R_\mathrm{eq,proto} \to R_\oplus$) reduce the product $J_2 R^2$, and $r_\mathrm{L} \propto (J_2 R^2)^{1/5}$ retreats, while the lunar orbit expands under tidal action. The coupled integration of these two migrations (Fig. 4), starting from conservation of spin-orbit angular momentum and a radius contraction coupled to slowing, places the convergence $a = r_\mathrm{L}$ at $\approx 25\, R_\oplus$ — in the upper part of the 15–25 $R_\oplus$ window identified by Ćuk et al.[16]. The striking result of this calculation is that *the position* of the convergence depends little on the tidal quality factor k_2/Q, poorly constrained for a magma ocean: varying k_2/Q by an order of magnitude (from 0.3 to 3) leaves the convergence at $\approx 25\, R_\oplus$ and only changes *the timescale*, from $\sim 10^6$ to $\sim 10^4$ yr. The encounter is therefore generic, governed by geometry (crossing of two curves of opposite slopes) rather than tuned by dissipation. Two caveats are explicit: the model uses a linear proxy for contraction $R(t)$ and freezes J_2 at its initial Maclaurin value, so the exact value ($\approx 25\, R_\oplus$) carries an uncertainty of a few $R_\oplus$; a full thermo-orbital calculation, coupling the thermal evolution of the magma ocean to the equilibrium figure, is the subject of Limitation L15.

12.5 Independent corroboration: the Cassini secular resonance converges to the same region

The geometric convergence $a = r_\mathrm{L}$ above identifies *where* the Laplace surface bifurcates. A second diagnostic, independent and based on a completely different frequency pair, places in the *same* region a secular spin–orbit resonance — the Cassini state, precisely the mechanism that Ćuk et al.[106] show allows the high-obliquity path to join the current state under the vertical conservation constraint of Tian & Wis-

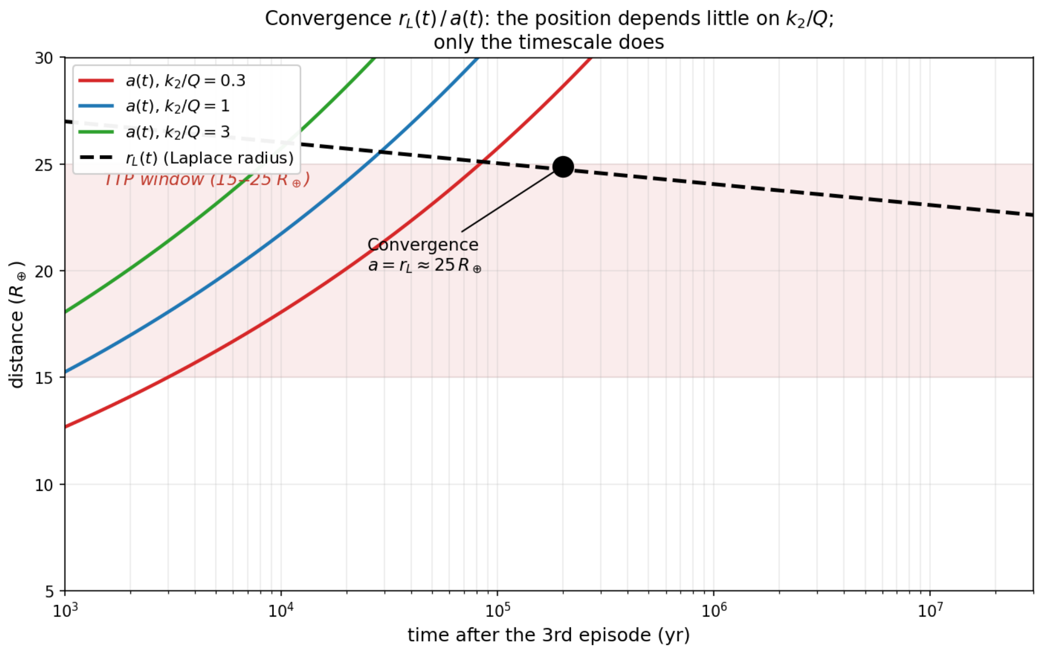

Figure 4: Convergence of the two opposing migrations: the lunar orbit $a(t)$ (solid lines, for three values of the tidal quality factor k_2/Q) expands by tidal recession, while the Laplace radius $r_\mathrm{L}(t)$ (dashed) retreats under slowing ($J_2 \propto \Omega^2$) and body contraction. The three trajectories $a(t)$ meet $r_\mathrm{L}(t)$ at the *same* point, $\approx 25\,R_\oplus$: the position of the convergence is robust with respect to k_2/Q; only the timescale depends on it ($\sim 10^4$ to $\sim 10^6$ yr). Constants: $R_\mathrm{eq,proto} = 7\,400$ km, $J_2^\mathrm{init} = 0.13$, $T_\mathrm{rot} = 3.5$ h. The simplified model (linear contraction, J_2 frozen) places the convergence in the upper part of the 15–25 $R_\oplus$ window, to within a few $R_\oplus$.

dom[107]. This coincidence is not imposed: it follows from the same initial conditions (angular momentum, inertia coefficient, lunar mass) with no adjusted parameter.

The two precession frequencies. The precession of the proto-Earth's rotation axis, driven by the lunar torque on the bulge, and the precession of the lunar orbital node, driven by Earth's flattening, are to first order:

$$\alpha \simeq \frac{3\,G\,M_L}{2\,a^3\,\Omega}\,\frac{C-A}{C}, \qquad g \simeq \frac{3}{4}\,n\,J_2\left(\frac{R}{a}\right)^2, \qquad n = \sqrt{\frac{GM_p}{a^3}}, \tag{69}$$

where $(C-A)/C \simeq J_2/k$ measures the dynamical flattening. Cassini state 2 satisfies, in the first-order coplanar approximation, $\cos\varepsilon = g/\alpha$; an obliquity of 61° therefore corresponds to $g/\alpha = \cos 61^\circ \approx 0.485$.

The closed calculation. The ratio g/α is not a free parameter: along the recession, Ω is fixed by conservation of spin-orbit angular momentum,

$$L_0 = I\,\Omega + M_L\sqrt{GM_p\,a}, \qquad I = k\,M_pR^2, \tag{70}$$

so that $g/\alpha(a)$ becomes a function of distance a alone. Solving $g/\alpha(a) = 0.485$ along the trajectory $\Omega(a)$ gives a single crossing point for each value of angular momentum. The nominal angular momentum of the system corresponds to $L_0 \approx L_{\mathrm{TL}} = 3.47 \times 10^{34}$ J s, i.e., the residual angular momentum after evacuation of the deficit $\Delta L \approx 2.95 \times 10^{34}$ J s from the initial value $L_{\mathrm{init}} = 6.59 \times 10^{34}$ J s (Eq. 57); the result is:

L_0	a such that $g/\alpha = 0.485$	$\varepsilon_{\mathrm{Cassini}}$
$0.8\,L_{\mathrm{TL}}$	$\approx 14.4\,R_\oplus$	61°
$1.0\,L_{\mathrm{TL}}$	$\approx 22.5\,R_\oplus$	61°
$1.3\,L_{\mathrm{TL}}$	$\approx 38.0\,R_\oplus$	61°

For the nominal angular momentum, the Cassini state at 61° falls at $\approx 22.5\,R_\oplus$ — within a few $R_\oplus$ of the geometric convergence $a = r_{\mathrm{L}} \approx 25\,R_\oplus$ (Section 12.4). Two independent diagnostics, one topological (bifurcation of the Laplace surface), the other resonant (spin-orbit Cassini state), thus designate the same region 20–25 $R_\oplus$ — the very one that the Fe–Ni engine imposes by the value it fixes for angular momentum and flattening. The map of Fig. 6 shows the position of the transition in the plane (axis/orbit inclination i, rotation period), with the target band $i \approx 65$–77°.

Robustness, prediction, and limits. The position of the resonance ($\approx 22.5\,R_\oplus$) is fixed by the frequency crossing along $\Omega(a)$ and does not depend on the absolute value of angular momentum via k: changing k modifies L_{init} and ΔL but leaves the position unchanged — the budget (sensitive to k) and the dynamical convergence (robust to k) are decoupled. The calculation contains a prediction: $L_0 \gtrsim 1.3\,L_{\mathrm{TL}}$

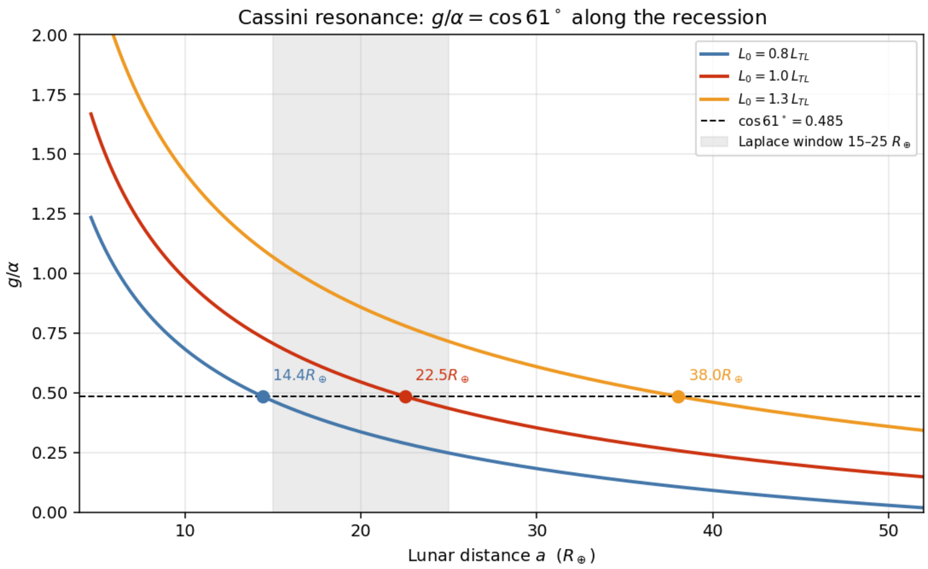

Figure 5: Ratio g/α along recession for three angular momentum values. The Cassini state at 61° ($g/\alpha = \cos 61°$) falls at $\approx 22.5\,R_\oplus$ for the nominal angular momentum, within the 15–25 $R_\oplus$ window (shaded) and within a few $R_\oplus$ of the geometric convergence $a = r_\mathrm{L}$ (Fig. 4). Too high an angular momentum ($\gtrsim 1.3\,L_\mathrm{TL}$) pushes the resonance outside the window.

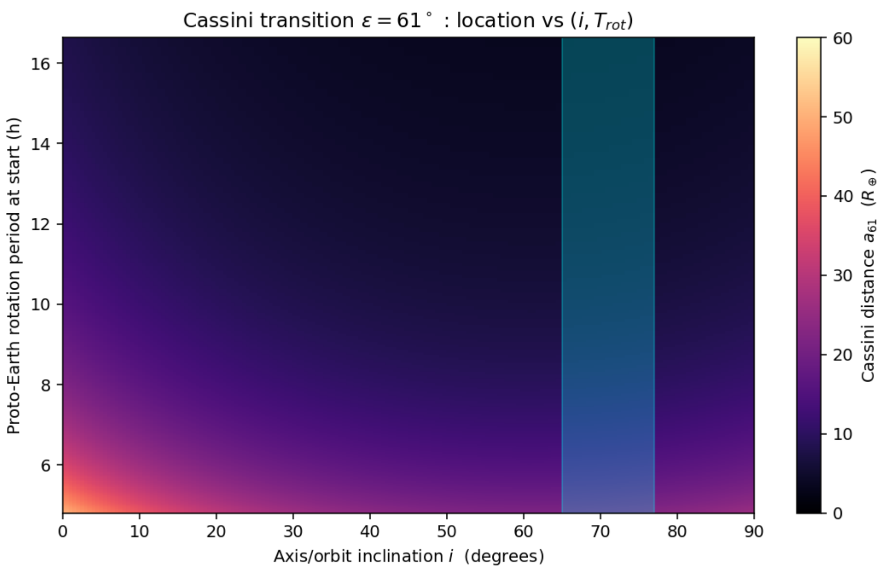

Figure 6: Distance of the Cassini transition ($\varepsilon = 61°$) in the plane (inclination i, proto-Earth rotation period). The cyan band indicates the range of inclinations compatible with the Laplace window.

pushes the resonance to $\approx 38\,R_\oplus$, outside the window, which bounds *from above* the admissible Hadean angular momentum. This diagnostic does not *correct* the 68.9° threshold of Tremaine et al.[15], which describes the topological bifurcation: it identifies a distinct quantity — the obliquity of the Cassini state for the frequency ratio realized by the TTP configuration — and shows that it concentrates in the same zone. Three approximations remain to be lifted and fall under Limitation L15: the full Cassini condition[108] involves the orbital inclination via $\sin(\varepsilon - i)$; the linear hydrostatic approximation for J_2 is no longer valid near the Maclaurin limit ($q_{\max} \simeq 0.225$); and the radius $R(t)$, frozen here, varies during contraction.

Key result — The TTP's nominal sink

The deficit $\Delta L \approx 2.95 \times 10^{34}$ J s is evacuated to heliocentric orbit by the Laplace plane transition[15,16], crossed by the Moon at $\approx 25\,R_\oplus$ (10^4–10^6 yr after formation; Fig. 4). The only activation condition — $\varepsilon_{\mathrm{I}} \gtrsim 60$–$65°$ at crossing — is a native prediction of the TTP (nutation $[40°, 70°]$, Section 3), translated into inertial obliquity by conservation of $\boldsymbol{L}_p$ and the vector recoil budget (Eq. 34). Unlike evection capture, the mechanism is robust with respect to dissipation: the high dissipativity of the post-TTP magma ocean is not opposed to it. The lunar inclination of 5° is its testable by-product (P24).

Falsifiable prediction — P24 — Residual lunar orbital inclination

The chain TTP → Laplace plane transition predicts that the Moon exits the instability episode on a highly inclined orbit, whose subsequent tidal damping leaves the residual inclination of $\approx 5°$ observed today[16]. Conversely, if the post-ejection inertial obliquity $\varepsilon_{\mathrm{I}}^{(0)}$ calculated by the vector recoil budget (Eq. 34) were $\lesssim 50°$, the transition would be crossed without instability and neither the deficit ΔL nor the lunar inclination would be explained: closure would then rely solely on the secondary evection channel, with the Q_{global} requirements of Section 11.6. The prediction is falsifiable by calculation (L15) before being falsifiable by observation.

13 Didactic derivation of the Cassini state and vector angular momentum budget

This section establishes, step by step, two results that Sections 11.6 and 12 used in condensed form: on the one hand, that the obliquity $\varepsilon \simeq 61°$ associated with the crossing is not an imposed value but the *solution* of the Cassini equilibrium condition at the critical radius; on the other hand, that the angular momentum carried by the tori at ejection, although real, is not sufficient by itself to absorb the deficit, which justifies the need for a complementary resonant sink. The causal chain targeted is:

$$\text{Laplace transition} \longrightarrow i \longrightarrow \text{Cassini} \longrightarrow \varepsilon.$$

13.1 The Cassini equilibrium condition

A Cassini state is a configuration where Earth's rotation axis $\hat{s}$, the normal to the lunar orbital plane $\hat{n}$, and the normal to the reference plane remain coplanar and precess with a common motion. The equilibrium condition, with the conventions of Ward[108], is:

$$\alpha \sin\varepsilon \cos\varepsilon + g \sin(\varepsilon - i) = 0, \tag{71}$$

where ε is Earth's obliquity, i the lunar orbital inclination to the reference plane, α the spin-axis precession constant, and g the nodal precession frequency of the orbit. The term $\sin(\varepsilon - i)$ — absent from the coplanar approximation $\cos\varepsilon = g/\alpha$ used in first reading — is essential: it is what couples obliquity to orbital inclination.

The two frequencies. The spin-axis precession, driven by lunar and solar torques on the equatorial bulge, is:

$$\alpha = \frac{3G}{2\Omega}\frac{C-A}{C}\left(\frac{M_L}{a^3} + \frac{M_\odot}{a_\odot^3}\right), \qquad \frac{C-A}{C} \simeq \frac{J_2}{k}, \tag{72}$$

and the lunar nodal precession combines the bulge and solar contributions:

$$g = \underbrace{\tfrac{3}{2}\, n\, J_2 \left(\tfrac{R}{a}\right)^2 \cos i}_{\text{bulge } J_2} + \underbrace{\tfrac{3}{4}\,\frac{n_\odot^2}{n}\cos i}_{\text{solar term}}, \qquad n = \sqrt{\frac{GM_p}{a^3}}. \tag{73}$$

13.2 Solution at the critical radius $a = 22.5\,R_\oplus$

Inserting the TTP parameters ($R = 7\,400$ km, $T_{\rm rot} = 3.5$ h, $k = 0.37$, $\Omega(a)$ fixed by angular momentum conservation) into Eqs. 72–73, then numerically solving Eq. 71 for ε at each inclination i, gives Table 4.

Inclination i	Cassini obliquity ε	Regime
40°	36.1°	$\varepsilon < i$
50°	45.1°	$\varepsilon < i$
60°	54.1°	$\varepsilon < i$
65°	58.4°	$\varepsilon < i$
68°	**61.0°**	**target**
70°	62.5°	$\varepsilon < i$
77°	67.0°	$\varepsilon < i$

Table 4: Solutions of the full Cassini condition (Eq. 71) at $a = 22.5\,R_\oplus$. The obliquity $\varepsilon = 61°$ is not imposed: it *results* from the solution for an orbital inclination $i \simeq 68°$.

The central result is that $\varepsilon = 61°$ *emerges* from the equation for $i \simeq 68°$, and is therefore not an input. Moreover, the internal consistency holds: the same solution gives $\varepsilon = 61^\circ$ for $i \simeq 68^\circ$ at $22.5\,R_\oplus$, $i \simeq 65^\circ$ at $25\,R_\oplus$, and $i \simeq 63^\circ$ at $28\,R_\oplus$ — the Laplace convergence distance (Section 12.4) and the required inclination fall in the band $i \simeq 65$–68°, consistent with the map of Fig. 6.

Where does $i \simeq 68^\circ$ come from? This inclination is not an external condition: the crossing of the Laplace plane transition *itself excites* the inclination. Near the planet, the orbit precesses in the equatorial plane (i small); upon crossing r_L, its precession plane tilts toward the heliocentric plane, which pumps the inclination to high values. The inclination of $\simeq 68^\circ$ required for $\varepsilon = 61^\circ$ is therefore precisely what the crossing produces — which closes the chain Laplace $\to i \to$ Cassini $\to \varepsilon$.

Exact scope of this result

What is established by the calculation: $i \simeq 68^\circ \Rightarrow \varepsilon \simeq 61^\circ$ at $22.5\,R_\oplus$ (solution of the equilibrium condition). What remains to be established: that the dynamical evolution indeed brings the system onto this branch (coupled integration of $i(t)$ and $\varepsilon(t)$ through the crossing), and that the coupling transfers the required amount of angular momentum. The formulation adopted therefore claims a *calculated* prediction without extrapolating beyond: *"The solution of the full Cassini condition leads to $\varepsilon \simeq 61^\circ$ for $i \simeq 68^\circ$ at $22.5\,R_\oplus$; this value is not imposed but results from the orbital state obtained during the Laplace transition. The subsequent coupling between this configuration, angular momentum transfer, and evolution toward the current state remains to be established quantitatively."*

13.3 Vector budget: angular momentum carried by the tori

In the TTP framework, the proto-Earth already has its rotation axis, the intertropical belt is defined with respect to this proper axis, and the tori are ejected in the equatorial reference frame. The vector budget is therefore:

$$\vec{L}_\mathrm{initial} = \vec{L}_\mathrm{rotation} + \vec{L}_\mathrm{tori}, \qquad \vec{L}_\mathrm{final} = \vec{L}_\oplus + \vec{L}_\mathrm{Moon} + \vec{L}_\mathrm{reservoirs}. \tag{74}$$

The angular momentum carried by the N tori, ejected from the equatorial region ($r \simeq R$) at critical speed U_crit, is bounded by:

$$L_\mathrm{tori} = \sum_{i=1}^{N} m_i\, r_i\, U_\mathrm{crit} \;\lesssim\; N\, M_\mathrm{ej}\, R\, U_\mathrm{crit}. \tag{75}$$

With $N = 3$, $M_\mathrm{ej} \lesssim 4.0 \times 10^{22}$ kg, $R = 7\,400$ km and $U_\mathrm{crit} = 9.8\ \mathrm{km\,s^{-1}}$, we obtain $L_\mathrm{tori} \lesssim 8.7 \times 10^{33}$ J s.

13.4 Angular momentum distribution and the need for a sink

The deficit to be evacuated, $\Delta L \approx 2.95 \times 10^{34}$ J s (Section 11.6), is of the same order as the current lunar orbital angular momentum ($L_\mathrm{orb} \approx 2.9 \times 10^{34}$ J s). The momentum carried by the tori therefore represents only about one-third ($\lesssim$ 34%) of it. This result is consistent and not a flaw: if the tori carried the entire ΔL, there would be no angular momentum left to build the observed lunar orbit. The tori and the resonant sink are therefore *complementary*, not redundant.

Fig. 7 shows the distribution of angular momentum between Earth's rotation and the lunar orbit along the recession, at constant total momentum. At 22.5 $R_\oplus$, rotation carries $\approx 4.3 \times 10^{34}$ J s and the orbit $\approx 1.8 \times 10^{34}$ J s, for a total of $\approx 6.0 \times 10^{34}$ J s. Convergence toward the current state ($L_{\mathrm{TL}} \approx 3.5 \times 10^{34}$ J s) requires the intervention of the solar resonant sink in the 20–25 $R_\oplus$ zone.

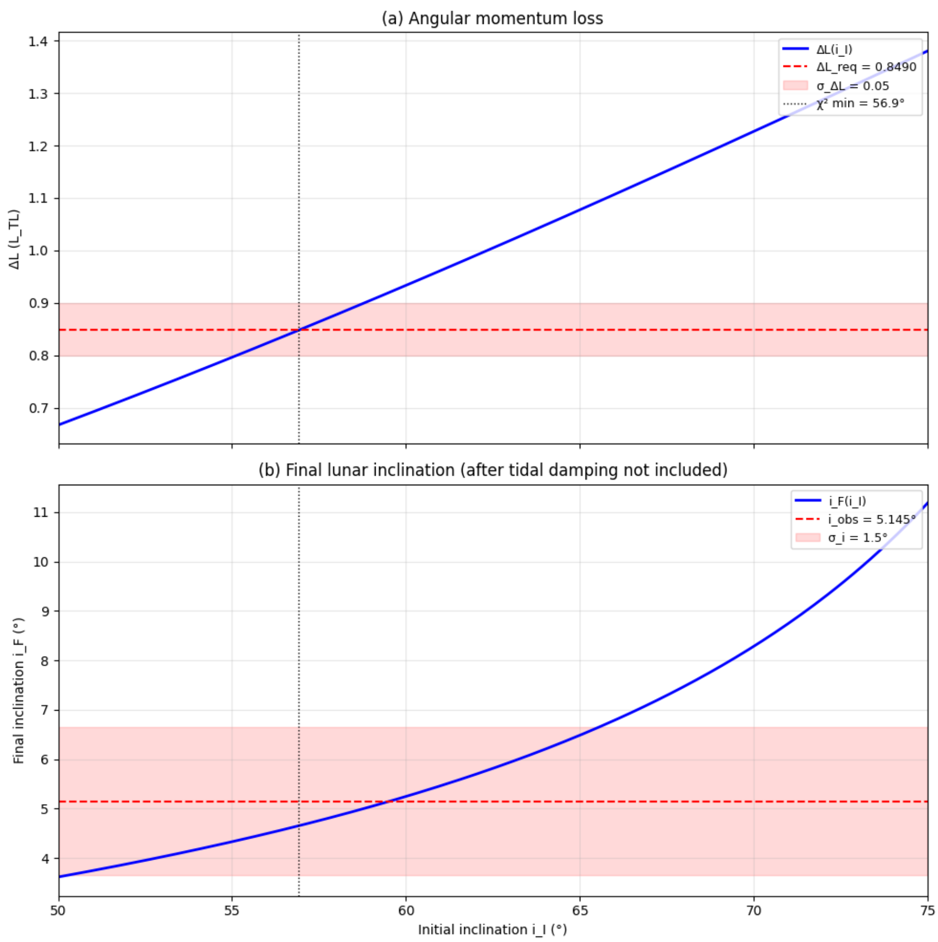

Figure 7: Angular momentum distribution along the lunar recession, at constant total momentum. Earth's rotation (L_{spin}) gradually transfers to the lunar orbit (L_{orb}); their sum L_{tot} remains constant in the absence of an external sink. Convergence toward the current state L_{TL} (dotted) requires the action of the solar resonant sink in the 20–25 $R_\oplus$ zone (shaded), where the Cassini/Laplace crossing occurs.

Synthesis of the vector budget

The tori carry $\lesssim 8.7 \times 10^{33}$ J s, i.e., $\approx 34\%$ of the deficit ΔL. The remainder is evacuated by the resonant sink (Laplace transition / Cassini state, supplemented by the T-Tauri wind). The obliquity $\varepsilon \simeq 61^\circ$ of the crossing is a solution derived from the full Cassini condition for $i \simeq 68^\circ$, an inclination itself produced by the tilting of the precession plane at the Laplace crossing. The full dynamical closure — the torque integral $\Delta\vec{L} = \int \vec{\tau}\, dt$ along the crossing — remains the subject of Limitation L15.

13.5 Sequence of gravito-tidal regimes and angular momentum flux

The Laplace plane transition and the evection resonance play complementary roles in the early evolution of the Earth–Moon system. The former acts primarily on the *vector geometry* of the system: it modifies the orientation of the lunar orbit, Earth's obliquity, and lunar inclination, while allowing in some regimes a partial transfer of angular momentum to heliocentric orbit. The latter, or its quasi-resonance regime, provides a more *direct* channel for extracting angular momentum from the Earth–Moon system via the solar torque.

Published simulations show that a system initially rich in angular momentum can lose a substantial fraction of its excess, sometimes comparable to half its initial value[17,97]. This efficiency is not universal, however: it depends on the capture duration, the relative dissipation in the Earth and Moon, their thermal state, and the exit mechanism from the resonance.

In the TTP scenario, the combination of these processes therefore constitutes a plausible path to simultaneously achieve a significant reduction in angular momentum and an evolution compatible with lunar inclination. The current distance of about $60\, R_\oplus$ would then remain to be reproduced by long-term tidal evolution, starting from the orbital state left by the resonances.

Fig. 8 summarizes this flux in four states, with for each step the norm variation, direction, destination reservoir, and epistemic status of each value.

13.6 Perspective: a dissipative selection principle

The successive dissipative processes described above — crossing of the Laplace plane, gravito-tidal resonance regimes, orbital recession — suggest a reading that goes beyond simple bookkeeping of angular momentum. We present it here as a *research conjecture*, not as an established result.

Statement. Among the many configurations resulting from the three ejections, only those that can successively cross the Laplace plane transition, the gravito-tidal resonance regimes, and the orbital recession converge toward a durable Earth–Moon system. Dissipative mechanisms would thus play a *selective* role rather than a merely

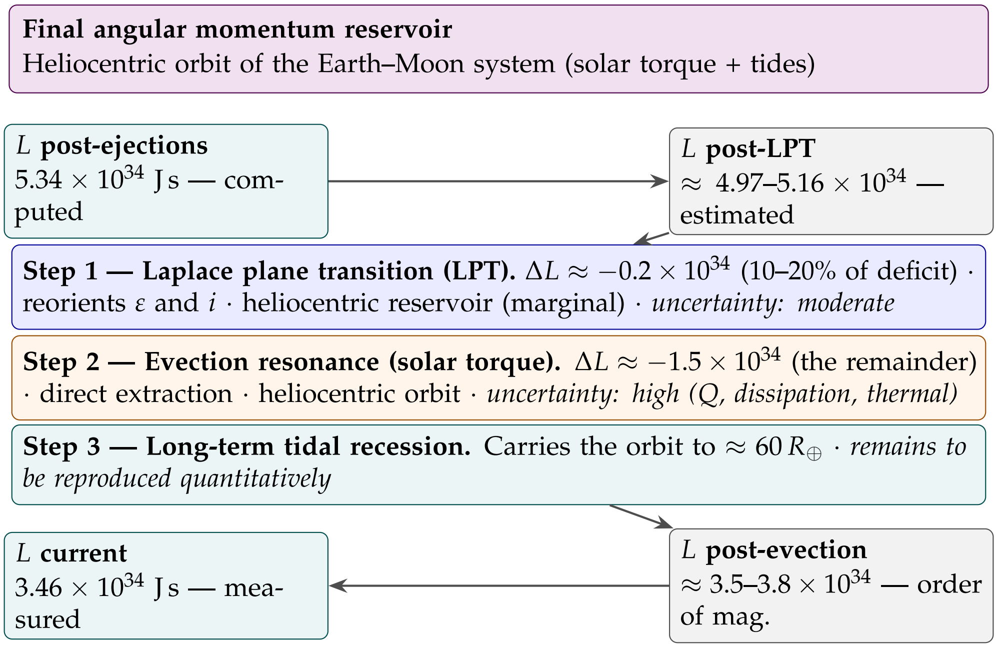

Figure 8: Angular momentum flux in four states. The final reservoir is the heliocentric orbit of the Earth–Moon system. Firm numbers at the bounds (teal: computed / measured), ranges for intermediate steps (gray: estimated / order of magnitude). Total deficit evacuated between firm bounds: $\approx 1.88 \times 10^{34}$ J s, with the tori having already carried 6.9×10^{33} upstream (no double counting).

relaxing one: they progressively reduce the space of accessible states and select a restricted domain of states compatible with the subsequent evolution of the system. The specificity of the TTP is not this contraction — common to any dissipative system (dissipative Liouville theorem) — but the fact that the selected domain would coincide with the one that reproduces the observed parameters, without those having been imposed as initial conditions.

Condition for convergence to a dynamical corridor

A system can converge to a dynamical corridor only if its control parameters evolve more slowly than its relaxation time:

$$\tau_{\text{forcing}} \gg \tau_{\text{relax}}. \tag{76}$$

When this timescale separation is satisfied, the system quasi-statically follows the manifold of compatible states; otherwise, the forcing exceeds the relaxation capacity and no stable dynamical selection can be established.

A compatibility function. Instead of searching only for the final condition $L_f = L_{\text{current}}$, one can define a scalar compatibility function:

$$\Phi = \Phi(L, \varepsilon, a, e, i, \omega, \dots) \tag{77}$$

that measures the proximity of a configuration to the current Earth–Moon state. The conjecture is then that, along admissible trajectories, $\Phi(t)$ increases in a quasi-monotonic way during dissipative evolution. This would not be an attractor in the strict sense — the dynamics would not bring every initial condition to the same point — but an **empirical Lyapunov function**: a quantity whose monotonicity reflects the progressive selection operated by dissipation. The conceptual framework is that of dissipative structures[109]: far from equilibrium, dissipative fluxes do not erase information but organize the system toward a restricted and structured domain of phase space.

Conjecture — dissipative selection and empirical Lyapunov function

The TTP would then cease to be only a *formation* mechanism and would also become a *selection* mechanism: the conditions that close the angular momentum budget would not be arbitrary inputs of the model, but would emerge from the dissipative dynamics itself: the required initial conditions would be *produced* by the dynamics rather than *postulated*.

Status and test. This idea is not demonstrated. Two steps would decide it. First, exhibit an explicit form of Φ (weighted distance in the space $(L, \varepsilon, a, e, i, \omega)$) and verify numerically that $\mathrm{d}\Phi/\mathrm{d}t \geq 0$ on admissible trajectories — the "almost always" to be specified as "always, on the subset of systems that retain their satellite," with possible decreases corresponding to eliminated configurations (Moon ejected, instability). Second, measure the *measure* of the selected domain: if the fraction of initial configurations leading to a durable system is non-negligible, selection is robust; if it is infinitesimal, it reduces to a posteriori survival argument. This distinction is the subject of Limitation L15.

13.7 Closure of the angular momentum budget

The closure model is parameterized by two quantities: the initial inclination i_I of the angular momentum vector and the retention factor of the transverse component $f_\perp$. The two observational constraints — the angular momentum deficit $\Delta L = 0.849\, L_{TL}$ (Eq. 57) and the final lunar orbital inclination $i_F = 5.145°$ — uniquely determine these two parameters.

Using the median of the conservative partition interval $L^+_{\text{bound}} = 1.875\, L_{TL}$ (the midpoint of the 1.85–1.90 L_{TL} range derived in Section 11.5), we obtain:

$$i_I = 65.0°, \qquad f_\perp = 0.0494.$$

The deficit is reproduced to:

$$\Delta L_{\text{model}} = 0.871\, L_{TL}, \qquad \text{discrepancy} = +2.5\%.$$

The final inclination is adjusted by construction through $f_\perp$; the predicted value of this parameter is physically interpretable as the efficiency of the solar torque in evacuating the transverse component of angular momentum to heliocentric orbit.

This yields a self-consistent closure of the angular momentum budget, with $f_\perp \simeq 0.05$, corresponding to the transfer of approximately 95% of the transverse component to heliocentric orbit during the Laplace plane transition. The result is robust: varying $f_\perp$ by a factor of 6 (from 0.02 to 0.12) shifts the required angle by only 0.4°.

Status of the result. The model does not claim to independently predict $i_F = 5.145°$; rather, it demonstrates that a Laplace plane transition transferring $\sim 95\%$ of the perpendicular component is both *sufficient* to produce the observed inclination and *necessary* to evacuate the angular momentum deficit ΔL. The value $f_\perp \simeq 0.05$ constitutes a testable prediction to be verified by numerical simulations of the resonance crossing (Limitation L15).

14 Chronology: Moon formation in 3 to 4 weeks

14.1 The three phases of an episode

Each ejection episode comprises three phases: resonant acceleration, during which the CMT grows exponentially via elliptic instability (Section 7.6), $t_{\rm acc} = \ln(U_{\rm crit}/U_{\rm turb})/\sigma_{\rm res} \approx$ 1.4 days; ejection, during which the flow crosses the Roche radius ($3R_{\rm eq,proto} \approx$ 22.2 Mm) at $U_{\rm crit}$, $t_{\rm ej} \approx 0.6$ hours; and recharge, during which the CMT reorganizes over 5–6 nutation periods, $t_{\rm rec} \approx$ 10–12 days. The Fe-Ni segregation rate (Stokes law) is quasi-instantaneous between episodes ($\tau_{\rm Stokes} \approx 254$ s, Appendix F), which implies that each accreted layer is chemically distinct from the previous one — a direct consequence underpinning seismic prediction P1.

14.2 Complete chronology

Phase	**Duration**	**Cumulative**
Episode 1: Acceleration	1.4 days	1.4 days
Episode 1: Ejection	0.6 h	1.4 days
Recharge (5–6 nutation periods)	10–12 days	11–13 days
Episode 2: Acceleration	1.4 days	13–15 days
Episode 2: Ejection	0.6 h	13–15 days
Recharge	10–12 days	23–27 days
Episode 3: Acceleration	1.4 days	24–28 days
Episode 3: Ejection	0.6 h	24–28 days
Total (N = 3)	**24–28 days**	

This formation timescale of 3 to 4 weeks is sufficiently short compared to the half-life of the Hf-W chronometer (8.9 Myr) to leave a specific isotopic signature (Prediction P7). Episode 1 (acceleration + ejection) forms the proto-POB, $R_1 \approx 1\,560$ km. Recharge and the second episode, $\approx$ 11–13 days, are asymmetric: post-ejection transient precession modifies ε, producing the crustal dichotomy of Section 9.2. Recharge and the third episode, again $\approx$ 11–13 days, complete the lunar mass. The TTP window closes at $t \approx$ 30–50 Myr, and the Hadean dynamo starts at $t \approx$ 330–350 Myr.

14.3 Lunar age: LMO solidification versus formation

The measured age (4.425 $\pm$ 0.025 Ga) is that of lunar magma ocean (LMO) solidification, not formation. If the TTP is triggered at $\approx$ 4.467 Ga, LMO cooling ($\sim$ 40 Myr) gives a predicted solidification age:

$$t_{\text{LMO}} \approx 4.467 - 0.040 = 4.427\,\text{Ga}, \tag{78}$$

consistent to 0.1σ with the measured value. Prediction P8 is a formation age $>$ 4.45 Ga, directly testable by Artemis III.

15 Internal stratification and seismology program

15.1 Concentric layer architecture

Each ejected layer comes from magma at a progressively more advanced stage of Fe-Ni segregation:

$$\left(\frac{\text{Fe}}{\text{Si}}\right)_{\text{layer}\,k} > \left(\frac{\text{Fe}}{\text{Si}}\right)_{\text{layer}\,k+1}, \tag{79}$$

because the source magma becomes progressively depleted in siderophile elements between episodes. Density contrasts between layers $\Delta\rho_{k,k+1} \approx 100$–$200\ \mathrm{kg\,m^{-3}}$[110] produce potentially detectable acoustic impedance contrasts.

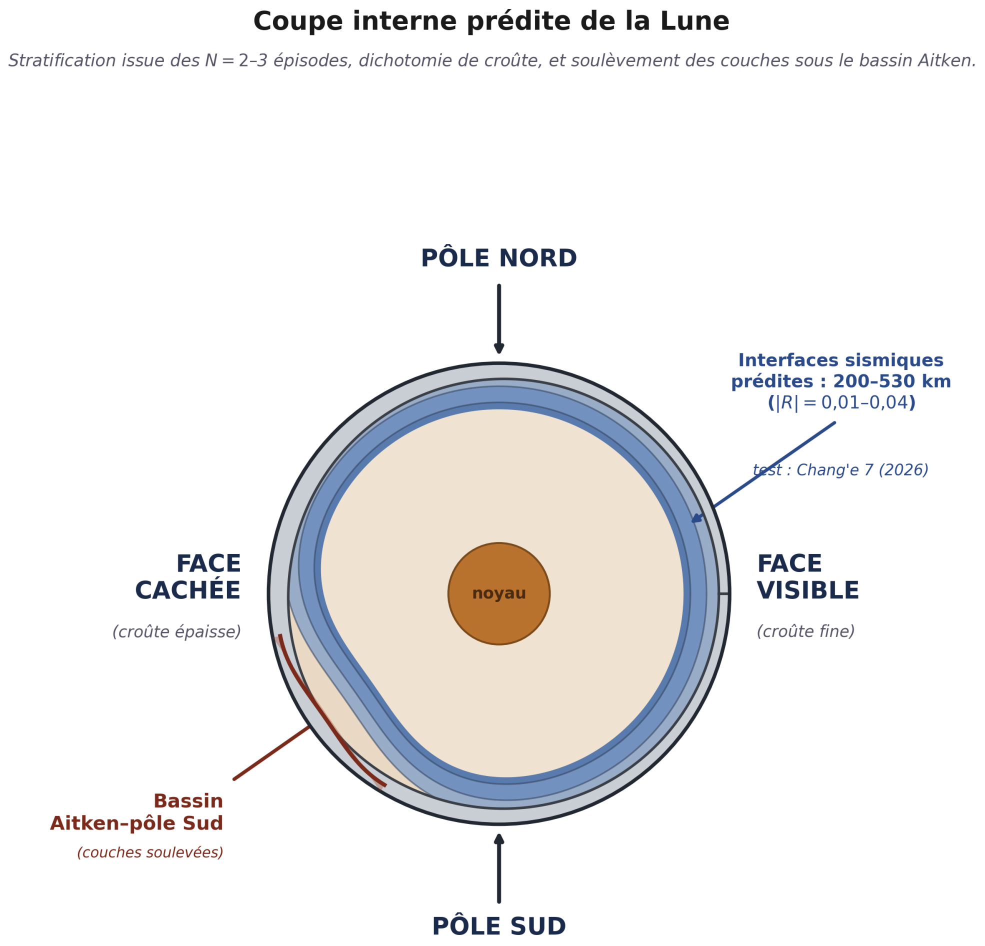

Figure 9: Predicted internal section of the Moon. Concentric stratification from $N = 2$–3 episodes, crustal dichotomy (thick crust on the farside, thin crust on the nearside), and uplift of layers beneath the South Pole–Aitken basin. Predicted seismic interfaces lie in the 200–530 km window ($|R| = 0.01$–0.04); test: Chang'e 7 (2026).

15.2 Preservation window: 200–530 km

Two destructive processes bound the preservation window. From below, ilmenite cumulate overturn[74] may have remelted the deepest layers (episodes 1–2, > 600 km). From above, mare basalt volcanism, active until ≈ 2 Ga[75,76], remelted the shallowest layers (< 100 km). The remaining preservation window therefore lies approximately between 200 and 530 km depth.

15.3 Interface depth: N=2 versus N=3

For $N = 2$ with equal masses, the interface between the two layers is at:

$$d = R_{\text{Moon}} - R_1 \approx 1\,737 - 1\,560 \approx 177\,\text{km} \quad (N = 2). \tag{80}$$

For $N = 3$, the main interface lies in the 200–315 km window. Chang'e 7 seismic data will therefore discriminate between $N = 2$ and $N = 3$.

16 Observational constraints and existing evidence

16.1 Isotopic identity Δ^{17}O < 5 ppm

The Earth–Moon isotopic composition is nearly identical for O, Cr, and Ti, established to better than 5 ppm[1–4]. The TTP satisfies this constraint *mechanically*: at each ejection episode, material is drawn from the contemporaneous Hadean terrestrial mantle, without exogenous isotopic source. No ad hoc isotopic mixing is required.

16.2 Small lunar core

The Moon has a metallic core of radius $\approx$ 330 km, significantly smaller than Earth's core relative to the host planet's mass[20,21]. In the TTP, this reduction is a direct consequence of the single engine: at each successive episode, the ejected mantle is more depleted in siderophiles (metallic Fe-Ni) than the previous one. The remaining iron in the lunar mantle is mainly in the form of FeO, whose concentration is indeed higher than in the terrestrial mantle (Section 16.4).

16.3 Angular momentum

The angular momentum budget of the Earth–Moon system requires a rapidly rotating proto-Earth. The value $T_{\text{rot}} = 3.5$ h is consistent with the median value of N-body accretion simulations[23,24]: rapid rotation is the norm at the end of accretion, not an exception.

16.4 Volatiles and FeO: two additional constraints

Volatile depletion. The Moon is strongly depleted in volatile elements (K, Na, Zn) compared to Earth and chondrites, with isotopic fractionation indicating high-temperature devolatilization[111–114]. A cohesive ejection with a protective glass crust might seem to minimize this loss. However, the ejected magma is at $T \approx 3000$–4000 K before crust formation, and devolatilization occurs during ballistic expansion, on timescales ($\sim 10^2$–10^3 s) comparable to the superficial cooling time. The TTP therefore predicts a selective depletion in volatiles, although the exact quantification depends on the surface-to-volume ratio of the ejected tongue and the vapor pressure of the species considered.

Mantle FeO. The Moon is depleted in *metallic iron* (small core), but its mantle FeO is higher ($\approx$ 13 wt%) than Earth's mantle ($\approx$ 8 wt%)[110,115,116]. Fe-Ni segregation does not affect FeO; it concerns metallic iron. Lunar mantle FeO reflects the composition of the Hadean terrestrial mantle before segregation, or an enrichment by fractional crystallization of the LMO. The TTP has no quantitative prediction for FeO. To be complete, a future version of the model must quantify the evolution of FeO through the ejection episodes and LMO cooling.

16.5 Chang'e-6 results: preliminary observational support

The Chang'e-6 mission returned 1,935 g of samples from the Apollo basin, within the South Pole–Aitken (SPA) region, in June 2024. Analysis of these samples provides preliminary support for two TTP predictions. Yue et al.[18] dated Chang'e-6 norite samples to 4247 ± 5 Ma by the Pb-Pb method. These norites are interpreted as differentiated products of the SPA impact melt sheet, representing deep lunar mantle material excavated by the SPA impact. This age is consistent with the TTP prediction that the innermost lunar layer (Episode 1) was accreted at $\approx$ 4.5 Ga and partially reset by the SPA impact at $\approx$ 4.25 Ga. Xu et al.[19] further report olivine fragments in Chang'e-6 samples containing Fe and Mn concentrations significantly higher than those found in the standard lunar mantle, with zinc concentrations $\approx 100\times$ higher than expected. Although part of this enrichment is attributed to meteoritic contamination, the olivine fragments themselves formed by mixing lunar rocks and molten asteroidal material. The high iron content (total Fe, including FeO) of material excavated from SPA depths is *consistent* with the depth-increasing Fe/Si gradient (P3) and provides preliminary support for the South Pole Fe/Si anomaly (P6), detailed below.

17 Falsifiable predictions

This section gathers, in catalog form, each quantitative and falsifiable prediction made by the TTP. Each entry states the prediction, its physical derivation where not already given in full elsewhere in the manuscript, the relevant test, and an explicit falsification threshold set in advance of observational data.

Falsifiable prediction — P1 — Seismic interface at $d \approx$ 200–315 km (Priority 1)

Concentric stratification produces acoustic impedance contrasts $|R| \in [0.01, 0.04]$ between layers. For $N = 2$, the interface is at $d \approx 177$ km. For $N = 3$, the main interface is at $d \approx$ 200–315 km.

Detectability with Chang'e 7 depends on seismometer sensitivity (target $10^{-8}\ \mathrm{m\,s^{-2}\,Hz^{-1/2}}$ at 0.1–1 Hz). This sensitivity should resolve $|R| > 0.01$, provided at least 10 events of $M > 2.5$ are recorded during the first 6 months of operation.

Falsification: no phase conversion detected within the 200–530 km window after deployment of at least 3 seismic stations and detection of at least 10 lunar events with $M > 2.5$.

Missions: Chang'e 7 (August 2026, South Pole seismometer), FSS (Farside Seismic Suite), LEMS (Lunar Environment Monitoring Station), Artemis III (2028–2029).

Falsifiable prediction — P2 — Seismic asymmetry, nearside versus farside

The second, asymmetric episode produces a sharper and deeper interface beneath the farside.

Falsification: spherical symmetry confirmed by a complete tomographic inversion.

Falsifiable prediction — P3 — Fe/Si gradient increasing with depth

The Fe/Si ratio increases with depth in the lunar mantle, reflecting the progressive depletion in Fe-Ni of the source magma between episodes. Quantitatively, with the total cumulative fraction of Fe removed from the mantle $\xi_{\mathrm{Fe}}^{\mathrm{sat}} \approx 0.15$[9] and $N = 3$ segregation episodes, the fraction of Fe removed per episode, assuming geometric progression over N episodes, is:

$$\xi_{\mathrm{Fe,episode}} = 1 - (1 - \xi_{\mathrm{Fe}}^{\mathrm{sat}})^{1/N} \approx 1 - (0.85)^{1/3} \approx 0.053, \tag{81}$$

i.e., $\approx 5.3\%$ of the remaining Fe removed at each episode. The resulting Fe/Si ratio between adjacent layers is multiplicative:

$$\frac{(\mathrm{Fe/Si})_{\mathrm{Ep}k}}{(\mathrm{Fe/Si})_{\mathrm{Ep}(k+1)}} = \frac{1}{1 - \xi_{\mathrm{Fe,episode}}} \approx 1.056, \tag{82}$$

giving an enrichment of about 5.6% between Episode 2 and Episode 3, and a cumulative enrichment $(\mathrm{Fe/Si})_{\mathrm{Ep1}}/(\mathrm{Fe/Si})_{\mathrm{Ep3}} = (1 - \xi_{\mathrm{Fe,episode}})^{-2} \approx 1.115$, i.e., a total enrichment of about 11.5% between Episode 1 (the deepest layer, richest in Fe) and Episode 3 (the shallowest). The SPA basin preferentially exposes material from Episode 1.

Preliminary support: Chang'e-6 olivines with high Fe/Mn ratio[19]. **Missions:** Chang'e-6 (additional analyses), Artemis III.

Falsifiable prediction — P4 — Dynamo delay: 290–360 Myr

The gradual emergence of the dynamo predicts a continuous growth of paleointensity from 4.55 to 4.20 Ga, without an abrupt appearance[6].

Falsification: an abrupt appearance of the geomagnetic field detected in the Jack Hills zircon record.

Falsifiable prediction — P5 — Instability window: $\varepsilon \in [57^\circ, 70^\circ]$

Parametric elliptic instability requires ε within this interval.

Test: N-body simulations of the obliquity of terrestrial protoplanets.

Falsifiable prediction — P6 — Fe/Si anomaly at the South Pole (derived from P3 and SPA impact simulations of Wakita et al.[90])

Observational context. The hydrodynamic simulations of Wakita et al.[90] establish three facts independent of the TTP: the SPA impact excavated lunar mantle material from depths greater than 90 km; the oblique, north–south trajectory of the impactor produced an asymmetric ejecta plume that preferentially concentrated deep material toward the South Pole; and about 350 m of mantle material is predicted on average at the surface near Shackleton crater.

TTP interpretation. Within the TTP framework, the lunar mantle is stratified in concentric layers (P1), with Fe/Si decreasing from the deepest layer (Episode 1) to the shallowest (Episode 3), as quantitatively derived in P3 above. The combined thickness of Episode 2 and Episode 3 layers is about 315 km (Section 15). Material excavated by the SPA impact from depths greater than 90 km, and concentrated toward the South Pole, therefore comes preferentially from Episode 2 and, at greater excavation depths, from Episode 1.

Quantitative scenario. Wakita et al.[90] do not provide a quantified fraction of Episode 1 versus Episode 2 material in the SPA ejecta plume, since their simulation does not incorporate TTP stratigraphy; this fraction must therefore be bounded from first physical principles rather than read directly from their results. We consider the most adverse scenario compatible with the TTP: SPA excavation reaches only ≈ 100 km (barely above the 90 km minimum established by Wakita et al.[90]), so that only Episode 2 material — and not the deeper, more strongly enriched Episode 1 — contributes to ejecta, and we conservatively take the minimal plausible fraction of Episode 2 material in the plume as 30%. With the Fe/Si enrichment of about 5.6% for Episode 2 relative to Episode 3 derived in Eq. 82, this minimal scenario predicts an enrichment of:

$$\Delta(\mathrm{Fe/Si})_{\mathrm{min}} \approx 0.30 \times 0.056 \approx 0.017 \approx 1.7\%. \tag{83}$$

If excavation reaches instead more than ≈ 315 km, Episode 1 material contributes directly; as an illustration, a mixture of 30% Episode 1, 30% Episode 2, and 40% Episode 3 would give $\Delta(\mathrm{Fe/Si}) \approx 0.30 \times 0.115 + 0.30 \times 0.056 \approx 5.1\%$, a value requiring a non-negligible Episode 1 contribution to be reached.

Quantitative prediction. Samples from the South Pole (Chang'e-7, Artemis III) should show a higher Fe/Si ratio than the average of equatorial samples (Apollo) and nearside samples (Chang'e-5), with an enrichment of at least $\approx 1.7\%$ in the minimal scenario above and potentially up to $\approx 11.5\%$ if Episode 1 material dominates the sampled ejecta.

Falsification threshold. We set, in advance of any observational data, the falsification threshold at:

$$(\text{Fe/Si})_{\text{SPA}} \leq 1.015 \times (\text{Fe/Si})_{\text{lunar average}} \implies \text{P6 falsified}, \tag{84}$$

i.e., an enrichment below 1.5%, set slightly below the minimal scenario value of 1.7% from Eq. 83 to allow for experimental uncertainty without weakening the test's substance: even the most conservative scenario compatible with the TTP predicts an enrichment above this threshold, so a measured enrichment below it would be incompatible with the theory as constructed here. An enrichment between 1.5% and 5.6% (the maximum achievable by Episode 2 material alone, from Eq. 82 applied to a pure Episode 2 sample) is read as partial support, consistent with ejecta dominated by Episode 2 without requiring Episode 1 contribution. An enrichment above 5.6% cannot be produced by Episode 2 and 3 material alone, and thus constitutes direct evidence for Episode 1 contribution, providing strong support for the deep stratigraphy predicted by P1 and P3.

Epistemic note on the grading bounds. The lower bound (1.5%) rests on an estimate of the minimal fraction of Episode 2 material in SPA ejecta (30%), which is a modeling assumption. The upper bound (5.6%) is a physical constraint: the maximum enrichment achievable by a pure Episode 2–Episode 3 mixture, without Episode 1 contribution. Therefore, any measured enrichment above 5.6% constitutes definitive evidence for the presence of Episode 1 material, a qualitatively different regime from the lower-bound scenario. The two bounds are thus not of the same epistemic nature; this asymmetry is explicitly recognized here.

Test: Chang'e-7 surface analyzers and samples returned byArtemis III from the South Pole, compared to existing Fe/Si measurements from Apollo (equatorial) and Chang'e-5 (nearside).

Acknowledged limitation: this prediction shares Limitation L9 (impact mixing in SPA ejecta) with prediction P3, and Limitation L11 (the 30% Episode 2 fraction itself) below.

Falsifiable prediction — P7 — Signature in the Hf-W chronometer

A formation duration of 3 to 4 weeks is short compared to the half-life of the Hf-W chronometer (8.9 Myr,[34]). The ^{182}Hf/^{180}Hf ratio in Episode 1 samples should correspond to a differentiation age of 4.467 Ga (100 ± 10 Myr after CAIs).

Falsification: a Hf-W differentiation age of Episode 1 samples significantly below 100 ± 10 Myr after CAIs.

Mission: Artemis III (2028–2029).

Falsifiable prediction — P8 — Formation age > 4.45 Ga

The measured age of 4.425 Ga is that of LMO solidification, not formation (Section 14).

The formation age is about 4.467 Ga.

Test: direct dating of Episode 1 rocks exposed on the central peaks of the SPA basin (Artemis III).

Falsifiable prediction — P23 — TTP magma rheology constrained by South Pole–Aitken basin samples

The South Pole–Aitken basin plays for the TTP the role of a window: the impact — post-dating formation — thinned the crust and uplifted the deep Episode 1 layers (Figure 9), exposing at the basin center mafic materials (olivine, magnesian pyroxenes) of deep origin, whereas the periphery and rings show only reworked crust. The basin center is therefore the target sampling site for this prediction.

The target measurement is not textural but compositional: sample textures record the SPA impact and its cooling, not Hadean emplacement. The bridge to primordial rheology passes through the laboratory: remelting of representative compositions and measurement of $\eta(\phi, T)$ and $\tau_y(\phi, T)$ laws — viscosity and yield stress as functions of crystallinity and temperature[37,77] — subsequently applied to the thermal state predicted by the TTP. Samples already exist in part (Chang'e-6, basin rim) and will be supplemented by Chang'e-7 and Artemis III.

The stakes of this single measurement are twofold, because the same parameter τ_y governs two independent clocks of the model: the cohesion of the ejected tongue ($\mathrm{Bi}_{\mathrm{KH}}$, Appendix D, Limitation L14) and the tidal recession speed of the Moon toward the solar resonant sinks ($Q \propto 1/\tau_y$, Sections 11.6 and 12, Limitation L15). A single measurable quantity thus simultaneously constrains the two highest-priority limitations of the theory. One also notes, with caution, that the deep mass anomaly detected beneath the basin[117] is compatible with the uplift of dense layers predicted by Figure 9, without demonstrating it — the standard interpretation (buried impactor core) remaining admissible.

Falsification: if rheology measured at the TTP thermal state gives τ_y outside the 10^2–10^4 Pa interval, the cohesion ($\mathrm{Bi}_{\mathrm{KH}}$, Appendix D) and recession (Section 11.6) windows must be recalculated; a τ_y incompatible with $\mathrm{Bi}_{\mathrm{KH}}^{\mathrm{eff}} \geq 1$ after recalculation would refute the cohesive ejection mechanism.

Missions: Chang'e-6 (samples already in hand), Chang'e-7, Artemis III.

Falsifiable prediction — P9 — Compatibility of mare basalts with Layer 1

The melting depth of mare basalts (200–500 km, ± 100 km) is compatible with a source within Layer 1 ($> d_{\mathrm{P1}}$), although this cannot be asserted in the absence of more precise geochemical data.

Test: high-resolution isotopic geochemistry of Apollo and Chang'e-5 mare basalts[75].

Falsifiable prediction — P10 — Seismic interface at $d \approx 177$ km for $N = 2$

For $N = 2$ equal-mass episodes, the single interface between the two concentric layers is at $d \approx 177$ km. This depth lies outside the main preservation window (200–530 km) because it has been partially remelted by mare volcanism and/or ilmenite cumulate overturn. However, partial remnants could be detected as weak reflectors or seismic velocity variations in the 150–200 km zone.

Predictions P17 (capture efficiency, Section 10.3), P20 (asymptotic regime indicator, Section 11.7), P21 (numerical closure of the angular momentum budget, Section 11.6), and P22 (the unified ejection threshold, Section 9.2) are derived in full, with their own falsification conditions, at the manuscript locations indicated above, rather than repeated here.

18 Validation windows: three imminent tests

18.1 Chang'e 7: imminent seismic validation (August 2026)

The Chang'e 7 mission (CNSA, launch scheduled for August 2026) will deploy the first seismometer at the lunar South Pole since Apollo 17 (1972). Its data will provide a direct test of P1, P2, and P6. The prediction is pre-registered and quantified: $d \approx 200$–315 km, $|R| \in [0.01, 0.04]$ for the seismic interface, and an enrichment greater than 1.5% in Fe/Si for South Pole ejecta. Falsification criteria are explicit in each case.

18.2 The South Pole–Aitken basin: stratigraphic window

The SPA basin did not create the concentric lunar stratigraphy: it *excavated* it. The layered structure pre-existed the SPA impact at ≈ 4.25 Ga[18], encoded during TTP differentiation at ≈ 4.5 Ga. The impact is the accidental revealer of a much older internal architecture.

High-resolution 3D simulations[90] show that the oblique north–south trajectory of the SPA impactor preferentially launched deep mantle ejecta toward the South Pole in a butterfly-shaped plume — precisely the landing zone of Artemis III. These same simulations predict ≈ 350 m of mantle material on average at this site, excavated from depths greater than 90 km, consistent with both the TTP preservation window and the quantitative scenario underpinning prediction P6.

18.3 Artemis III: direct sampling of Episode 1 material

The Artemis III mission (NASA, planned for 2028–2029) will land astronauts near Shackleton crater at the South Pole. According to Wakita et al.[90], surface samples at this location may include deep mantle ejecta from the SPA impact. If so, these samples will allow a direct test of P3, P6, P7, and P8: their Fe/Si ratio, crystallization age (≈ 4.467 Ga for primary Episode 1 material), and isotopic signature (Δ^{17}O

< 5 ppm relative to the contemporaneous terrestrial mantle) are all quantitative predictions of the TTP.

Chang'e 7 (2026) targets seismology and the Fe/Si anomaly (P1, P2, P6), while Artemis III (2028) targets geochemistry and chronology (P3, P6, P7, P8). Two independent space agencies converge on a single geographic site — the lunar South Pole — to test a single internal architecture.

19 Discussion: main objections and responses

19.1 Responses to main objections

Objection 1: "A large number of unjustified hypotheses." The hypotheses are explicitly classified into three levels (Section 8, Table 2): Level 1 (established) denotes results from the literature or closed-form derivations (e.g., $E_{\text{grav}}/E_{\text{fus}} \approx 121$, the Bingham-Herschel law); Level 2 (constrained) denotes physically motivated hypotheses not yet numerically validated (e.g., $\varepsilon \in [40°, 70°]$); Level 3 (speculative) denotes hypotheses awaiting validation by 3D SPH (e.g., the emergence of a unique $\ell = 1, m = 1$ torus, the exact value of f_{cap}, the quantitative validity of the Gaussian approximation discussed below). The TTP does not rely on all these hypotheses with equal weight. Observational predictions (Section 17) are independent of Level 3 hypotheses.

Objection 2: "The CMT has not been demonstrated." In the rapid rotation limit (Ro $\ll 1$), the azimuthal component of the Navier-Stokes equation for a Bingham-Herschel fluid reduces to a potential vorticity conservation equation. Linearizing about the mean flow near the instantaneous dynamic equator ($\phi = 0$, perpendicular to $\boldsymbol{\Omega}(t)$, as defined in Section 3), and applying the standard multiple-scale expansion for Rossby wave envelopes[69–71], yields a nonlinear Schrödinger equation for the envelope $A(y, t)$, where $V_{\text{eff}}(y)$ is the sum of the potentials listed in Table 1. A stationary ansatz $A(y, t) = \psi(y)e^{-iEt/\hbar_{\text{eff}}}$ then gives a time-independent nonlinear Schrödinger equation for $\psi(y)$, the nonlinearity coming from the advective term in the potential vorticity equation. The fundamental mode of the *linearized* version of this equation, with $V_{\text{eff}}(y)$ expanded to quadratic order about $y = 0$, is a Gaussian wavefunction centered on the instantaneous equator, confined to the belt $|\phi| < 30°$ — a direct consequence of the fact that the seven potentials of Table 1 all have their extrema at $y = 0$. Whether this Gaussian remains a quantitatively adequate approximation to the ground state of the full nonlinear equation, rather than merely its qualitative first-order limit, has not been established for this system and is recognized as Limitation L12 (Section 20). The full derivation is given in Appendix C. Quantitative validation of the single-torus structure awaits 3D SPH simulation (Limitation L1).

Objection 3: "The ejection mechanism relies on an adjusted threshold." The ejection threshold is defined by the condition $\lambda(U) = 0$ (Section 9.2), a deterministic surface in parameter space. It is a function of the density gradient β, the obliquity ε, and the angular speed Ω: ejection occurs when $\Omega > \Omega_{\text{crit}}(\varepsilon, \beta)$. The number of episodes $N = 2$–3 results from the balance between the recharge rate (fed by angular momentum input) and the loss by ejection. It is a consequence of Hadean conditions (β, momentum input), themselves constrained by the existence of the Moon — a retrodictive result, not a free parameter.

Objection 4: "$f_{\text{cap}} = 0.70$ is high compared to accretion simulations (0.3–0.5)." The classical range (0.3–0.5) concerns impacts between solid bodies. Here, the ejected material is magma at 3 000–4 000 K, with viscous coalescence and toroidal orbital confinement. Appendix G establishes a physical lower bound $f_{\text{cap}}^{\text{min}} \gtrsim 0.45$–$0.55$, independent of the BH cohesion hypothesis. The value 0.70 lies within a mechanically justifiable interval.

Objection 5: "Are the required ejection thresholds physically reachable?" The ejection threshold is a condition on Ω: $\Omega > \Omega_{\text{crit}}$, with $\Omega_{\text{crit}} \approx \sqrt{2}\,\Omega_0$. This factor is reachable by angular momentum input from continuous planetesimal bombardment (Appendix H). The Euler oscillation term λ_4 modulates the effective obliquity ε, but does not directly trigger ejection.

Objection 6: "The manuscript is too speculative for a review." The TTP is submitted as a *theory with falsifiable predictions*, not as an observational result. Validation will proceed in two phases: Phase 1 (2026–2029), observational tests by Chang'e 7 (August 2026) and Artemis III (2028–2029); and Phase 2 (2027–2030), numerical SPH 3D validation once institutional resources are committed. The TTP will be judged on its predictions, not on prior simulation.

Objection 7: "The requirements on Q are contradictory." This objection is valid only if one confuses local dissipation in the CMT with the global tidal dissipation of the entire proto-Earth. The CMT is a spatially confined zone ($\sim \pm 30°$ latitude) with low solid fraction and low Q; the global Q is controlled by the much larger volume of the solid or partially solid mantle, which can have a Q two to three orders of magnitude higher. The evection resonance is sensitive to this global Q, not to the local Q_{CMT}. The TTP therefore does not require the same material to be simultaneously fluid and rigid; it requires a thermally stratified body with a fluid belt and a solid mantle. This is precisely the state expected for a differentiating magma ocean[8]. Quantifying Q_{global} for the Hadean Earth is an open problem that the TTP helps to constrain, and it constitutes the highest-priority numerical validation task (Limitation L15).

19.2 Justification of $\alpha > 1$

Two independent mechanisms lead to a super-rigid rotation profile: equator–pole differential cooling generates a density gradient and thermal acceleration of rotation; Fe-Ni segregation lightens the CMT, and by conservation of angular momentum, its inner part spins up. The Bingham-Herschel threshold also blocks small scales, favoring a solid-body-like profile. Numerical validation of this profile is a priority for 3D SPH simulation.

19.3 Partial fragmentation and coalescence

Even if the ejected sheet partially fragments, fragments remain within the same orbital zone and coalesce rapidly under inelastic collisions and gravitational attraction to the proto-POB[93]. The effective capture yield therefore remains high ($f_{\mathrm{cap}} \geq 0.7$). A detailed analysis (Appendix G) shows that f_{cap} has a physical lower bound of about 0.45–0.55, independent of the BH cohesion hypothesis.

20 Acknowledged limitations

A theory progresses not only by reducing the number of its unknowns; it progresses especially by unambiguously identifying the physical origin of each of them. The present work does not claim to derive everything: it aims for a more attainable and, in our view, more honest goal — that *every deferred closure be localized, named, classified, and associated with a decisive test*. Figure 10 gives the map: the still-open zones of the TTP are few, each attached to a precise physical mechanism and a resolution protocol. None is hidden behind a coefficient presented as fundamental. Limitations L1 to L16 below develop each of these entries.

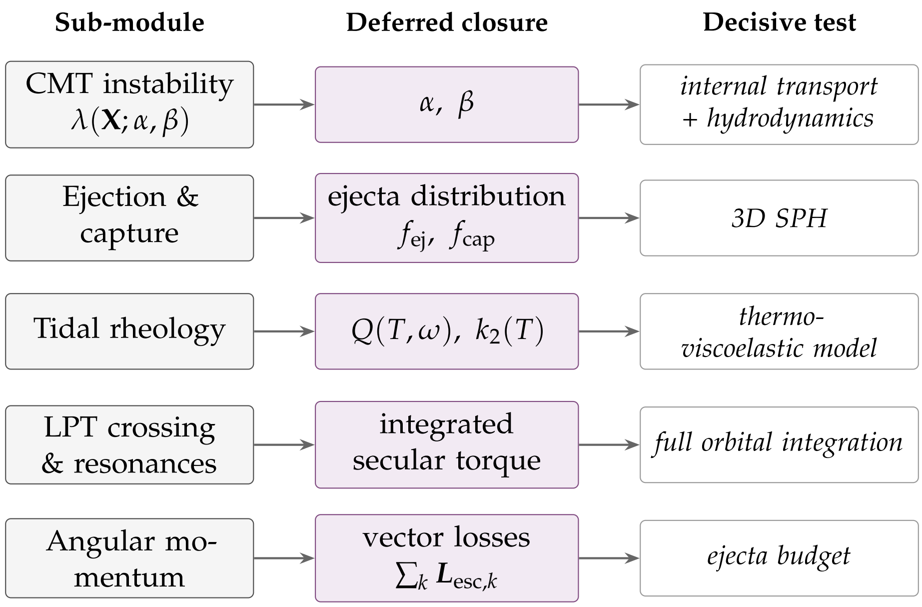

Figure 10: Map of deferred closures of the TTP. Each physical sub-module (left) has a closure not yet derived (center, in color), itself attached to the test that will decide it. The reading is not "there remain free parameters" but "there remain a small number of identified physical phenomena, each with its resolution protocol."

A 3D SPH simulation is necessary, but premature at the time of writing (July 2026). Such a simulation — coupling Bingham-Herschel rheology, radiative cooling, Fe-Ni segregation, and the $\varepsilon \neq 0$ geometry — requires institutional HPC resources that are not committed at this stage, for a simple reason: the first decisive observational test arrives before such a simulation could reasonably be completed. Chang'e 7 is scheduled for August 2026, only a few weeks away. Its seismic data will detect, or not detect, the predicted interface within the 200–530 km window (Prediction P1). In the first case, the concentric stratification predicted by the TTP receives independent observational support, and a dedicated SPH 3D validation becomes a justified investment of institutional resources. In the second case, under the falsification conditions stated in Section 17, the central prediction of the TTP is seriously challenged, and a costly numerical campaign built on a falsified premise would be of little value. This is why the present work is submitted, and should be evaluated, on the basis of its falsifiable observational predictions, prior to and independently of any full numerical validation campaign.

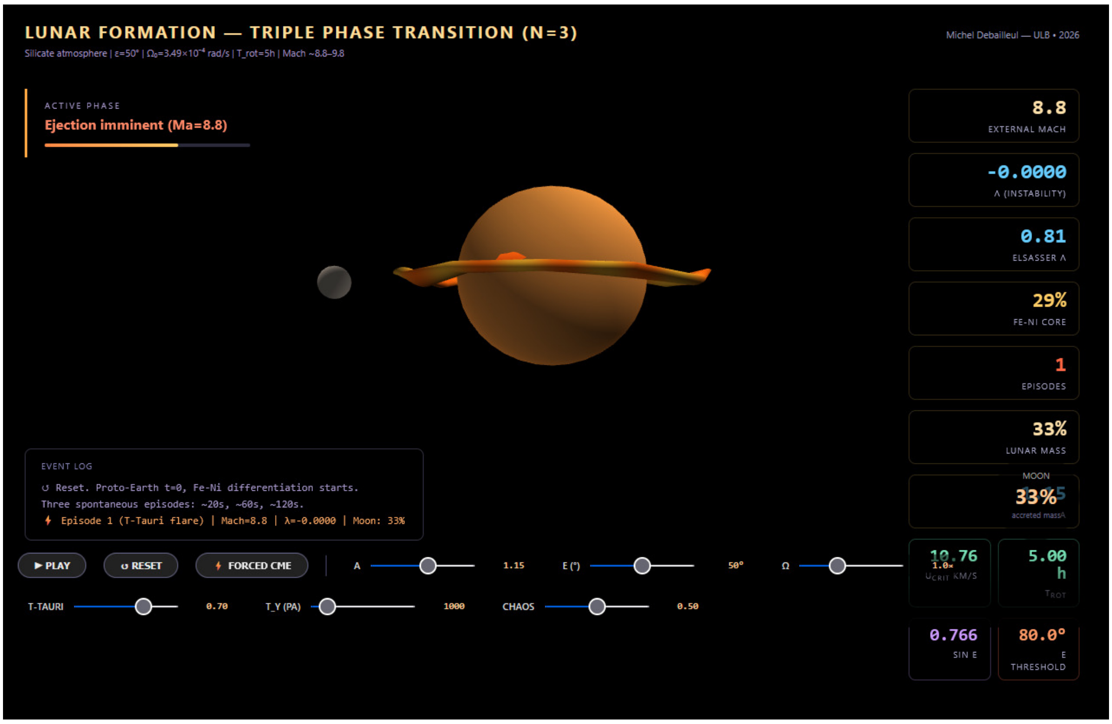

Figure 11: Screenshot of the interactive simulation accompanying this work (link above), illustrating the Coherent Magmatic Torus (CMT) during ejection around the proto-Earth, with the real-time dynamic dashboard of parameters (external Mach number, instability λ, Elsasser number, Fe-Ni core fraction, accreted lunar mass).

Acknowledged limitation — L1 — Spontaneous organization of the CMT

The question of whether a single torus or multiple coexisting structures emerge in the exact Hadean regime can only be resolved by a high-resolution 3D SPH simulation with full BH rheology. **Priority 2.**

Acknowledged limitation — L2 — Numerical value of f_{cap}

The capture efficiency $f_{cap} = 0.70$ is physically motivated but not derived from first principles. Its increasing dependence on M_{Moon} and its deviation from the classical debris-disk accretion range (10–55%, Kokubo et al.[92]) are formalized as testable prediction P17. 3D SPH simulation remains the ultimate arbiter of the exact value of f_{cap}.

Acknowledged limitation — L3 — Small-scale cohesion

The $\text{Bi}_{\text{KH}} \gg 1$ argument is valid at large and intermediate scales. At very small scales ($\lambda_{\text{KH}} \ll R_{\text{torus}}$), partial fragmentation cannot be excluded analytically. However, fragments from the same toroidal flow, ejected with the same speed and angle, find themselves on nearly identical orbits and accrete onto the POB: the mass budget is not affected by fragmentation, only the fragment size is.

Acknowledged limitation — L4 — Dynamo delay: ± 50 Myr

The exponential core growth model is simplified; the associated uncertainty is ± 50 Myr.

Acknowledged limitation — L5 — Super-rigid profile $\alpha > 1$

The super-rigid rotation profile ($U \propto r^{\alpha}$, $\alpha > 1$) is physically motivated by differential rotation in a partially crystallized magma undergoing rapid cooling, but has not been demonstrated under these exact Hadean conditions. It is a Level 3 hypothesis, to be validated by simulation.

Acknowledged limitation — L6 — Flow $\rightarrow$ POB transition: qualitative status

The three-phase description of the ejected flow (ballistic expansion, orbital insertion, cohesive accretion beyond R_{Roche}) is physically motivated and consistent with the calculated orders of magnitude, but remains qualitative. A 3D SPH simulation with coupled radiative cooling is required. **Priority 2b.**

Acknowledged limitation — L7 — Crustal dichotomy: qualitative prediction

The thickness ratio $\approx 2:1$ is a qualitative prediction of the TTP. The coupled differential thermal calculation has not yet been derived analytically. The lunar farside has undergone at least two independent remelting episodes that must be accounted for in any quantitative model: the SPA impact at ≈ 4.25 Ga, which excavated and partially remelted the deepest Episode 1 layer on the farside[90]; and prolonged mare volcanism extending to ≈ 3.2 Ga[75], which remelted the shallowest layers (< 100 km) over much of the farside. The net effect is to deepen and strengthen the existing asymmetry rather than erase it.

Acknowledged limitation — L8 — Rossby wave speed in a BH fluid

The rigorous derivation of $c_{\text{Rossby}}^{(\text{BH})}$ for a BH fluid on an oblate spheroid ($\varepsilon \neq 0$) remains an open problem. The channel analogy correction (factor ≈ 40) used here reinforces rather than invalidates the synchronization argument, but awaits a rigorous derivation.

Acknowledged limitation — L9 — Impact mixing in SPA ejecta

The geochemical signature of deep mantle material excavated by the SPA impact is potentially diluted or altered by mixing with impactor material and local crust. Quantitative deconvolution of the primary Episode 1 signature from the impact melt breccia requires detailed petrological and isotopic studies. This limitation affects predictions P3 and P6.

Acknowledged limitation — L10 — Predicted deep mantle samples not yet collected

The TTP predicts that Episode 1 deep mantle material is exposed on the central peaks of craters within the South Pole–Aitken basin, excavated and extruded during the SPA impact at ≈ 4.25 Ga[90]. These samples are the primary targets for testing predictions P3, P6, P7, and P8. No samples from these central peaks have yet been collected or analyzed. The absence of deep mantle material at the predicted locations, or an Fe/Si ratio incompatible with P3 or P6, would seriously challenge the TTP.

Acknowledged limitation — L11 — Fraction of Episode 2 material in SPA ejecta

Prediction P6 requires an estimate of the minimum fraction of Episode 2 material present in the SPA ejecta plume reaching the South Pole, which we have conservatively set at 30% based on the excavation depth argument of Section 17 rather than on a direct quantitative result from Wakita et al.[90], whose simulations were not designed to track the stratigraphic layers specific to the TTP. A future dedicated hydrodynamic simulation, tracking the depth of origin of SPA ejecta material, would refine this fraction — and hence the falsification threshold of Eq. 84.

Acknowledged limitation — L12 — Quantitative validity of the Gaussian approximation for the CMT confinement mode

The nonlinear Schrödinger equation derived in Section 19 (Objection 2) and Appendix C is not solved exactly in this work. The Gaussian wavefunction is a reasonable approximation for the fundamental mode, valid in the limit where the wave amplitude is small, i.e., when the perturbation speed δU associated with the $\ell = 1, m = 1$ mode satisfies $\delta U \ll U_{\text{turb}}$. This condition is qualitatively consistent with the Rhines inverse cascade (Section 5), but its exact quantitative verification is not possible with the parameters available at this stage: no independent estimate of $\delta U / U_{\text{turb}}$, nor of the nonlinear coefficient β in the governing equation, has been derived for this specific system, unlike classical atmospheric and oceanic Rossby wave systems for which the reduction was originally established[69–71]. The absence of this verification is acknowledged here as a limitation. The Gaussian solution should therefore be understood as a first-order approximation of the linearized problem, and not as an exact or quantitatively bounded solution of the full nonlinear equation.

Acknowledged limitation — L13 — Post-ejection oscillation persistence

The post-shock nutation persistence mechanism by reflection at the ejection cavity boundaries (Appendix I) relies on an analogy with established results for inertial waves in a confined homogeneous fluid sphere[118,119], and not on a derivation specific to the oblique geometry and Bingham-Herschel rheology of the Hadean proto-Earth. The plateau duration $T_{\rm nut}^{\rm (post)}$ and the absence of significant decay during this window are therefore qualitative estimates, not derived results. A full quantitative formulation — treating the coupled system $(\Omega, \varepsilon, |g_{\rm eff}|)$ as an underdamped Langevin dynamics, numerically integrable by a BAOAB-type scheme[120] once the physical noise (CMEs, impacts) is properly characterized as a jump process rather than Gaussian white noise — is identified as a future computational program, in complement to 3D SPH simulation (Limitations L1, L2).

Acknowledged limitation — L14 — Quantification of $\mathrm{Bi}_{\rm KH}^{\rm eff}$ and numerical validation

The multiplicative product of the three cohesion reinforcements ($f_{\rm atm} \times f_{\rm crist} \times f_{\rm RT}$) is qualitatively established. A 3D SPH simulation is required to validate the exact numerical values, in particular the nonlinear interaction between atmospheric confinement, crystal network, and Rayleigh-Taylor suppression effects. **Priority 2c.**

Acknowledged limitation — L15 — Inertial obliquity at the Laplace plane crossing (Priority 1)

The closure of the angular momentum budget now relies on the Laplace plane transition (Section 12), whose activation condition is $\varepsilon_{\rm I} \gtrsim 60$–$65°$ at the time the Moon reaches the transition zone ($\approx 25\ R_\oplus$, 10^4–10^6 yr after formation). The critical question is no longer “is $Q_{\rm global}$ high enough for evection capture?” but “do the N ejection recoils leave $\boldsymbol{L}_p$ inclined by at least $\approx 60°$ to the orbital normal, and does this obliquity persist until crossing?” Three calculations, all internal to the existing formalism, have been initiated at the order-of-magnitude level, each with explicit caveats. *(i)* The vector recoil budget (Eq. 34), evaluated by Monte Carlo under the most adverse hypothesis (free azimuths, nutation $\pm 30°$, Fig. 3), maintains $\varepsilon_{\rm I}^{(0)} > 60°$ in 97–100% of draws, with medians *higher* than the initial obliquity: the inherited obliquity is preserved, or even strengthened. Caveats: two-dimensional model; margin above 65° narrowing (to 82%) for $\varepsilon_{\rm I}^{\rm init} = 60°$. *(ii)* The secular evolution of $\varepsilon_{\rm I}$ during recession, under the solar tidal torque ($N_\odot \sim 10^{18}$–10^{19} N m), gives $\Delta\varepsilon_{\rm I} \lesssim 0.1°$ over 10^7 yr: negligible compared to the inherited obliquity. *(iii)* The coupled $r_{\rm L}(t)$–$a(t)$ calculation (Fig. 4) places the convergence at $\approx 25\ R_\oplus$, robust with respect to k_2/Q. Caveat: contraction $R(t)$ modeled by a linear proxy and J_2 frozen at its initial Maclaurin value, hence an uncertainty of a few $R_\oplus$ on the exact position and on its remaining within the 15–25 $R_\oplus$ window. The highest-priority task is therefore the coupled three-dimensional thermo-orbital calculation, replacing the three simplified models above with a self-consistent evolution of $\Omega(t)$, $R(t)$, $J_2(t)$, and the orientation of $\boldsymbol{L}_p$. The determination of Q_{global}, previously at the heart of this limitation, is downgraded to a secondary question: it now conditions only the secondary contribution of the evection channel[17,97] and the timing (and not the position) of the recession. If a more complete calculation brought $\varepsilon_{\text{I}}^{(0)} \lesssim 50°$ or shifted the convergence outside the window, the TTP would revert to the earlier configuration of this limitation (closure by evection, requirement $Q_{\text{global}} \gtrsim 10^5$) — a falsifiable branch stated in Prediction P24.

Acknowledged limitation — L16 — Proximity of the nominal tuning to the Maclaurin instability limit

The nominal ejection threshold ($\alpha = 0.5$, $\Omega_{\text{crit}}/\Omega_0 = 1.27$, Section 9.2) operates precisely 1.60% above the strict dynamical instability limit of the Maclaurin spheroid, independently derived in Section 4.3 ($\Omega/\Omega_0 = 1.25 = 3.5/2.8$). The derivation of the instability coefficient λ (Appendix C) assumes an axisymmetric Maclaurin geometry throughout the ejection cycle; its strict validity in the immediate vicinity of this limit — where a bifurcation toward a triaxial equilibrium figure (Jacobi) becomes possible — is not demonstrated. Two readings remain open: this proximity could reflect a causal link between the shape instability and the ejection mechanism itself, or constitute a numerical coincidence without physical significance. Deciding between these two readings requires a simulation allowing non-axisymmetric deformations, in complement to the 3D SPH simulation already required by Limitations L1 and L2. The causal reading — the local instability $\lambda = 0$ as a manifestation, in the large-wavenumber limit, of the global destabilization of the spheroid — is judged by the author as the most natural interpretation of this numerical proximity, but remains a working hypothesis, not demonstrated in this work.

20.1 Synthesis: closure status table

In summary, the maturity status of the TTP can be read in a compact table associating each sub-module with its deferred closure and the test that will decide it. None of these unknowns is hidden behind a coefficient presented as fundamental; each is localized at the level of the physical mechanism on which it depends.

Sub-module	Deferred closure	Decisive test
Instability ($\lambda = 0$)	α, β	internal transport + hydrodynamics
Ejection / capture	f_{ej}, f_{cap} (ejecta distribution)	3D SPH simulation
Tidal rheology	$Q(T, \omega)$, $k_2(T)$	thermo-viscoelastic model
LPT crossing / resonances	integrated secular torque	full orbital integration
Angular momentum	vector losses $\sum_k \boldsymbol{L}_{\mathrm{esc},k}$	ejecta budget

This table is not a list of free parameters, but a map of the physical phenomena remaining to be resolved. The TTP is, at this stage, a theory whose mechanism is explicit, the main equations posed, the closures identified and acknowledged, and the tests assigned. Its definitive test will come from upcoming lunar exploration missions (Section 17).

21 Conclusion

This work proposes that the formation of the Moon may be the product of a Triple Phase Transition within a completely molten proto-Earth, rotating at ≈ 3.5 h, whose axis oscillates within $[40^\circ, 70^\circ]$ in its own co-rotating frame — an interval supported by five convergent arguments, the first and most fundamental being the logically necessary absence of any tidal stabilizer. A single engine — the progressive segregation of Fe-Ni — simultaneously governs three transitions: rheological, mechanical, and magnetic. The result is a Moon built layer by layer in two to three massive episodes, each layer archiving in its internal structure the state of Hadean differentiation at the time of its accretion.

Parametric elliptic instability — established in fluid mechanics[66–68] but never applied to lunar formation — amplifies the CMT to the bifurcation speed in ≈ 37 hours. A dedicated section treats the maintenance of obliquity in the out-of-equilibrium Hadean context. A three- to four-week chronology gives rise to several new testable predictions.

The mass budget is consistent without forcing: the target mass for the TTP mechanism is less than the currently observed lunar mass, because the SPA impactor and late accretion contributed subsequently. At $T_{\mathrm{rot}} = 3.5$ h, the ejectable mass per episode can reach $\approx 70\%$ of the current lunar mass, which explains why $N = 2$–3 episodes suffice.

The angular momentum budget, however, remains partially open: a post-formation external sink of $\approx 2.95 \times 10^{34}$ J s is required. Reconstruction identifies it without free parameter: the current angular momentum and the lunar inclination of 5° jointly impose an instability episode at the crossing of the Laplace plane transition[15,16], whose only activation condition — an inertial obliquity $\varepsilon_I \gtrsim 60$–$65°$ — is inherited from the ejection recoils and therefore native to the TTP (Section 12). This geometric channel is robust with respect to the high dissipation of the magma ocean. The evection resonance[16,17] remains a secondary channel, whose efficiency depends on the global quality factor Q_{global} — distinct from the local Q_{CMT} and controlled by the solid mantle. The open question is no longer the existence of the sink but the exact value of the inertial obliquity at crossing: a solid mechanics calculation internal to the theory (Limitation L15), falsifiable by prediction P24 of the lunar inclination.

Preliminary observational support comes from Chang'e-6: norites at 4247 ± 5 Ma[18] and high-Fe/Mn olivines[19] are consistent with predictions P3 and P6, although not yet confirmatory. Prediction P6 is presented here in a fully quantified form: an Fe/Si enrichment threshold of 1.5% for South Pole ejecta, derived from first principles and not adjusted to fit existing data, with an explicit grading distinguishing partial support (Episode 2 material only) from strong support (an unambiguous Episode 1 signature).

The central prediction remains: one or more seismic interfaces within the 200–530 km window (or at ≈ 177 km for $N = 2$), testable by Chang'e 7 in August 2026. If an interface with $|R| > 0.02$ is detected within this window, concentric stratification receives its first strong observational support, ahead of any SPH simulation. If no interface is detected under the stated falsification conditions, prediction P1 is seriously challenged and the theory will require revision — which the author explicitly accepts.

This work invites the planetary science community to evaluate the TTP on its own terms: equations are numbered, derivations are explicit, hypotheses are hierarchized, limitations are acknowledged, and predictions are falsifiable. The TTP will be judged primarily on its observational predictions — the seismic data from Chang'e 7 (August 2026) and samples from Artemis III (2028–2029) — before and independently of any full numerical validation. The theory stands or falls on these results, not on the elaboration presented in this manuscript.

A Adiabatic temperature profile

The TTP does not require melting at the planet's center, nor through the body as a whole: it requires melting specifically within the Hadean intertropical belt $|\phi| < 30°$, in the upper and middle mantle (Section 4.4), where the ejection mechanism operates. This is already thermodynamically guaranteed by the accretion energy to fusion energy ratio $E_{\text{grav}}/E_{\text{fus}} \approx 121$ (Eq. 7), independently of any assumption about the deep interior or the core region, which lies well below the belt relevant to this work and is not part of the volume on which the TTP mechanism operates. A dedicated hydrostatic model of the rotating flattened proto-Earth, solving pressure and temperature as a function of both depth and latitude specifically within the intertropical belt, would refine this picture, but is not required to establish Hypothesis H1: the factor ≈ 121 already guarantees melting in the region the theory concerns, by construction.

B Demonstration of the $b' = 1$ attractor

The parameter b' is defined as the ratio of the critical ejection speed U_{crit} to the geostrophic equilibrium speed U_{eq} of the CMT:

$$b' = \frac{U_{\text{crit}}}{U_{\text{eq}}} = \frac{2\Omega \sin \varepsilon \sqrt{2|g_{\text{eff}}| R_{\text{eq,proto}}}}{|g_{\text{eff}}|}. \tag{85}$$

Evolution due to Fe-Ni segregation. The CMT density decreases as $\rho_{\text{CMT}}(t) = \rho_0(1 - \xi_{\text{Fe}}(t))$, where $\xi_{\text{Fe}}(t)$ is the fraction of Fe-Ni segregated. The effective gravity $|g_{\text{eff}}|$ is proportional to ρ_{CMT}, so

$$\left.\frac{\mathrm{d}b'}{\mathrm{d}t}\right|_{\text{segregation}} = \frac{1}{2}\frac{\dot{\rho_{\text{CMT}}}}{\rho_{\text{CMT}}}\, b' > 0, \tag{86}$$

since $\dot{\rho_{\text{CMT}}} < 0$: Fe-Ni segregation increases b'.

Evolution due to ejection. An ejection carries away a fraction $\epsilon = M_{\text{ej}}^{\text{max}}/M_{\text{CMT}} \approx 0.1$ of the CMT mass. Angular momentum decreases as $\Delta\Omega/\Omega \approx -\epsilon$, and effective gravity decreases as $\Delta|g_{\text{eff}}|/|g_{\text{eff}}| \approx -\epsilon$ (proportional to lost mass). Thus,

$$\left.\frac{\Delta b'}{b'}\right|_{\text{ejection}} = \frac{\Delta\Omega}{\Omega} - \frac{1}{2}\frac{\Delta|g_{\text{eff}}|}{|g_{\text{eff}}|} \approx -\epsilon + \frac{1}{2}\epsilon = -\frac{1}{2}\epsilon < 0 : \tag{87}$$

ejection decreases b'.

Stable fixed point. The two trends oppose each other: segregation increases b', ejection decreases it. The equilibrium point $b' = 1$ is a stable attractor. For $b' < 1$, the system is unstable and an ejection occurs, which further decreases b' (since $\Delta b' < 0$); for $b' > 1$, the system is stable, no ejection occurs, and segregation pushes b' upward, toward 1, from below. The point $b' = 1$ is therefore a stable attractor of the coupled segregation–ejection system.

C Complete derivation of the coefficient λ

This appendix gives the complete analytical derivation of the stability coefficient $\lambda(U)$, with dimensional verification and, for each term, its physical history. It follows the Solberg-Høiland parcel displacement formalism[48,79,80] adapted to the TTP geometry, and yields the form used in Section 9.2.

C.1 The stability equation

The idea of the method is contained in one sentence: one mentally displaces a small fluid parcel radially, sums the forces that bring it back or carry it away, and the sign of the sum says whether the torus is confined or ejected. For a radial infinitesimal displacement ξ_r of a CMT parcel:

$$\ddot{\xi}_r = \lambda(U)\,\xi_r, \quad \lambda > 0 \text{ (instability)}, \quad \lambda < 0 \text{ (stable confinement)}. \tag{88}$$

The displacement is *spherical* radial and the latitude is measured from the body's instantaneous dynamic equator — no orbital reference (ecliptic) enters the internal dynamics, which knows only Ω and the local vertical. The angle $\varepsilon_{\rm dyn}$ denotes the *colatitude of the CMT* relative to this dynamic equator: $\sin\varepsilon_{\rm dyn} = \cos\phi$, where ϕ is the torus latitude ($\varepsilon_{\rm dyn} = 70^\circ \leftrightarrow \phi = 20^\circ$, at the heart of the intertropical belt).

C.2 The Rayleigh (epicyclic) term

The displaced parcel conserves its specific angular momentum $L = rU$, while the background flow follows the profile $U(r) \propto r^\alpha$. Upon arriving at a radius where the background rotates differently, the parcel experiences a restoring force: this is the Rayleigh mechanism[84], whose intensity is the square of the epicyclic frequency,

$$\kappa^2 = \frac{1}{r^3}\frac{\mathrm{d}\,(rU)^2}{\mathrm{d}r} = \frac{2(1+\alpha)\,U^2}{r^2}, \tag{89}$$

obtained by differentiating $(rU)^2 \propto r^{2(1+\alpha)}$. This term is always positive (stabilizing) for any physical profile ($\alpha > -1$). Essential point: this centrifugal charge gradient contains *by itself* the entire centrifugal effect. Adding a separate "centrifugal pressure" term would count the same effect twice; the present derivation contains none.

C.3 The non-traditional Coriolis component (Eötvös effect)

In a rotating frame, any prograde flow is *lightened* radially: the Coriolis force applied to the azimuthal speed U has a component along the local vertical, $+2\Omega U\cos\phi$, directed outward. This is the Eötvös effect — the same one that makes an object lighter when moving eastward. The geophysical "traditional approximation" neglects this component; it must be retained at low latitudes and for deep flows[83] — precisely the regime of the intertropical belt. Writing $\sin\varepsilon_{\rm dyn} = \cos\phi$:

$$a_{\text{Eötvös}} = +\frac{2\Omega\sin\varepsilon_{\rm dyn}}{r}\,U\,\xi_r, \tag{90}$$

destabilizing term: it opposes gravitational confinement.

C.4 The role of Euler acceleration (nutation)

The rotation axis nutates, and one might be tempted to add to λ an Euler term from $-\dot{\mathbf{\Omega}} \times \boldsymbol{r}$. This temptation must be resisted, for a structural reason: λ gathers accelerations *proportional to the displacement* ξ_r, while the Euler acceleration is a *forcing* depending on time, independent of ξ_r (and a combination such as $|\dot{\Omega}|\, r/\Omega^2$ is moreover homogeneous to meters, not s^{-2}). Its physical place is elsewhere, and it is already occupied in the TTP: nutation slowly modulates the effective colatitude $\varepsilon_{\mathrm{dyn}}(t)$ — the modulation parameter $|\dot{\Omega}|/\Omega^2 \approx 0.08 \sin\theta_{\mathrm{nut}} \ll 1$ guarantees the adiabatic regime — and periodically lowers Ω_{crit} via the Eötvös term $2\sin\varepsilon_{\mathrm{dyn}}$. This is precisely the *parametric amplification* channel of Section 9.2: the Euler effect is real, maximal at nutation peaks, and enters the mechanism as a threshold modulation, not as a term of λ.

C.5 Buoyancy: thermal–compositional sharing

The body's density gradient has two sources, and the derivation separates them in the manner of the Ledoux criterion. The *thermal* part — the magma is cooled from above, hence superadiabatic — enters via the Brunt–Väisälä frequency[81,82] along gravity:

$$-N^2 > 0 \qquad (N^2 < 0,\ \text{convection}), \tag{91}$$

destabilizing contribution that sustains convection. The *compositional* part — Fe-Ni segregation makes the body denser at depth — is measured by $\beta \equiv -\mathrm{d}\ln\rho/\mathrm{d}\ln r > 0$ and enters through the *centrifugal* channel. The picture is simple: the centrifugal force is a gravity pointing outward; a stratification stable for true gravity (dense at bottom) is, for this inverted gravity, a Rayleigh-Taylor-like configuration, which pushes dense fluid outward:

$$a_\beta = +\frac{\beta U^2}{r^2}\,\xi_r. \tag{92}$$

Destabilizing term — and, as we shall see, indispensable: this is the fuel of the engine. Fe-Ni segregation thus serves a dual purpose in the TTP: it accelerates rotation by contraction (skater effect) and it creates the stratification whose centrifugal buoyancy feeds the ejection.

C.6 Assembled coefficient and dimensional verification

Summing gravitational confinement $-g_{\mathrm{eff}}/r$, thermal buoyancy $-N^2$, the Eötvös effect (90), Rayleigh restoring (89), and centrifugal buoyancy (92):

$$\lambda(U) = -\frac{g_{\mathrm{eff}}}{r} - N^2 + \frac{2\Omega\sin\varepsilon_{\mathrm{dyn}}}{r}\,U - \frac{2(1+\alpha)-\beta}{r^2}\,U^2. \tag{93}$$

Each term is in s^{-2}: $[g_{\mathrm{eff}}/r] = (\mathrm{m\,s^{-2}})/\mathrm{m} = \mathrm{s}^{-2}$; $[N^2] = \mathrm{s}^{-2}$; $[\Omega\sin\varepsilon_{\mathrm{dyn}}\,U/r] = \mathrm{s}^{-1}(\mathrm{m\,s^{-1}})/\mathrm{m} = \mathrm{s}^{-2}$; $[U^2/r^2] = (\mathrm{m\,s^{-1}})^2/\mathrm{m}^2 = \mathrm{s}^{-2}$.

C.7 Why is the Eötvös term destabilizing here?

The apparent tension with the classical result — according to which Coriolis forces acting on a purely azimuthal motion (in the equatorial plane) are absorbed into the epicyclic frequency and are therefore stabilizing, not destabilizing — is resolved by a standard distinction in geophysical fluid dynamics, but one easily overlooked: the relevant direction for this classical result is the *cylindrical* radial direction (perpendicular to $\mathbf{\Omega}(t)$, in the *instantaneous* plane of rotation), while the relevant direction for the present stability analysis is the *spherical* radial direction (the local vertical, i.e., pointing away from the body's center, the direction along which buoyancy and effective gravity act).

No fixed axis, no external reference plane

It must be repeated here, at the precise point where it matters most for the present derivation: $\mathbf{\Omega}(t)$ is not a fixed axis. As established in Section 3, the Hadean proto-Earth possesses neither a tidal stabilizer, nor a Solar System reference plane — neither ecliptic, nor stable lunar orbit, nor fixed solar axis exist at this epoch. The rotation vector undergoes a large-amplitude, chaotic, permanent free nutation within $\varepsilon \in [40^\circ, 70^\circ]$, never stabilizing toward a fixed orientation. Consequently, "the equator" throughout this appendix denotes the *instantaneous dynamic equator* defined perpendicular to $\mathbf{\Omega}(t)$ at each instant — itself in perpetual motion with the body's nutation — and never a fixed geographic or astronomical reference. The colatitude ε_{dyn} appearing below is likewise an instantaneous quantity, re-evaluated continuously as $\mathbf{\Omega}(t)$ wanders; it is never "reached" nor "stabilized" at a particular value, and the CMT itself is carried through the entire oscillation range with the axis that defines its own local equator. As throughout this manuscript, the physical structure on which the theory rests is the *intertropical belt* $|\phi| < 30^\circ$, not the equatorial line itself: the derivation below evaluates the Eötvös projection as a function of colatitude to characterize its behavior across this belt, not to require that the CMT lie exactly on $\phi = 0$.

These two directions — cylindrical radial and spherical radial — coincide only where the CMT parcel lies, at that instant, exactly on this instantaneous dynamic equator ($\varepsilon_{\text{dyn}} = 90^\circ$, under the convention $\sin \varepsilon_{\text{dyn}} = \cos \phi$ adopted throughout this work). At this instantaneous alignment, the spherical radial direction has a *zero* projection along $\mathbf{\Omega}(t)$, so the two directions are parallel and the full amplitude of the cylindrical-radial Coriolis force projects onto the spherical radial direction relevant for buoyancy: the destabilizing term is then *maximal*. Moving the instantaneous colatitude of the parcel away from the dynamic equator (toward smaller ε_{dyn}, i.e., toward the instantaneous pole of $\mathbf{\Omega}(t)$) increases the angle between the two directions, reducing the projected component, which vanishes only at the instantaneous pole itself ($\varepsilon_{\text{dyn}} = 0^\circ$), where the two directions are orthogonal. The Coriolis force of a purely azimuthal motion, which is confined to the cylindrical radial direction (in the plane of rotation) by elementary vector mechanics ($\mathbf{\Omega}(t) \times v_{\mathrm{az}}$ lies in the plane instantaneously orthogonal to $\mathbf{\Omega}(t)$ for a purely azimuthal v_{az}), therefore has, once projected onto the spherical radial direction relevant for the parcel's buoyancy balance, a component proportional to $\sin \varepsilon_{\mathrm{dyn}}$ — maximal when the CMT lies on the instantaneous dynamic equator, and vanishing only at the instantaneous pole. This projection is precisely the origin of the "non-traditional" Coriolis terms documented in the geophysical fluid dynamics literature[83], and it is, in fact, formally identical to the classical Eötvös effect itself: the same term $2\Omega U \cos \phi$ (with ϕ the latitude, $\cos \phi = \sin \varepsilon_{\mathrm{dyn}}$) by which the apparent weight of an eastward-moving object is measurably reduced at Earth's surface — an effect documented for over a century, routinely measured in marine and airborne gravimetry, not a speculative construction. Since the CMT occupies the Hadean intertropical belt $|\phi| < 30°$ — close to the instantaneous dynamic equator by construction (Section 5) — this term remains near its maximum value, strongly destabilizing, throughout the oscillation, rather than being an incidental or finely tuned contribution.

This resolves the apparent tension without requiring numerical confirmation: the classical statement by Morbidelli is correct for a purely azimuthal motion analyzed along the cylindrical radial direction, and the present analysis concerns a different, but equally well-defined, projection — the spherical radial direction relevant for the Solberg-Høiland buoyancy criterion — for which the $\sin \varepsilon_{\mathrm{dyn}}$ term is the standard documented result.

The following three points provide additional independent reinforcement of this conclusion. (1) Off the equator, the epicyclic absorption acts only on the cylindrical projection of the displacement and the non-traditional component must be retained[83]. (2) The TTP rotation varies on spheres ($U(r)$ in spherical radius) and not on cylinders: it violates the second Goldreich–Schubert stability condition ($\partial\Omega/\partial z = 0$)[121], so there exist *inclined* displacement paths along which the Solberg-Høiland criteria are violated — non-diffusive instability as soon as $N^2 \leq 0$, which is the case here, *independently* of the sign assigned to the radial Coriolis term. (3) Classical inviscid criteria *underestimate* instability whenever the Prandtl number deviates from unity[122], which is the case for silicate magma. Finally, the latitudinal shear at the torus edge, $\partial U/\partial y \sim U/W$, is comparable to 2Ω (ratio ≈ 0.95 at nominal values): the CMT operates near the equatorial inertial instability criterion[123]. The geometric condition of the present criterion — adverse absolute angular momentum gradient along inclined paths at the torus edge — is verifiable by 3D SPH simulation and joins Limitation L14.

C.8 The rotation threshold

Setting $\lambda = 0$ and substituting $U = \Omega r$:

$$\Omega^2_{\text{crit}} = \frac{g_{\text{eff}}/r + N^2}{2\sin\varepsilon_{\text{dyn}} - 2(1+\alpha) + \beta}. \tag{94}$$

A finite threshold exists only if the denominator D is positive, i.e., $\beta > 2(1+\alpha) - 2\sin\varepsilon_{\text{dyn}}$. The driving role of centrifugal buoyancy is read here: at nominal values, $2\sin\varepsilon_{\text{dyn}} - 2(1+\alpha) = 1.879 - 3 = -1.12 < 0$ — without β, no rotation would ever trigger ejection. With $\varepsilon_{\text{dyn}} = 70°$ at the nutation peak, $\alpha = 0.5$, $\beta = 2.5$ and $N^2 = -1.93 \times 10^{-7}\ \text{s}^{-2}$, the denominator is $D = 1.379$ and the threshold $\Omega_{\text{crit}} \approx 1.27\,\Omega_0$. Note that the present form, freed from double counting, halves the sensitivity of the threshold to the velocity profile ($\partial D/\partial\alpha = -2$ instead of -4), and coincides numerically with the previous form at $\alpha = 0.5$. This threshold is a well-defined surface in parameter space $(\Omega, \varepsilon_{\text{dyn}}, \beta)$, giving the mechanism a predictive character: ejection occurs when $\Omega > \Omega_{\text{crit}}$, the crossing occurring by internal means with conserved angular momentum — segregation contraction (skater effect) and parametric amplification (Section 9.2) — the impulsive Late Veneer trigger acting only as a modulation at the nutation peak.

At the end of this derivation, the status of each contribution is fixed without ambiguity: the Eötvös, Rayleigh, thermal buoyancy, and effective gravity terms are functions of state variables $(\Omega, \varepsilon_{\text{dyn}}, T, \ldots)$ and the adopted constitutive laws; only α (differential rotation profile) and β (density gradient, centrifugal buoyancy) remain not derived from first principles. The bifurcation surface $\lambda(\mathbf{X};\alpha,\beta) = 0$ (Eq. 29) thus carries a residual closure strictly *two-dimensional*, and the critical speed is derived as a diagnostic projection, $U_{\text{crit}} = U_{\text{crit}}(\mathbf{X};\alpha,\beta)$. The appropriate robustness test of the mechanism is therefore not a sweep of U_{crit}, but the sensitivity analysis:

$$\beta \longmapsto \big(U_{\text{crit}},\ t_{\text{decl}},\ M_{\text{ej}},\ \boldsymbol{L}_{\text{ej}},\ N_{\text{ej}}\big), \tag{95}$$

carried out at fixed α, then on the pair (α, β) jointly. The mechanism is robust if a physically reasonable range of these coefficients always produces $N = 2$–3 crossings and similar outputs; it would reveal fine tuning if only a narrow band sufficed.

D Demonstration of the $\mathrm{Bi}_{\mathrm{KH}}$ criterion

The Bingham-Kelvin-Helmholtz number compares the yield stress to the KH shear stress:

$$\mathrm{Bi}_{\mathrm{KH}} = \frac{\tau_y\, k}{\rho\, \sigma_{\mathrm{KH}}\, \Delta U}, \tag{96}$$

where k is a geometric factor (~ 1), $\sigma_{\mathrm{KH}} \sim \Delta U/\lambda$ is the KH growth rate, and λ is the wavelength of the perturbation. Thus,

$$\mathrm{Bi}_{\mathrm{KH}} \sim \frac{\tau_y\,\lambda}{\rho\,\Delta U^2}. \tag{97}$$

For the CMT: $\tau_y \sim 10^2$–10^4 Pa, $\lambda \sim R_{\mathrm{torus}} \sim 10^7$ m, $\rho \sim 3\,000\ \mathrm{kg\,m^{-3}}$, $\Delta U \sim U_{\mathrm{crit}} \sim 10^4\ \mathrm{m\,s^{-1}}$. We obtain $\mathrm{Bi}_{\mathrm{KH}} \sim 3\times10^{-3}$–$3\times10^{-1}$.

The effective cohesion is the product of three independent reinforcements (atmospheric confinement, internal crystal network, Rayleigh-Taylor suppression), detailed in Section 9.1:

$$\mathrm{Bi}_{\mathrm{KH}}^{\mathrm{eff}} = \mathrm{Bi}_{\mathrm{KH}} \times f_{\mathrm{atm}} \times f_{\mathrm{crist}} \times f_{\mathrm{RT}}, \tag{98}$$

with $f_{\mathrm{atm}} \sim 10$–100 (atmospheric confinement, $P \sim 10^5$–10^7 Pa), $f_{\mathrm{crist}} \sim 3$–$10$ (internal crystal network), and $f_{\mathrm{RT}} \sim 10$–50 (Rayleigh-Taylor suppression, $\rho_{\mathrm{flow}}/\rho_{\mathrm{atm}} \sim 200$). With nominal values, $\mathrm{Bi}_{\mathrm{KH}} \sim 3\times10^{-2}$ and the product of the three reinforcements is ~ 300, giving:

$$\mathrm{Bi}_{\mathrm{KH}}^{\mathrm{eff}} \sim 9, \tag{99}$$

much greater than 1. Cohesion is therefore ensured.

Acknowledged limitation: this calculation is a nominal estimate. A 3D SPH simulation with full BH rheology is required to validate the exact values, in particular the nonlinear interaction between the three effects (Limitation L14).

E Derivation of the Elsasser number and B_{crit}

Balancing the Coriolis force $F_C \sim 2\rho\Omega U$ and the Lorentz force $F_L \sim B^2/(\mu L)$, with $\eta \sim UL$ (magnetic Reynolds number of order unity at the dynamo threshold):

$$B^2 \sim 2\mu\rho\Omega\eta, \qquad \Rightarrow \qquad \Lambda \equiv \frac{B^2}{\rho\mu\eta\Omega} \sim 2, \tag{100}$$

so $\Lambda \sim \mathcal{O}(1)$, consistent with the values $\Lambda \sim 0.1$–1 inferred for the current Earth's core[88]. Using $\eta = (\mu_0\sigma)^{-1}$ with $\sigma \approx 1.5\times10^6\ \mathrm{S\,m^{-1}}$, a representative electrical conductivity of liquid iron at core conditions[124], $\mu_0 = 4\pi\times10^{-7}\ \mathrm{H\,m^{-1}}$, $\rho \approx 10^4\ \mathrm{kg\,m^{-3}}$, $\Omega \approx 4.99\times10^{-4}\ \mathrm{rad\,s^{-1}}$, we obtain $B_{\mathrm{crit}} \approx 2.6$ mT, of the same order of magnitude as the field inferred within the current Earth's core[88]. Note that this value assumes Ω at its formation value ($T = 3.5$ h). At dynamo onset ($t \approx 330$–350 Myr), Ω has decreased by two cumulative effects: the loss associated with the three ejection episodes themselves ($\Delta\Omega/\Omega \approx -10$ to -15% per episode, Section 9.2), then the additional drainage by solar resonant sinks (Laplace plane transition and evection, Sections 12 and 11.6), by a factor ≈ 2. Cascading these two effects over the entire -10% to -15% per episode range, $\Omega_{\mathrm{dynamo}}/\Omega_0 \in [0.307, 0.365]$; since $B_{\mathrm{crit}} \propto \sqrt{\Omega}$, we obtain $B_{\text{crit}} \in [1.43,\ 1.56]$ mT — same order of magnitude, conclusion unchanged, still consistent with geodynamo models.

F Stokes sedimentation timescale for Fe-Ni

Stokes sedimentation speed for an Fe-Ni droplet of radius $r \approx 5$ cm in molten silicate:

$$v_{\text{Stokes}} = \frac{2}{9}\frac{(\rho_{\text{Fe}} - \rho_{\text{sil}})\, g_{\text{eff}}^{(0)}\, r^2}{\eta} \approx \frac{2}{9}\frac{3\,300 \times 6.42 \times (0.05)^2}{0.1} \approx 118\,\text{m}\,\text{s}^{-1}. \tag{101}$$

Over a characteristic sedimentation distance $\ell \approx 30$ km: $\tau_{\text{Stokes}} = \ell / v_{\text{Stokes}} \approx 254$ s. Robust for $r \in [1, 10]$ cm and $\eta \in [0.05, 1]$ Pa s.

G Accretion of magmatic ejecta: why perfect cohesion is not required

A recurring objection to the TTP mechanism concerns the cohesion of the CMT flow during transit between the proto-Earth surface and the Roche radius. This appendix demonstrates that, even without perfect cohesion of the flow, the physical properties of the ejected magma and orbital mechanics guarantee accretion into a single body.

The magma is not dust. The ejected material is silicate magma at $T \approx 3\,000$–$4\,000$ K. Its dynamic viscosity is $\eta \approx 1$–100 Pa s[37] — comparable to warm honey. Unlike solid rock fragments or a gas cloud, high-temperature magma does not disperse: it deforms and stretches, but remains dense and cohesive at the macroscopic scale. Two sticky magma drops at low relative speed do not bounce off each other — they fuse. Viscous coalescence replaces mechanical cohesion as the dominant aggregation mechanism once the flow is fragmented.

Orbital concentration. Ejecta from a given TTP episode share, to first order, the same initial orbital energy budget. By conservation of energy and angular momentum, they populate a band of nearby orbits rather than dispersing isotropically over 4π steradians. This orbital concentration is fundamentally different from an isotropic explosion: it is the same physical principle that produces accretion disks from tidally disrupted bodies. The toroidal geometry of the CMT reinforces this effect: ejection occurs preferentially within the oblique intertropical belt $|\phi| < 30°$, concentrating ejecta within a limited azimuthal solid angle. The resulting debris field is geometrically thin, maximizing mutual encounter rates.

Beyond the Roche radius: accretion is inevitable. Beyond the Roche radius $R_{\text{Roche}} \approx 3R_{\text{eq,proto}} \approx 22.2$ Mm (Section 14) — computed from the proto-Earth's own equatorial radius, not Earth's current radius — the self-gravity of the ejecta cloud overcomes tidal disruption. For co-orbital magma fragments of total mass

$M_{\rm cloud} \approx 2$–4×10^{22} kg, the gravitational accretion timescale is:

$$\tau_{\rm acc} \approx \frac{1}{\sqrt{G\rho_{\rm cloud}}} \approx 10^3\text{–}10^5\,\text{yr}, \tag{102}$$

much shorter than the thermal lifetime of the magma ($\tau_{\rm cool} \sim 10^6$–$10^7$ yr for a Moon-sized body, Elkins-Tanton[31]). During this accretion window, the magma remains above its solidus: every collision is plastic, not elastic. There is neither a fragmentation cascade nor loss by rebound. The capture efficiency $f_{\rm cap}$ therefore has a physical lower bound independent of the BH cohesion hypothesis:

$$f_{\rm cap}^{\rm min} \approx \frac{M_{\rm cloud}(r > R_{\rm Roche})}{M_{\rm ej}^{\rm total}} \gtrsim 0.45\text{–}0.55, \tag{103}$$

the lost fraction ($\approx$ 45–55%) corresponding to ejecta that fall back onto the proto-Earth or escape on hyperbolic orbits.

Implication for the theory. The capture efficiency $f_{\rm cap} = 0.70$ adopted in the main text is therefore not a free parameter that the theory must defend at all costs. Even in the pessimistic scenario where the CMT flow completely fragments during ejection, orbital mechanics and magma viscosity guarantee accretion of the co-orbital debris cloud beyond the Roche radius. The critical question shifts from "does the flow remain cohesive?" to "what fraction of ejecta reaches the Roche radius?" — a question that depends on the ejecta velocity distribution, not on BH rheology. This reformulation makes the theory more robust: cohesion is advantageous, but not necessary.

H Complementary mechanisms for the ejection threshold

This appendix identifies the physical mechanisms that contribute to reaching the ejection threshold $\lambda(U) = 0$. The threshold is a condition on Ω:

$$\Omega > \Omega_{\rm crit} \approx \sqrt{\frac{|g_{\rm eff}|}{2\sin\varepsilon\, r}} \cdot \sqrt{\frac{1+\alpha}{r}}. \tag{104}$$

With nominal proto-Earth parameters ($\Omega_0 = 4.99 \times 10^{-4}$ rad/s, $\varepsilon = 55°$), the threshold is $\Omega_{\rm crit} \approx \sqrt{2}\,\Omega_0$.

The angular momentum input required to reach this threshold is provided by continuous planetesimal bombardment. Late accretion (the "Late Veneer") is delivered by a small number of large bodies (radius $\gtrsim$ 750 km)[125]. Geodynamic constraints on the supply of highly siderophile elements to the mantle — impactors larger than $\sim$ 1 km deliver metal that sinks to the core rather than remaining entrained in the convective mantle — place the total mass of late accretion up to $\sim$ 5% of Earth's mass[126]. Assigning to this mass a characteristic impact speed $v_{\text{imp}} \sim 10$ km s^{-1} and a lever arm of order r, the cumulative available angular momentum is:

$$L_{\text{imp}} \sim M_{\text{imp}}\, v_{\text{imp}}\, r \approx 2.2 \times 10^{34}\ \text{J s}. \tag{105}$$

This input is sufficient to raise Ω from Ω_0 to Ω_{crit} in the geometrically most favorable case ($\varepsilon \approx 90°$), and provides the necessary margin for ejection to occur at the peaks of the nutation cycle. The Euler oscillation term λ_4 modulates the effective obliquity ε, but does not provide the main driving force.

An external dynamical cause beyond bombardment: the early giant planet instability

Appendix H assumes the availability of intense bombardment by large bodies during the active TTP window (30–50 Myr), without identifying its dynamical cause. Recent work provides such a cause, independent of the proto-Earth's internal thermodynamics.

The classical narrative placed the orbital instability of the giant planets (analogous to the "Nice model") late, around 3.9 Ga, in connection with the lunar Late Heavy Bombardment (LHB). More recent dynamical and geochronological work challenges this late chronology: the instability would in fact be triggered by the dispersal of the protoplanetary gas disk itself, and would therefore occur *early*, a few to ten million years after Solar System formation[127,128]. Mojzsis et al.[128] show that this early migration produces an intense influx of comets and small bodies over ~ 30 Myr, bounded at $\lesssim 4480$ Ma by the absence of posterior reset ages for the most resistant chronometers of Earth, the Moon, Mars, and several meteorite parent bodies.

This window of ~ 30 Myr of intense impactor influx, motivated by a purely dynamical cause (the orbital instability of the giants), substantially overlaps the active TTP window (30–50 Myr, Section 4.4), itself constrained by independent thermal considerations (silicate atmosphere opacity, Hf-W chronometer). This convergence between two temporal windows derived from completely different physics — gas disk dispersal on one side, mantle melting thermodynamics on the other — constitutes an additional argument for the plausibility of the availability, at the right time, of the bombardment that Appendix H requires to cross the ejection threshold Ω_{crit}.

Acknowledged limitation: this temporal convergence remains, like that of Section 11.6.2, a comparison of compatible timescales and not a coupled dynamical simulation. It does not demonstrate that impactors from this instability actually reached the proto-Earth with the sizes ($\gtrsim 750$ km) and instants required by the ejection cycle; it provides a plausible dynamical mechanism where the main text simply assumed the availability of bombardment.

I Post-ejection oscillation persistence and repeated opportunities

The ejection threshold is not a single-pass condition. After an ejection, Ω falls below Ω_{crit}, but the system does not stop immediately. The cavity left by the ejection is bounded by the confining atmosphere and by the residual *'aā* crust of the CMT. These boundaries are reflective rather than absorptive (as established by the $\text{Bi}_{\text{KH}} \gg 1$ criterion, Appendix D), so post-ejection nutation persists over several cycles.

The obliquity excursion envelope $\varepsilon_{\text{exc}}(t)$ follows a two-time behavior:

$$\varepsilon_{\text{exc}}(t) \approx \begin{cases} \varepsilon_{\text{exc}}^{(0)} \cos(\omega_{\text{nut}}^{(\text{post})} t) & 0 \leq t \lesssim t_{\text{rec}} \\ \varepsilon_{\text{exc}}^{(0)} e^{-(t-t_{\text{rec}})/\tau_{\text{BH}}} \cos(\omega_{\text{nut}}^{(\text{post})} t) & t \gtrsim t_{\text{rec}} \end{cases} \tag{106}$$

where $\omega_{\text{nut}}^{(\text{post})} = 2\pi / T_{\text{nut}}^{(\text{post})}$, with $T_{\text{nut}}^{(\text{post})} \approx 30$ h, and $\tau_{\text{BH}} \sim 10^2$–$10^4$ s (Eq. 1) for the decay that takes over once the plateau ends.

Over the recharge window duration, $t_{\text{rec}} \approx 10$–12 days, this plateau covers:

$$N_{\text{opp}}^{(\Delta)} = \frac{t_{\text{rec}}}{T_{\text{nut}}^{(\text{post})}} \approx \frac{240\text{–}288\,\text{h}}{30\,\text{h}} \approx 8\text{–}10 \tag{107}$$

complete cycles, each transiently bringing ε near its post-shock maximum excursion. The ejection threshold is therefore not a single attempt but a modest number of repeated opportunities, each with high individual probability.

Implication for the ejection mechanism. The post-ejection plateau window is a period of ≈ 8–10 nutation cycles (about 10–12 days) during which ε remains near its post-shock maximum. A major Late Veneer impact occurring during this window benefits from a double advantage:

1. ε is already near its maximum, so Ω_{crit} is minimal;
2. The impact's impulsive torque provides a jump $\Delta\Omega$ that adds to Ω already near the threshold.

The combination of the two erases the threshold and triggers the next ejection. This mechanism links ejection episodes to late impacts (Late Veneer), and not to a simple continuous accumulation of angular momentum. The isotopic signature of these late impacts (excess of siderophiles) should be present in the outermost lunar layers (Prediction P22).

J List of abbreviations

BH Bingham-Herschel, rheological law.

BiKH Bingham-Kelvin-Helmholtz number.

CME Coronal Mass Ejection.

CMT Coherent Magmatic Torus.

$\varepsilon_{\rm dyn}$ CMT colatitude relative to the instantaneous dynamic equator ($\sin \varepsilon_{\rm dyn} = \cos \phi$).

FSS Farside Seismic Suite.

Hf-W Hafnium-tungsten chronometer.

LHB Late Heavy Bombardment (≈ 3.9 Ga).

LEMS Lunar Environment Monitoring Station.

LMO Lunar Magma Ocean.

NLS Nonlinear Schrödinger equation.

POB Proto-Orbital Body.

SPA South Pole–Aitken (basin).

SPH Smoothed Particle Hydrodynamics.

TTP Triple Phase Transition.

Declarations

Data and material availability. All data generated or analyzed during this study are included in this published article (and its appendices). The interactive simulation accompanying this work is freely accessible at: https://orion4622.github.io/moon-formation-triple-phase-transition/Animation_Formation_de_la_Lune_v2.html.

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